TFPT: Carrier Algebra, Closure Theorems, and Intrinsic Observation

An overdetermined rank-5 spectral theory from the admissible reduced datum

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Abstract

From the admissible reduced datum

$$\mathfrak{T}^{\mathrm{adm}}_{\min} = (\mathcal{A}, \mathcal{H}, D, J, \Gamma, \iota, \nu),$$

the carrier-first architecture reconstructs the primitive layer, the unique minimal carrier $E_3 \oplus E_2$, the global Standard Model quotient $S(U(3)\times U(2)) \cong (SU(3)\times SU(2)\times U(1)_Y)/\mathbb{Z}_6$, one chiral family with ν^c from the positive half-spinor, the hypercharge–seam inversion $6Y^2 - Y - \mathbf{1} = 0$, and the exact electromagnetic closure equation whose canonical admissible branch reproduces $\alpha^{-1}(0) = 137.035\,999\,2$.

The same closed theorem surface then generates the canonical admissibility selector P_{adm}, the family bundle with $\mathrm{Ind}^{\mathrm{orb,APS}}(D^+_{\mathrm{rel}} \otimes L_F) = 3$, full operator reconstruction of (X, Σ, g, σ) from the admissible operator, exact Yukawa transport and holonomy phases, hadronic closure, anomaly cancellation with seam inflow, strong-CP protection, reflection positivity, Lorentzian reconstruction, a regulator-independent UV Bernstein-cone class, the cosmology block $(\Omega_{\mathrm{DM}}, \eta_B, \Lambda_{\mathrm{obs}}, n_s, r, T_R)$, and the intrinsic observation functor $\mathcal{R}_{\mathrm{TOE}}$.

Compression identities remain structural rather than decorative: the rank-lock $2^{g-3} = g-1$ fixes $g = 5$, the hard 41-chain $10\,b_1 = \Omega_{\mathrm{adm}}\gamma + 1 = 41$ overdetermines the canonical branch, and wrong discrete worlds still fail the electromagnetic closure decisively. Optional E_8 arithmetic continuations, asymptotic expansions, and horizon language are retained only as appendix-level extrapolations and are not part of the closed theorem surface.

Reading Status for the Present Paper

- **Reduced datum:** $\mathfrak{T}^{\mathrm{adm}}_{\min} = (\mathcal{A}, \mathcal{H}, D, J, \Gamma, \iota, \nu)$ with admissible spectral regularity, seam collar, and positive complete-Bernstein spectral measure.
- **Derived closure:** primitive layer, minimal carrier $E_3 \oplus E_2$, global quotient, one chiral family, admissibility selector P_{adm}, family/Higgs/local-scale closure, full operator reconstruction, exact transport closure, hadronic closure, strong CP, Lorentz reconstruction, UV Bernstein-cone closure, and cosmology closure.
- **Intrinsic observation layer:** solved global closure functional, intrinsic threshold outputs, and the observation functor $\mathcal{R}_{\mathrm{TOE}}$.
- **Optional downstream grammar:** E_8 scale grammar, asymptotic expansions, and horizon semantics, none of which are load-bearing for the closed theory surface.

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1 Reading guide and claim map

The paper is organized in three reading layers:

- **Big picture:** one section explains what TFPT claims in plain language before the detailed datum and conventions begin.
- **Detail:** the main body then runs from primitive core and hard carrier kernel through kernel numerics, the two closure-architecture sections, the closure theorem stack, the particle/mass compiler, and the benchmark interface.
- **Proof sketches:** the main text gives only the central proof sketches, while dense technical derivations remain in appendices or companion material.

The status language is also fixed once and for all:

- **Reduced datum** means the admissible spectral starting package from which the primitive layer is reconstructed.
- **Derived** means theorem-level output on the closed TFPT surface.
- **Intrinsic observation** means output generated by the unique global closure solution through $\mathcal{R}_{\mathrm{TOE}}$.
- **Optional downstream grammar** means appendix-level arithmetic continuations, asymptotic expansions, or horizon semantics not needed for closure.

To keep theorem language, benchmark semantics, and appendix bookkeeping lexically stable, all prose, captions, and status labels are kept in English throughout.

Datum Hierarchy for the Present Draft

- **Reduced datum:** $\mathfrak{T}^{\mathrm{adm}}_{\mathrm{min}}$ with admissible center, involution, seam collar, and positive spectral measure.
- **Derived kernel and closure:** reconstructed primitive layer, minimal carrier $E_3 \oplus E_2$, global quotient, family bundle, P_{adm}, geometry, Higgs, Yukawa transport, hadronic sector, UV closure, and cosmology closure.
- **Intrinsic observation layer:** global minimizer $x_\star$, intrinsic thresholds, and observable generation through $\mathcal{R}_{\mathrm{TOE}}$.
- **Optional downstream grammar:** E_8 continuations, asymptotic transport expansions, and horizon semantics.

The APS family closure used throughout the main text is now read as canonically induced by the doubled admissible branch; no separate family datum remains outside the theorem surface.

2 Big picture: what TFPT actually claims

TFPT is presented here as a carrier-first theory of admissible vacua on the oriented double cover. The primitive datum fixes the relative spectral starting package $(X, \Sigma, \iota, D_{\mathrm{rel}}, \mathcal{H}_{\mathrm{int}}, f, \chi_{\mathrm{seed}})$ before any carrier decomposition is imposed. The later hard carrier theorem then enforces the minimal split $E_3 \oplus E_2$ and hence the global Standard Model group, hypercharge, one chiral family, seam normalization, and the carrier-form electromagnetic closure equation. Under the declared admissibility, $\mathrm{APS}_{\mathrm{fam}}$, Bernstein-kernel positivity, and reflection-positivity axioms, the same architecture closes further: P_{adm} becomes an operative selector, the imported APS family theorem fixes F and $N_{\mathrm{fam}} = 3$, the Callias index selects the Higgs, the local spectral-scale channel generates both the Planck and electroweak scales and closes the gravity/cosmology

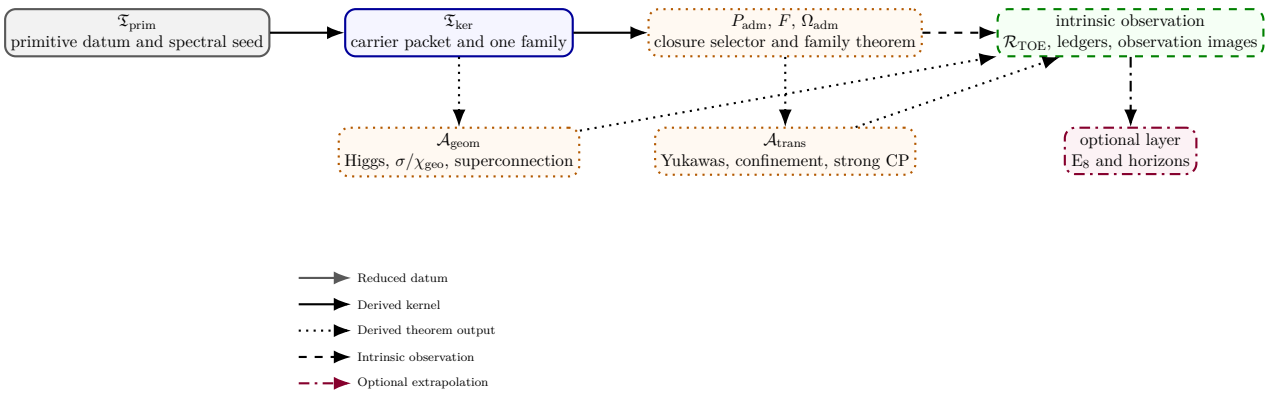


Figure 1: Claim map for the carrier-first compiler form across reduced datum, derived kernel, derived theorem output, intrinsic observation, and optional extrapolations.

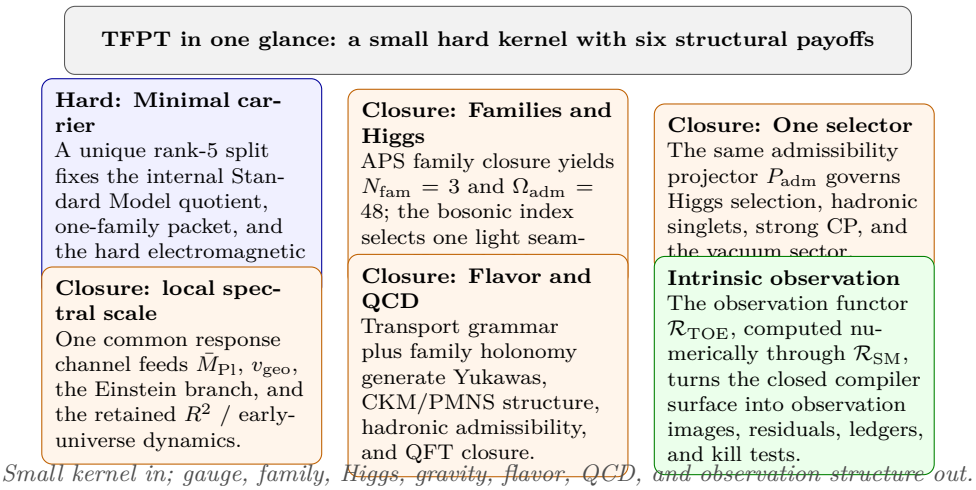


Figure 2: Headline claims of the present draft. The figure is intentionally simpler than the dependency maps that follow: its job is to put the manuscript's core payoff on one page, while still keeping the hard / closure / intrinsic-observation split visible.

branch, discrete transport emits Yukawas and mixings in matrix-holonomy form, and the same selector grammar controls hadrons, anomaly inflow, strong CP, nonperturbative QFT closure, Lorentz reconstruction, and the admissible vacuum with renormalized seam energy $\Lambda_{\Sigma,\rm ren}$. The comparison map then translates this closed compiler surface into PDG-style observables.

Figure 2 compresses that statement into six headline claims. The point is not that every downstream extension is already proved without qualification, but that the present draft claims a small carrier-first kernel with unusually broad structural reach.

TFPT master equation and selectors

$$S_{\rm TFPT} = \mathrm{Tr}\, f\left(\frac{D_{\rm rel} + A_\Sigma^{\rm seed}}{\chi_{\rm seed}}\right) - \mathrm{Tr}\, f\left(\frac{D_{\rm ref}}{\chi_{\rm seed}}\right) + \frac{i\pi}{2}\Delta\eta_\Sigma, \qquad A_\Sigma^{\rm seed} := \mathcal{B}_{\rm rel} \oplus \nabla_{\chi_{\rm seed}}. \tag{1}$$

$$\begin{gathered} \operatorname*{Stat}_{\alpha,\chi_{\rm seed},e,\varpi_{\rm spin}} S_{\rm TFPT} = 0, \qquad P_{\rm adm} := P_{\Sigma,+} P_{\rm sing} P_\Theta P_{\rm gap}, \\ \mathrm{Ind}_{APS}(D_{\rm rel}^+ \otimes L_F) = 3, \qquad \mathrm{Ind}_{\rm rel}^{P_{\rm adm}}(\mathcal{B}_{E_2}^+) = 1, \qquad \mathrm{Ind}_{\rm rel}^{P_{\rm adm}}(\mathcal{B}_{E_3}^+) = 0. \end{gathered} \tag{2}$$

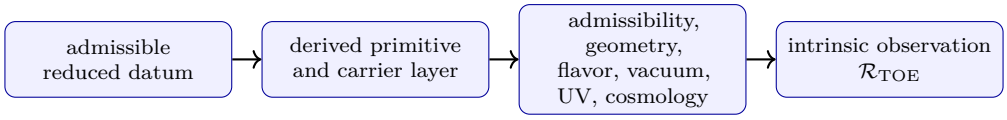
Layer or statement	Role	Proof source or remaining note	Main empirical or structural test
Admissible reduced datum and Bernstein spectral action	Reduced datum	admissible datum definition and primitive reconstruction theorem in the main text	failure would collapse the operator-level starting point
Oriented double cover, seam, involution	Derived	full operator-reconstruction theorem on the admissible branch	structural consistency of all later layers
Minimal carrier $E_3 \oplus E_2$	Derived	proof sketch given in Section 3.3	failure would collapse the whole packet
One-family packet $S^+ = \Lambda^{\mathrm{even}} E$	Derived	decomposition and proof sketch given in the main text	group/matter mismatch would be global
Carrier coefficient function $b_1(N_{\mathrm{fam}}, N_\Phi)$	Derived	abelian-index formula in the main text	hypercharge-trace mismatch would be structural
Canonical seam/kernel constants $(c_3, \varphi_{\mathrm{base}}, \delta_{\mathrm{top}})$	Derived	seam normalization, exact seam opening, and carrier-form closure equations summarized in the main text	scheme discipline for observable comparison
Canonical closure value $b_1 = 41/10$	Derived	canonical family/Higgs closure of the abelian index on the admissible branch	failure would break the 41-chain and the canonical packet count
Operative admissibility selector P_{adm}	Derived	closure theorem stack in Section 8	unified electroweak / QCD / CP sector selection
Family closure, $F \cong \mathbb{C}^3$, and $\Omega_{\mathrm{adm}} = 48$	Derived	master closure theorem and occupancy corollary in Section 8	flavor multiplicity and family lifting
Bosonic relative index selecting Φ	Derived	Callias Higgs-selection theorem and bosonic closure theorem in Section 8	unique light doublet / doublet-triplet split
Local spectral-scale / gravity closure for $\bar{M}_{\mathrm{Pl}}$ and v_{geo}	Derived	bosonic closure theorem, geometry reconstruction, and two-sheet zero-mode lemma	Planck / electroweak / gravity relation
Yukawa holonomy matrix and mass closure	Derived	exact transport closure and exact family-holonomy phase control	m_f, CKM / PMNS, hierarchy grammar
Hadronic space, anomaly inflow, and strong CP	Derived	hadronic / anomaly closure theorems in Section 8	hadron spectrum and $\bar{\theta}$ null tests
Nonperturbative QFT closure	Derived	gap lift, Callias positivity bounds, and admissible QFT theorem	mass gap, clustering, and confinement route
Reflection positivity, Lorentz reconstruction, record algebra, and vacuum closure	Derived	relative generating functional, admissible reflection-positivity lemma, gap/clustering record construction, split OS / vacuum theorems, and primitive Lorentzian corollary	causal dynamics, stable records, and vacuum energy map
Observation functor $\mathcal{R}_{\mathrm{TOE}}$, threshold outputs, and benchmark rows	Intrinsic observation	global observation-closure theorem with scheme-covariant numerical preconditioner $\mathcal{R}_{\mathrm{SM}}$	convention-complete theory-versus-data translation
E_8 and horizon semantics	Optional extrapolation	deferred to appendices and the auxiliary appendix continuation	downstream extension rather than main-text hard claim

Table 1: Status matrix for the carrier-first compiler form with reduced datum, derived theorem outputs, intrinsic observation rows, and optional extrapolations kept visibly distinct.

The hard carrier theorems below do not require the full closure stack behind Equations (1) and (2). In the present paper the master equation is the organizing compiler form, the carrier theorems are hard, and the later closure theorems are internal statements on the admissible reduced datum rather than extra primitive ontology. At this overview level the seed package is written as A_Σ^{seed}; on the later geometric branch it is reorganized as A_Σ^{geo} with $\chi_{\mathrm{seed}} \mapsto \chi_{\mathrm{geo}} = \Lambda e^\sigma$.

Closed TFPT Chain

$$\begin{aligned} \mathfrak{T}_{\mathrm{min}}^{\mathrm{adm}} &\Rightarrow (\mathfrak{T}_{\mathrm{prim}},\ E_3 \oplus E_2,\ G_{\mathrm{SM}}^{\mathrm{glob}},\ S^+) \\ &\Rightarrow (P_{\mathrm{adm}},\ F,\ g,\ \sigma,\ \Phi,\ \nabla_F^\star) \\ &\Rightarrow (\Gamma_\star,\ \Omega_{\mathrm{DM}},\ \eta_B,\ \Lambda_{\mathrm{obs}}) \\ &\Rightarrow (\mathcal{R}_{\mathrm{TOE}},\ O_{\mathrm{th}},\ \mathrm{Pred}). \end{aligned} \tag{3}$$



Closed main-text stack only; optional continuations are deferred to appendices.

Figure 3: Closed theorem surface in the present draft. The main text runs from the admissible reduced datum through the derived closure stack to the intrinsic observation functor, with no parallel theorem ladder carried in parallel.

Closed Theorem Stack in the Present Version

$$\mathfrak{T}_{\mathrm{min}}^{\mathrm{adm}} \Longrightarrow \mathfrak{T}_{\mathrm{prim}} \Longrightarrow \text{carrier/kernel closure} \Longrightarrow \text{admissibility} + \text{geometry} + \text{flavor} + \text{vacuum} + \text{UV} + \text{cosm}$$

There are no unresolved theorem obligations in the main text. Optional arithmetic continuations, asymptotic expansions, and horizon language are kept only in appendices.

The remainder of the main text sharpens each arrow of this chain in turn.

3 Primitive core and hard carrier kernel

3.1 Primitive core and conventions

We use natural units $c = \hbar = 1$. Relative operators are formulated in Euclidean signature and compared through a declared Wick-rotation bridge when Lorentzian language is needed. Higgs fields are denoted by Φ, Hamiltonians by $\mathcal{H}$, and the cosmological Hubble rate by $H_{\mathrm{hub}}(t)$; this avoids overloading the letter H. Newton's constant is always G_N, while $G_{\mathrm{SM}}^{\mathrm{glob}}$ is reserved for the global Standard Model gauge quotient. The seam is denoted by Σ, the involution by ι, and word parity by $\varepsilon(w)$.

Here $S \to \widetilde{M}$ denotes the spinor bundle on the oriented double cover, and by the same symbol we denote its restriction to

$$X = \widetilde{M} \setminus \Sigma.$$

When color-center notation is needed later, we use multiplicative notation

$$z(w) \in \{1, \omega, \omega^2\}, \qquad \omega = e^{2\pi i/3}.$$

Thus center neutrality means $z(w) = 1$.

$$\mathfrak{T}_{\mathrm{core}} := \big(\widetilde{M} \to M,\ \Sigma,\ S,\ \mathcal{H}_{\mathrm{int}},\ \iota,\ D_{\mathrm{rel}},\ f,\ \chi_{\mathrm{seed}}\big).$$

Here "relative" always means comparison with a declared reference operator or reference-subtracted action. The admissibility predicate Adm and its operator realization $(K_{\mathrm{adm}}, P_{\mathrm{adm}})$ belong to the later closure layer rather than to this primitive bookkeeping package. The technical tables for relative objects, layered closure/benchmark data, and the constants atlas are collected in Appendix A and Appendix B; they are kept out of the early main-text flow so that the carrier kernel appears before the technical bookkeeping apparatus.

Remark (Reduced datum versus derived structure). In the present version the oriented double cover, seam, involution, reference/relative split, and primitive seam response are not additional starting ontology. They are reconstructed from the admissible reduced datum and then fed into the hard carrier, closure, and intrinsic observation layers.

3.2 Primitive relative spectral dynamics

The carrier theorem is not intended to float above an unspecified microscopic dynamics. The starting point is therefore the admissible reduced datum from which the primitive layer is reconstructed before the later closure theorems are invoked.

Definition 3.1 (Admissible reduced theorem datum)**.** The admissible reduced datum is

$$\mathfrak{T}_{\min}^{\mathrm{adm}} := (\mathcal{A}, \mathcal{H}, D, J, \Gamma, \iota, \nu)$$

with the following built-in admissibility conditions:

(i) $Z(\mathcal{A})$ is spectrally regular and its Gelfand spectrum is a smooth compact oriented manifold $\widetilde{M}$;

(ii) ι acts by a smooth order-two involution on $Z(\mathcal{A})$, commutes with J and Γ, and has smooth fixed locus $\Sigma := \mathrm{Fix}(\iota)$;

(iii) in a collar neighborhood of Σ, the Dirac operator has canonical product form

$$D = \gamma_n(\partial_n + B_\Sigma);$$

(iv) the cutoff profile is generated by a positive complete-Bernstein measure

$$f(x) = \int_0^\infty e^{-tx^2}\, d\nu(t), \qquad d\nu(t) \geq 0; \tag{4}$$

(v) the seam collar is reflection-regular so that the doubled operator and admissibility complex defined later are well posed on the same datum.

Theorem 3.2 (Primitive layer reconstructed canonically from the admissible reduced datum)**.** *Let $\mathfrak{T}_{\min}^{\mathrm{adm}}$ be as above. Then the primitive TFPT layer is not additional ontology. It is reconstructed canonically as*

$$\widetilde{M} = \operatorname{Spec} Z(\mathcal{A}), \qquad M = \widetilde{M}/\iota, \qquad \Sigma = \mathrm{Fix}(\iota), \qquad X = \widetilde{M} \setminus \Sigma,$$

$$D_{\mathrm{ref}} := \frac{1}{2}(D + \iota D \iota), \qquad D_{\mathrm{rel}} := \frac{1}{2}(D - \iota D \iota), \qquad \chi_{\mathrm{seed}} := \lambda_1^+(|B_\Sigma|)^{-1}.$$

Hence the primitive package

$$\mathfrak{T}_{\mathrm{prim}} := (X,\ S,\ \mathcal{H}_{\mathrm{int}},\ D_{\mathrm{rel}},\ \iota,\ f,\ \chi_{\mathrm{seed}}), \qquad \mathcal{H}_{\mathrm{prim}} := L^2(X, S) \otimes \mathcal{H}_{\mathrm{int}},$$

is derived, not assumed.

Corollary 3.3 (No residual primitive ontology)**.** *Within TFPT the oriented double cover, the physical quotient, the seam, the reference/relative operator pair, and the primitive seam response are theorem-level outputs of $\mathfrak{T}_{\min}^{\mathrm{adm}}$. No independent primitive datum remains in the main-text ontology.*

Definition 3.4 (Primitive Euclidean action)**.** The primitive Euclidean action is the relative spectral seed

$$S_{\mathrm{prim}}[\chi_{\mathrm{seed}}] := \mathrm{Tr}\, f\left(\frac{D_{\mathrm{rel}}}{\chi_{\mathrm{seed}}}\right) - \mathrm{Tr}\, f\left(\frac{D_{\mathrm{ref}}}{\chi_{\mathrm{seed}}}\right) + \frac{i\pi}{2}\Delta\eta_\Sigma.$$

Gauge fixing, ghosts, carrier-compatible bosonic fluctuations, and the admissibly compressed net

$$\mathfrak{A}_{\mathrm{adm}}(\mathcal{O}) := P_{\mathrm{adm}}\, \mathfrak{A}(\mathcal{O})\, P_{\mathrm{adm}}.$$

are introduced only after the hard carrier theorem and the later admissibility closure package are in place.

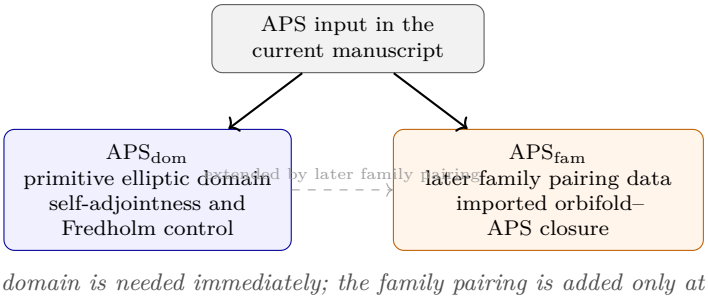


Figure 4: APS split in the present draft. $\mathrm{APS}_{\mathrm{dom}}$ handles primitive elliptic well-posedness, while $\mathrm{APS}_{\mathrm{fam}}$ is the later family-pairing layer needed for the imported orbifold–APS family theorem.

Theorem 3.5 (Primitive complete-Bernstein positivity)**.** *Let the admissible reduced datum carry a positive complete-Bernstein spectral measure*

$$f(z) = \int_0^\infty e^{-tz^2}\, d\nu(t), \qquad d\nu(t) \geq 0.$$

Then on the admissible seam-even bosonic branch the associated heat-kernel semigroup is positivity preserving and reflection stable. Consequently Bernstein positivity is a derived property of the admissible reduced datum and not an additional assumption.

Proof sketch. Each heat kernel $e^{-tD_{\mathrm{rel}}^2}/\chi_{\mathrm{seed}}^2$ is positivity preserving on the admissible seam-even branch. Positive superposition with $d\nu(t) \geq 0$ preserves positivity and reflection stability, so the complete Bernstein profile inherits the same property. □

Theorem 3.6 (Well-posed primitive dynamics)**.** *Assume X is compact up to the elliptic boundary domain $\mathrm{APS}_{\mathrm{dom}}$, the coefficients of D_{rel} are smooth, and the Bernstein measure in Equation* (4) *has finite moments of every order. Then:*

(a) D_{rel} is essentially self-adjoint on the $\mathrm{APS}_{\mathrm{dom}}$ domain;

(b) D_{rel} has compact resolvent on the finite-cutoff Euclidean branch;

(c) the spectral action trace is finite for every finite cutoff;

(d) and the finite-cutoff relative Euclidean seed defines a well-posed primitive dynamics before carrier and admissibility closure are imposed.

Proof sketch. Dirac-type operators with $\mathrm{APS}_{\mathrm{dom}}$ boundary data are Fredholm and essentially self-adjoint on the corresponding Sobolev domain. Compactness of the resolvent follows on compact X with $\mathrm{APS}_{\mathrm{dom}}$ boundary, while the Bernstein representation with finite moments makes the spectral trace finite at every finite cutoff. This yields a well-defined primitive relative spectral seed before any later carrier-compatible fluctuation package is turned on. □

Remark (APS roles on the admissible branch)*.* The primitive layer needs only the elliptic domain choice $\mathrm{APS}_{\mathrm{dom}}$ for self-adjointness and Fredholm control of D_{rel}. On the closed admissible branch, the later family pairing data are induced canonically by the doubled geometry and are therefore no longer an additional input package outside the theorem surface.

Corollary 3.7 (Carrier-compatible Lorentzian dynamics after closure)**.** *Once the later carrier-compatible fluctuation package and the admissibility closure hypotheses of Section 8 are imposed, the resulting admissible Euclidean theory reconstructs a self-adjoint Lorentzian Hamiltonian H_{adm} and a one-parameter automorphism group*

$$\alpha_t(O) = e^{itH_{\mathrm{adm}}} O e^{-itH_{\mathrm{adm}}}.$$

3.3 Minimal carrier theorem

Theorem 3.8 (Minimal carrier theorem)**.** *Let $E = E_b \oplus E_s$ be a split carrier with traceless diagonal generator*

$$Y_{b,s} = \operatorname{diag}\left(\underbrace{-\frac{1}{b}, \dots, -\frac{1}{b}}_{b \text{ times}}, \underbrace{\frac{1}{s}, \dots, \frac{1}{s}}_{s \text{ times}}\right).$$

If one simultaneously requires

$$\dim \Lambda^{\text{even}} E = 16, \qquad \operatorname{Tr}_E Y_{b,s}^2 = \frac{5}{6}, \tag{5}$$

then necessarily $\{b, s\} = \{3, 2\}$.

Proof. Write $g = b + s$. Since

$$\dim \Lambda^{\text{even}} E = 2^{g-1},$$

the first condition gives $g = 5$. For the diagonal generator above one has

$$\operatorname{Tr}_E Y_{b,s}^2 = b\left(\frac{1}{b^2}\right) + s\left(\frac{1}{s^2}\right) = \frac{1}{b} + \frac{1}{s} = \frac{b+s}{bs} = \frac{5}{bs}.$$

Imposing $\operatorname{Tr}_E Y_{b,s}^2 = 5/6$ yields $bs = 6$. Together with $b + s = 5$ this forces the unordered solution $\{b, s\} = \{3, 2\}$. □

Hence from now on

$$E = E_3 \oplus E_2, \qquad P_3 = \frac{\mathbf{1} - \iota}{2}, \qquad P_2 = \frac{\mathbf{1} + \iota}{2},$$

and the hypercharge operator is read concretely as

$$Y = \operatorname{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right) = -\frac{1}{3}P_3 + \frac{1}{2}P_2 = \frac{1}{12}\mathbf{1} + \frac{5}{12}\iota. \tag{6}$$

Lemma 3.9 (Hypercharge–seam inversion)**.** *On the minimal carrier, the seam involution is recovered from hypercharge by*

$$\iota = \frac{12Y - \mathbf{1}}{5}, \tag{7}$$

$$(12Y - \mathbf{1})^2 = 25\,\mathbf{1},$$

$$6Y^2 - Y - \mathbf{1} = 0. \tag{8}$$

Consequently the carrier projectors are

$$P_2 = \frac{\mathbf{1} + \iota}{2} = \frac{2(3Y + \mathbf{1})}{5}, \qquad P_3 = \frac{\mathbf{1} - \iota}{2} = \frac{3(\mathbf{1} - 2Y)}{5}.$$

Proof. The affine relation between Y and ι is inverted algebraically. Squaring the result and using $\iota^2 = \mathbf{1}$ gives the quadratic carrier polynomial. The projector formulas then follow from $P_2 = (\mathbf{1} + \iota)/2$ and $P_3 = (\mathbf{1} - \iota)/2$. □

The hard carrier package is summarized visually in Figure 5. The point is that the same minimal data are readable in two equivalent ways: geometrically as an oriented double cover with seam involution, and algebraically as the $3 + 2$ carrier split with two spectral hypercharges.

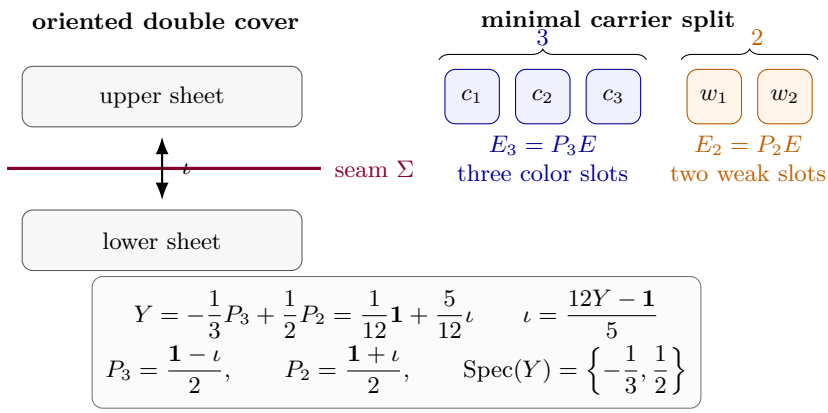


Figure 5: Minimal carrier at a glance. The seam involution and the hypercharge operator carry the same hard two-point information: geometrically they separate the two sheets, and algebraically they separate the $3 + 2$ carrier split through the projectors P_3 and P_2.

Remark (Two-point carrier algebra). Because the minimal carrier satisfies $6Y^2 - Y - \mathbf{1} = 0$, every analytic function of Y reduces to

$$f(Y) = A_f\,\mathbf{1} + B_f\,Y, \qquad A_f = \frac{2f(1/2) + 3f(-1/3)}{5}, \qquad B_f = \frac{6(f(1/2) - f(-1/3))}{5}.$$

Thus the hard carrier is effectively a two-point algebra supported on the spectral values $-1/3$ and $1/2$, and the projectors P_3, P_2 are already polynomial functions of Y.

Remark (Factorized carrier polynomial and carrier discriminant). For a general split carrier $E = E_b \oplus E_s$ with total rank $g := b + s$ and eigenvalues $-1/b$ and $1/s$, the carrier generator satisfies

$$\left(Y + \frac{1}{b}\right)\left(Y - \frac{1}{s}\right) = 0 \quad \Longleftrightarrow \quad bs\,Y^2 + (s - b)Y - \mathbf{1} = 0.$$

Its discriminant is

$$\Delta_Y = (s - b)^2 + 4bs = (b + s)^2 = g^2.$$

Thus the square root of the carrier discriminant is the carrier rank itself. Moreover,

$$Y = \frac{b - s}{2bs}\,\mathbf{1} + \frac{g}{2bs}\,\iota, \qquad \iota = \frac{2bs\,Y - (b - s)\,\mathbf{1}}{g}.$$

On the minimal branch $(b, s) = (3, 2)$, this reduces to

$$6Y^2 - Y - \mathbf{1} = (2Y - \mathbf{1})(3Y + \mathbf{1}), \qquad \Delta_Y = 25 = 5^2, \qquad \iota = \frac{12Y - \mathbf{1}}{5}.$$

Hence the denominator 5 in the hypercharge–seam inversion is exactly $\sqrt{\Delta_Y}$, not an additional inserted constant.

This does not replace the stabilizer proof of

$$G_{\mathrm{SM}}^{\mathrm{glob}} = S(U(3) \times U(2)),$$

which is the actual source of the global Standard Model quotient in the present manuscript. What the factorization and discriminant do show exactly is that the hard carrier already supports the distinguished spectral values

$$Y = \frac{1}{2}, \qquad Y = -\frac{1}{3},$$

and that the inversion denominator is fixed algebraically by the carrier discriminant.

The two carrier invariants used repeatedly below are

$$\gamma = \mathrm{Tr}_E\, Y^2 = \frac{5}{6}, \qquad B = \frac{\operatorname{rank} E_3}{\operatorname{rank} E_2} = \frac{3}{2}.$$

Corollary 3.10 (Carrier norm from the carrier polynomial)**.** *The carrier polynomial already fixes*

$$\gamma = \mathrm{Tr}_E\, Y^2 = \frac{1}{6}\,\mathrm{Tr}_E(Y + \mathbf{1}) = \frac{5}{6},$$

since $\mathrm{Tr}_E\, Y = 0$ *and* $\operatorname{rank} E = 5$.

Corollary 3.11 (Minimal carrier compression)**.** *On the minimal carrier branch,*

$$B\gamma = \frac{3}{2} \cdot \frac{5}{6} = \frac{5}{4}.$$

We package the hard kernel as

$$\mathcal{C}_{\min} = (E_3 \oplus E_2,\ Y,\ S^+,\ \gamma,\ B).$$

3.4 Global gauge group and one-family packet

Theorem 3.12 (Global Standard Model theorem)**.** *The determinant-preserving connected stabilizer of the minimal carrier split is*

$$G_{\mathrm{SM}}^{\mathrm{glob}} = S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)_Y}{\mathbb{Z}_6}.$$

Proof. An automorphism preserving the split $E_3 \oplus E_2$ acts by a pair $(u_3, u_2) \in U(3) \times U(2)$. Requiring determinant preservation on the full carrier gives

$$\det(u_3)\det(u_2) = 1,$$

so the connected stabilizer is exactly $S(U(3) \times U(2))$.

Define

$$\Phi : SU(3) \times SU(2) \times U(1)_Y \longrightarrow S(U(3) \times U(2)), \qquad \Phi(g_3, g_2, z) = (z^2 g_3,\ z^{-3} g_2).$$

This is a well-defined group homomorphism because the exponents are integers, and

$$\det(z^2 g_3)\det(z^{-3} g_2) = z^6 z^{-6} = 1.$$

To prove surjectivity, take $(u_3, u_2) \in S(U(3) \times U(2))$. Choose $z \in U(1)$ with

$$z^6 = \det(u_3) = \det(u_2)^{-1}.$$

Then

$$g_3 := z^{-2} u_3 \in SU(3), \qquad g_2 := z^3 u_2 \in SU(2),$$

and therefore

$$\Phi(g_3, g_2, z) = (u_3, u_2).$$

If $\Phi(g_3, g_2, z) = (\mathbf{1}_3, \mathbf{1}_2)$, then

$$g_3 = z^{-2}\mathbf{1}_3, \qquad g_2 = z^3 \mathbf{1}_2.$$

The conditions $g_3 \in SU(3)$ and $g_2 \in SU(2)$ imply $z^6 = 1$. Hence

$$\ker \Phi = \left\{ (z^{-2}\mathbf{1}_3,\ z^3 \mathbf{1}_2,\ z) : z^6 = 1 \right\} \cong \mathbb{Z}_6.$$

Therefore

$$S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)_Y}{\mathbb{Z}_6}.$$

□

This is an internal carrier theorem: it identifies the connected stabilizer of the minimal carrier split. The paper does not claim that the global physical gauge group has already been experimentally fixed at the same level of closure.

Matter is read from the positive half-spinor

$$S^+ = \Lambda^{\text{even}} E, \qquad \dim S^+ = 16.$$

Theorem 3.13 (One carrier family)**.** *The positive half-spinor of the minimal carrier decomposes as*

$$S^+ \cong (1,1)_0 \oplus (3,2)_{1/6} \oplus (\bar{3},1)_{-2/3} \oplus (\bar{3},1)_{1/3} \oplus (1,2)_{-1/2} \oplus (1,1)_1,$$

that is, exactly one chiral Standard Model family with ν^c. Equivalently, in the familiar unified packaging language,

$$S^+ \cong \mathbf{1} \oplus \mathbf{10} \oplus \overline{\mathbf{5}}.$$

Proof. One has

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E.$$

The $\Lambda^0 E$ sector is immediately

$$\Lambda^0 E \cong (1,1)_0,$$

which is identified with ν^c in left-chiral notation. For the minimal carrier,

$$\Lambda^2 E \cong \Lambda^2 E_3 \oplus (E_3 \otimes E_2) \oplus \Lambda^2 E_2.$$

Here

$$\Lambda^2 E_3 \cong \bar{3}, \qquad E_3 \otimes E_2 \cong (3,2), \qquad \Lambda^2 E_2 \cong 1,$$

and the hypercharges are obtained by adding the carrier charges from

$$Y = \operatorname{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right).$$

Therefore

$$\Lambda^2 E \cong (\bar{3},1)_{-2/3} \oplus (3,2)_{1/6} \oplus (1,1)_1.$$

Because the full determinant is trivial in $S(U(3) \times U(2))$, one has $\Lambda^5 E \cong 1$ and hence

$$\Lambda^4 E \cong E^*.$$

Equivalently,

$$\Lambda^4 E \cong (\Lambda^3 E_3 \otimes E_2) \oplus (\Lambda^2 E_3 \otimes \Lambda^2 E_2).$$

Using $\Lambda^3 E_3 \cong 1$, $\Lambda^2 E_2 \cong 1$, and $\Lambda^2 E_3 \cong E_3^* \cong \bar{3}$, one gets

$$\Lambda^4 E \cong (1,2)_{-1/2} \oplus (\bar{3},1)_{1/3}.$$

Collecting the three even-degree sectors yields

$$S^+ \cong (1,1)_0 \oplus (\bar{3},1)_{-2/3} \oplus (3,2)_{1/6} \oplus (1,1)_1 \oplus (\bar{3},1)_{1/3} \oplus (1,2)_{-1/2},$$

with total multiplicity $1+3+6+1+3+2 = 16$. The $\Lambda^2 E$ sector is the standard left-chiral $\mathbf{10}$ of $SU(5)$, the $\Lambda^4 E$ sector is the standard left-chiral $\overline{\mathbf{5}}$, and the $\Lambda^0 E$ sector is the singlet $\mathbf{1}$. □

In left-chiral notation the one-family decomposition takes the explicit form shown in Table 2. This is the place where the manuscript must be explicit: the carrier theorem is meaningful only if the reader can see how the sixteen states are organized.

Exterior component	SM representation	Hypercharge	Field content	Mult.
$\Lambda^0 E$	$(1,1)_0$	0	ν^c	1
$\Lambda^2 E_3$	$(\bar{3},1)_{-2/3}$	$-2/3$	u^c	3
$E_3 \otimes E_2$	$(3,2)_{1/6}$	$1/6$	Q_L	6
$\Lambda^2 E_2$	$(1,1)_1$	1	e^c	1
$\Lambda^3 E_3 \otimes E_2$	$(1,2)_{-1/2}$	$-1/2$	L_L	2
$\Lambda^2 E_3 \otimes \Lambda^2 E_2$	$(\bar{3},1)_{1/3}$	$1/3$	d^c	3
			Total multiplicity	16

Table 2: One chiral family from $S^+ = \Lambda^{\text{even}} E$, written in left-chiral notation with charge-conjugate singlets.

Proposition 3.14 (Exact parity-code realization of the family packet)**.** Identifying each wedge monomial in ΛE with its occupation word

$$x = (x_1, \ldots, x_5) \in \{0,1\}^5,$$

the positive half-spinor is exactly the even-parity subspace

$$S^+ \cong \operatorname{span}\Big\{ |x\rangle : \sum_{i=1}^{5} x_i \equiv 0 \pmod 2 \Big\}.$$

Equivalently,

$$S^+ = (\Lambda^{\text{even}} E_3 \otimes \Lambda^{\text{even}} E_2) \oplus (\Lambda^{\text{odd}} E_3 \otimes \Lambda^{\text{odd}} E_2),$$

and each summand has dimension 8.

Proof. Choose a basis of E adapted to the split $E_3 \oplus E_2$. Exterior monomials are then in bijection with occupation words $x \in \{0,1\}^5$, and the exterior degree is exactly the Hamming weight $\sum_i x_i$. Hence $\Lambda^{\text{even}} E$ is the span of the even-parity words. Splitting the degree into the E_3 and E_2 contributions gives the direct-sum decomposition above. Since

$$\dim \Lambda^{\text{even}} E_3 = 1 + \binom{3}{2} = 4, \qquad \dim \Lambda^{\text{odd}} E_3 = \binom{3}{1} + \binom{3}{3} = 4,$$

and

$$\dim \Lambda^{\text{even}} E_2 = 1 + \binom{2}{2} = 2, \qquad \dim \Lambda^{\text{odd}} E_2 = \binom{2}{1} = 2,$$

each tensor-product sector has dimension 8. □

Remark (Exact parity lock, not a loose code analogy)*.* The family packet is therefore not merely code-like: at the level of occupation kinematics it is exactly the classical even-parity $[5,4,2]$ code on five bits. What the present paper still does *not* claim is a dynamical error-correction theorem for realistic noise channels. The exact statement is the parity lock itself. In the maximally mixed state on the 16 occupation words, eight states lie in the $(\text{even}, \text{even})$ sector and eight in (odd, odd), so the coarse parity observables of the 3-side and 2-side carry exactly one bit of mutual information.

The family packet can also be read as a two-dimensional occupancy map, shown in Figure 6. This is where the one-family table, the even-parity code, the split weight enumerator, and the $X = 6Y$ arithmetic all meet in one glance.

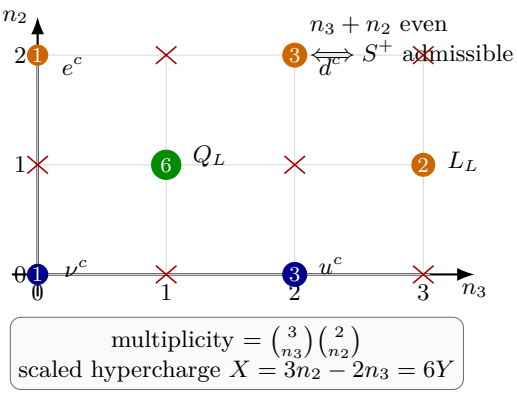


Figure 6: Occupancy map of the family packet. The admissible sectors are exactly the even lattice points in (n_3, n_2), with multiplicities $(1, 3, 6, 1, 2, 3)$ and field labels matching Table 2. This is the visual meeting point of parity lock, split weight enumerator, and family hypercharge arithmetic.

3.5 Character, trace identities, and the abelian index

Proposition 3.15 (Split weight enumerator and one-family character)**.** For the present paragraph let

$$\chi_{S^+}(t) := \mathrm{Tr}_{S^+}(e^{tY}).$$

Define also the split weight enumerator

$$W(u,v) := \sum_{\substack{0\le n_3\le 3\\ 0\le n_2\le 2\\ n_3+n_2 \text{ even}}} \binom{3}{n_3}\binom{2}{n_2} u^{n_3} v^{n_2}.$$

Then

$$W(u,v) = 1 + 3u^2 + 6uv + v^2 + 2u^3v + 3u^2v^2 = \frac{1}{2}\Big[(1+u)^3(1+v)^2 + (1-u)^3(1-v)^2\Big]. \quad (9)$$

Moreover,

$$\chi_{S^+}(t) = \frac{1}{2}\Big[\big(1+e^{-t/3}\big)^3\big(1+e^{t/2}\big)^2 + \big(1-e^{-t/3}\big)^3\big(1-e^{t/2}\big)^2\Big] = W(e^{-t/3}, e^{t/2}). \quad (10)$$

Consequently,

$$\chi_{S^+}(0) = 16, \qquad \chi'_{S^+}(0) = \mathrm{Tr}_{S^+} Y = 0, \qquad \chi''_{S^+}(0) = \mathrm{Tr}_{S^+} Y^2 = \frac{10}{3}, \qquad \chi'''_{S^+}(0) = \mathrm{Tr}_{S^+} Y^3 = 0.$$

Proof. The pairs (n_3, n_2) allowed by even total degree are

$$(0,0),\ (2,0),\ (1,1),\ (0,2),\ (3,1),\ (2,2),$$

with multiplicities

$$1,\ 3,\ 6,\ 1,\ 2,\ 3.$$

This gives the explicit polynomial formula for $W(u,v)$. Equivalently, the parity projector

$$\frac{1}{2}(1 + (-1)^{n_3+n_2})$$

acting on $(1+u)^3(1+v)^2$ yields Equation (9). Substituting

$$u = e^{-t/3}, \qquad v = e^{t/2}$$

gives Equation (10). For an even exterior algebra one also has

$$\mathrm{Tr}_{\Lambda^{\mathrm{even}}E}(e^{tY}) = \frac{1}{2}\Big[\prod_{j=1}^{5}(1+e^{tq_j}) + \prod_{j=1}^{5}(1-e^{tq_j})\Big],$$

with carrier charges $q_j \in \{-1/3, -1/3, -1/3, 1/2, 1/2\}$. Differentiating at $t = 0$ yields the stated dimension and trace identities. □

Remark (Six admissible occupancy sectors). The six monomials in Equation (9) are exactly the six allowed occupancy pairs (n_3, n_2) of the family packet. This is the shortest combinatorial route from the split carrier to the six Standard Model multiplet sectors listed in Table 2.

The carrier packet therefore reproduces the standard trace cancellations

$$\mathrm{Tr}_{S^+} Y = 0, \qquad \mathrm{Tr}_{S^+} Y^3 = 0,$$

and packages them together with the quadratic trace in one formula. Since $\mathrm{Tr}_{S^+} Y^2 = 4\gamma = 10/3$ and $\mathrm{Tr}_{\Phi} Y^2 = 1/2$ for one weak doublet with hypercharge $1/2$, the hypercharge trace coefficient organizes as

$$b_1(N_{\mathrm{fam}}, N_\Phi) = \frac{2}{5}N_{\mathrm{fam}}\,\mathrm{Tr}_{S^+} Y^2 + \frac{1}{5}N_\Phi\,\mathrm{Tr}_\Phi Y^2 = \frac{8}{5}N_{\mathrm{fam}}\gamma + \frac{1}{10}N_\Phi = \frac{4}{3}N_{\mathrm{fam}} + \frac{1}{10}N_\Phi. \tag{11}$$

On the canonical branch $(N_{\mathrm{fam}}, N_\Phi) = (3, 1)$ this gives

$$b_1 = \frac{41}{10}.$$

The abelian index is therefore not inserted as an isolated Standard Model datum; it is a carrier-level output once the family and Higgs counts are declared.

Remark (Canonical-branch Diophantine lock). If one imposes $b_1 = 41/10$ on nonnegative integer data, then

$$40N_{\mathrm{fam}} + 3N_\Phi = 123.$$

The physically nontrivial solution is $(N_{\mathrm{fam}}, N_\Phi) = (3, 1)$, so the canonical branch is already sharply constrained at the level of the trace arithmetic.

Proposition 3.16 (Family hypercharge arithmetic and compression)**.** Let N_3 and N_2 be the occupation-number operators on the E_3 and E_2 slots, restricted to S^+, and define the scaled hypercharge operator

$$X := 6Y = 3N_2 - 2N_3.$$

Then on S^+ one has

$$X \equiv N_2 \equiv N_3 \pmod 2, \tag{12}$$

$$X \equiv N_3 \pmod 3, \tag{13}$$

$$X \equiv 3(N_2 + N_3) \pmod 5. \tag{14}$$

In particular, the even exterior degree $k = N_2 + N_3 \in \{0, 2, 4\}$ is read off from the residue class of X modulo 5: if $X \equiv r \pmod 5$ with $r \in \{0, 1, 2\}$, then $k = 2r$, so

$$X \bmod 5 = 0, 1, 2 \qquad \Longleftrightarrow \qquad \Lambda^0 E,\ \Lambda^2 E,\ \Lambda^4 E.$$

Moreover,

$$\mathrm{Spec}(X|_{S^+}) = \{0, -4, 1, 6, -3, 2\},$$

and therefore

$$X(X+4)(X-1)(X-6)(X+3)(X-2) = 0 \qquad \text{on } S^+. \tag{15}$$

Proof. Each occupied E_3 mode contributes hypercharge $-1/3$ and each occupied E_2 mode contributes $+1/2$, so multiplying by 6 gives $X = 3N_2 - 2N_3$. Modulo 2, the term $-2N_3$ vanishes, so $X \equiv N_2$. Because S^+ has even total degree, $N_2 \equiv N_3 \pmod 2$, which proves Equation (12). Modulo 3, one has $3N_2 \equiv 0$ and $-2 \equiv 1$, giving Equation (13). Modulo 5, the congruence $-2 \equiv 3$ yields Equation (14). Since on S^+ the even degree is $k = N_2 + N_3 \in \{0, 2, 4\}$, the possible residues are $3k \equiv 0, 1, 2 \pmod 5$, from which $k = 2r$ with $r = X \bmod 5$ follows. Evaluating $X = 3n_2 - 2n_3$ on the six admissible occupancy pairs (n_3, n_2) gives the listed spectrum. The polynomial Equation (15) is then the spectral polynomial with these six simple roots. □

Corollary 3.17 (Five-fold phase filters for even degree)**.** *Let*

$$\zeta_5 = e^{2\pi i/5}, \qquad U_5 := e^{2\pi i X/5}.$$

Then the even-degree projectors are the discrete Fourier filters

$$P_{\Lambda^{2m}E} = \frac{1}{5}\sum_{r=0}^{4} \zeta_5^{-mr} U_5^r, \qquad m = 0, 1, 2.$$

Proof. By Proposition 3.16, the eigenspaces $\Lambda^{2m}E$ carry the residue class $X \equiv m \pmod 5$. Hence U_5 has eigenvalue ζ_5^m on $\Lambda^{2m}E$, and the standard Fourier projector formula on the cyclic group $\mathbb{Z}_5$ gives the claim. □

Remark (From the two-point carrier to the six-point family algebra)*.* The minimal carrier itself is a two-point algebra, but the family packet lifts the scaled hypercharge to a six-point algebra. Therefore every analytic function of X on S^+ reduces to a polynomial of degree at most five. If

$$M_n := \mathrm{Tr}_{S^+}(X^n),$$

then Equation (15) implies the finite recursion

$$M_{n+6} = 2M_{n+5} + 31M_{n+4} - 20M_{n+3} - 156M_{n+2} + 144M_{n+1}.$$

In this sense the whole hypercharge statistics of one family is finitely compressed into six spectral values.

4 Kernel numerics and electromagnetic closure

The carrier-first presentation should not leave the impression that TFPT is hard only at the level of group theory and representation bookkeeping. The numerical kernel already fixes a nontrivial output layer. Here that layer is pulled forward and read as the first established output package of the carrier/compiler architecture rather than as a later list of disconnected constants.

Proposition 4.1 (Seam normalization and base opening)**.** Let the seam loop carry one unit of relative spectral flow,

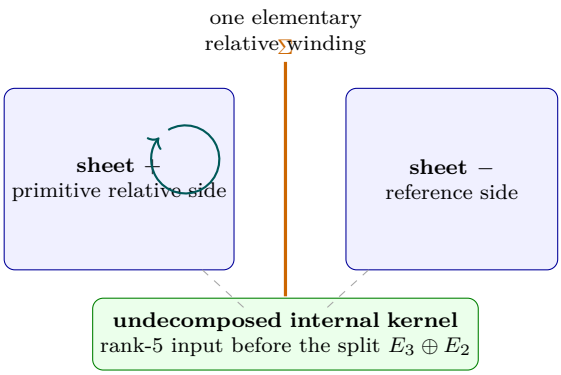
$$SF(U_\Sigma) = 1, \qquad \Delta_\Sigma = 2\pi.$$

In the relative Chern normalization, the seam constant is

$$c_3 := \frac{\Delta_\Sigma}{16\pi^2} = \frac{1}{8\pi}.$$

Moreover the seam-even base opening is the normalized complement of the carrier quadratic norm,

$$\varphi_{\text{base}} := \frac{1-\gamma}{\pi} = \frac{1 - \mathrm{Tr}_E Y^2}{\pi} = \frac{1}{6\pi}.$$



Interpretive only: the cover, seam, and primitive response are reconstructed from the reduced datum in the present version.

Figure 7: Interpretive primitive start sketch. The early seam constants are attached to a minimal relative picture with two sheets, one seam, one elementary winding, and an internal rank-5 kernel before the split into $E_3 \oplus E_2$ is derived.

Proof sketch. One unit of seam spectral flow fixes the topological normalization of the relative one-form coefficient. The carrier theorem gives $\gamma = \mathrm{Tr}_E\, Y^2 = 5/6$. The seam-even opening is therefore the normalized complement of the carrier quadratic norm rather than a new independent numerical input. □

Remark (Interpretive picture of the primitive start state). At the level of physical intuition, the primitive spectral seed may be read as a minimal nontrivial relative defect state: two sheets, one seam, one elementary relative winding, and an undecomposed internal rank-5 kernel before the carrier split is imposed. This is only an interpretive picture, not a new theorem. Its role is to explain why the earliest hard numbers $(c_3, \varphi_{\mathrm{base}})$ already look like seam-normalized responses of a genuinely relative starting configuration.

4.1 Carrier form of the electromagnetic closure

On the canonical admissible branch the inherited topological corrections are

$$\delta_{\mathrm{top}} = \frac{3}{256\pi^4}, \qquad \delta_2 = \frac{5}{4}\,\delta_{\mathrm{top}}^2. \tag{16}$$

Numerically,

$$c_3 = \frac{1}{8\pi} \approx 0.039\,788\,735\,77, \qquad \delta_{\mathrm{top}} \approx 1.203\,044\,8 \times 10^{-4}, \qquad \delta_2 \approx 1.809\,145\,97 \times 10^{-8}.$$

Definition 4.2 (Exact seam generating function)**.** Let

$$x := e^{-2\alpha}.$$

Define the exact seam opening by

$$\varphi_\Sigma(\alpha) := \varphi_{\mathrm{base}} + \delta_{\mathrm{top}} x (1 - \delta_{\mathrm{top}} x)^{-5/4}. \tag{17}$$

Its Taylor expansion on the admissible branch begins as

$$\varphi_\Sigma(\alpha) = \varphi_{\mathrm{base}} + \delta_{\mathrm{top}} e^{-2\alpha} + \frac{5}{4}\delta_{\mathrm{top}}^2 e^{-4\alpha} + O(e^{-6\alpha}),$$

so the former quadratic opening is retained only as the $n \le 2$ asymptotic truncation of the exact seam generating function.

Theorem 4.3 (Exact electromagnetic closure)**.** *The fine-structure constant on the canonical admissible branch is the unique positive solution of*

$$\alpha^3 - 2c_3^3\alpha^2 - 8c_3^6\, b_1(N_{\mathrm{fam}}, N_\Phi)\ln\big((\varphi_\Sigma(\alpha))^{-1}\big) = 0. \tag{18}$$

Corollary 4.4 (Second-order asymptotic form)**.** *The former opening*

$$\varphi_{\mathrm{base}} + \delta_{\mathrm{top}} e^{-2\alpha} + \delta_2 e^{-4\alpha}$$

is the second-order asymptotic expansion of the exact seam generating function $\varphi_\Sigma(\alpha)$ and should be used only for asymptotic discussion, not for the benchmark definition.

On the canonical family branch these hard coefficients admit the bridge reinterpretations

$$\Omega_{\mathrm{adm}} = 48, \qquad b_1 = \frac{41}{10}, \qquad \delta_{\mathrm{top}} = \Omega_{\mathrm{adm}}\, c_3^4, \qquad \delta_2 = B\gamma\, \Omega_{\mathrm{adm}}^2 c_3^8 = \frac{5}{4}\,\delta_{\mathrm{top}}^2.$$

On the canonical branch this is the explicit scalar equation

$$\alpha^3 - \frac{\alpha^2}{256\pi^3} - \frac{41}{327680\pi^6}\ln\big((\varphi_\Sigma(\alpha))^{-1}\big) = 0.$$

The dependency structure of the electromagnetic closure is displayed in Figure 8. What matters is not one formula in isolation, but the fact that the same carrier-side inputs fix both the exact seam opening and the self-consistent root.

Remark (Logarithmic scale closure of the electromagnetic equation). Define the internally generated logarithmic scale

$$\mu_\Sigma(\alpha) := (\varphi_\Sigma(\alpha))^{-1}.$$

Then Equation (18) is equivalent to

$$\ln \mu_\Sigma(\alpha) = \frac{\alpha^2(\alpha - 2c_3^3)}{8c_3^6\, b_1(N_{\mathrm{fam}}, N_\Phi)}.$$

Thus the observed fine-structure constant is selected by a balance between a geometric cubic term and a self-generated logarithmic scale parameter. This is RG-like and transmutation-like, because b_1 is the abelian trace coefficient and the logarithm is attached to an internally produced scale. In the present paper it should be read as *logarithmic scale closure* or a geometric fixed-point equation rather than as an externally prescribed one-loop running law. The symbol $\mu_\Sigma(\alpha)$ here is local to the electromagnetic closure discussion and should not be confused with the later matching scale M_Σ.

A related appendix continuation records that

$$\gcd(\Omega_{\mathrm{adm}}, D_{\mathrm{start}}) = \dim \mathfrak{g}_{\mathrm{SM}}, \qquad \mathrm{lcm}(\Omega_{\mathrm{adm}}, D_{\mathrm{start}}) = |R(E_8)|.$$

4.2 Kernel output package

The benchmark row for $\alpha^{-1}(0)$ is defined by the positive self-consistent root of Equation (18), not by freezing the opening at its retained seed value. On the canonical branch that positive root remains

$$\alpha^{-1}(0) = 137.035\,999\,216\,158\,43.$$

For the remaining one-seed relations we keep the retained branch opening

$$c_3 = \frac{1}{8\pi}, \qquad \varphi_0^{\mathrm{ret}} = \frac{1}{6\pi} + \frac{3}{256\pi^4}, \tag{19}$$

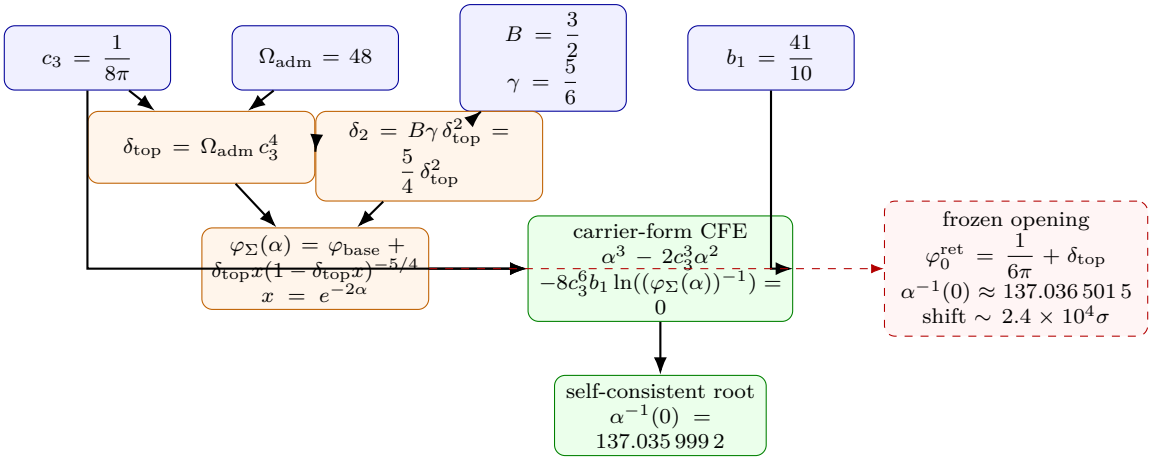


Figure 8: α closure flow. Carrier-side integers and minimal-carrier invariants feed the defect terms, those defect terms feed the exact seam generating function $\varphi_\Sigma(\alpha)$, and only the self-consistent loop gives the benchmark root. The dashed inset shows why a frozen opening is not a harmless simplification.

Remark (Exact opening versus retained seed). The two symbols belong to the same admissible branch but play different jobs. The electromagnetic α row uses the exact seam generating function $\varphi_\Sigma(\alpha)$ inside the carrier-form closure equation, whereas the one-seed bridge relations freeze that branch to the retained canonical seed $\varphi_0^{\rm ret}$. Thus $\varphi_\Sigma(\alpha)$ belongs only to the exact α benchmark line, while $\varphi_0^{\rm ret}$ is the seed used for $(\beta_{\rm rad}, \Omega_b, \lambda_C, \sin^2\theta_{13})$.

and the same retained kernel package emits the compact seed chain

$$C_{a\gamma\gamma} = -4c_3 = -\frac{1}{2\pi}, \qquad \beta_{\rm rad} = \frac{\varphi_0^{\rm ret}}{4\pi}, \qquad \Omega_b = (4\pi - 1)\beta_{\rm rad}, \qquad \lambda_C = \sqrt{\varphi_0^{\rm ret}(1-\varphi_0^{\rm ret})}, \qquad \sin^2\theta_{13} = \varphi_0^{\rm ret} e^{-\gamma} \tag{20}$$

Remark (Why the α row must stay self-consistent). The benchmark number $\alpha^{-1}(0) = 137.035\,999\,216\,158\,43$ comes from the exact self-consistent equation with $\varphi_\Sigma(\alpha)$. Its second-order truncation reproduces the first two correction coefficients, but it is not itself the benchmark definition. If one instead freezes the opening at

$$\varphi_0^{\rm ret} = \frac{1}{6\pi} + \frac{3}{256\pi^4},$$

the root shifts to $\alpha^{-1}(0) \approx 137.036\,501\,465$, i.e. by about 5.02×10^{-4}, which is roughly $2.4 \times 10^4 \sigma$ on the CODATA/NIST uncertainty. The wording of the α benchmark must therefore remain explicit: the benchmark uses the exact seam generating function, while the quadratic formula is only its controlled asymptotic shadow.

Kernel-output package

The point of this section is not to claim more numerics than TFPT currently owns. It is to show that the established kernel already fixes a coherent electromagnetic package before the later completion theorems begin: the seam constant c_3, the retained canonical seed $\varphi_0^{\rm ret}$, the self-consistent carrier-form CFE for α, the dimensionless axion–photon anomaly coefficient, the birefringence seed, and the Cabibbo anchor.

5 Compression identities and carrier exclusion

This section collects the algebraic compressions that turn the carrier packet and kernel numerics from a list of internally consistent outputs into an overdetermined structure with genuine exclusion power. The compression identities are not new theorems; they are exact algebraic consequences of the already established carrier data. The exclusion theorems are the sharpest available evidence that the carrier packet actively selects the physical world.

5.1 Rank lock and the 2-3-5 arithmetic

Proposition 5.1 (Carrier rank lock)**.** If the carrier identity $\dim S^+ = 2^{g-1}$, the canonical family data $N_{\text{fam}} = 3$, and the optional occupancy compression $\Omega_{\text{adm}} = (g-1)\dim \mathfrak{g}_{\text{SM}} = 12(g-1)$ are combined, then

$$3 \cdot 2^{g-1} = 12(g-1) \qquad \Longleftrightarrow \qquad 2^{g-3} = g-1. \tag{21}$$

The unique integral solution is $g = 5$.

Proof. Set $f(g) := 2^{g-3} - (g-1)$. Then

$$f(1) = \frac{1}{4}, \qquad f(2) = -\frac{1}{2}, \qquad f(3) = -1, \qquad f(4) = -1, \qquad f(5) = 0.$$

For $g \geq 4$,

$$f(g+1) - f(g) = 2^{g-3} - 1 > 0,$$

so f is strictly increasing on $g \geq 4$. Since $f(4) < 0$ and $f(5) = 0$, there is no solution with $g \geq 6$. A direct check for $g \leq 4$ shows no other solution. Hence the unique integral solution is $g = 5$. □

Remark (The 2-3-5 arithmetic)*.* The hard carrier packet rests on exactly three discrete numbers: 2 (the double cover), 3 (the color rank and family count), and 5 (the carrier rank). From these three atoms one recovers

$$\gamma = \frac{5}{6} = \frac{5}{2 \cdot 3}, \qquad B = \frac{3}{2}, \qquad B\gamma = \frac{5}{4} = \frac{5}{2^2},$$

$$\Omega_{\text{adm}} = 3 \cdot 2^4 = 48, \qquad D_{\text{start}} = 5 \cdot 12 = 60, \qquad \frac{D_{\text{start}}}{\Omega_{\text{adm}}} = \frac{5}{4} = B\gamma,$$

and the defect coefficients $\delta_{\text{top}} = 48\, c_3^4$ and $\delta_2 = \frac{5}{4}\, \delta_{\text{top}}^2 = 2880\, c_3^8$ with $2880 = 60 \cdot 48$. The persistent appearance of 5/4 as carrier asymmetry × norm, second defect stage, and ladder-to-occupancy ratio is therefore not three coincidences but one rational shadow of the 2-3-5 triad.

Lemma 5.2 (Universal 5/4 compression quotient on the minimal branch)**.** *Define*

$$\rho_{5/4} := B\gamma.$$

On the canonical minimal branch,

$$\rho_{5/4} = B\gamma = \frac{\delta_2}{\delta_{\text{top}}^2} = \frac{D_{\text{start}}}{\Omega_{\text{adm}}} = \frac{g_{\text{car}}}{g_{\text{car}} - 1} = \frac{5}{4}.$$

Thus 5/4 *is the universal compression quotient of the minimal branch rather than a recurring numerical motif.*

Proof. The identity $B\gamma = \frac{5}{4}$ is Corollary 3.11. The defect relation gives

$$\frac{\delta_2}{\delta_{\text{top}}^2} = \frac{5}{4}.$$

On the canonical branch,

$$\frac{D_{\text{start}}}{\Omega_{\text{adm}}} = \frac{60}{48} = \frac{5}{4},$$

and with $g_{\text{car}} = 5$ one also has

$$\frac{g_{\text{car}}}{g_{\text{car}} - 1} = \frac{5}{4}.$$

Hence all displayed ratios agree on the same exact value. □

Remark (Interpretive status of the quotient). The quotient itself is exact. Any holographic or spacetime-dimensional reading of the denominator $g_{\text{car}} - 1 = 4$ remains interpretive in the present manuscript rather than theorem-level.

Remark (Rank-5 master compression). If one accepts the carrier rank as the master parameter, then on the canonical branch

$$N_{\text{fam}} = \frac{g_{\text{car}} + 1}{2} = 3, \qquad \Omega_{\text{adm}} = N_{\text{fam}}\, 2^{g_{\text{car}}-1} = \frac{(g_{\text{car}} + 1)}{2}\, 2^{g_{\text{car}}-1} = (g_{\text{car}} + 1)\, 2^{g_{\text{car}}-2} = 48,$$

$$\Omega_{\text{adm}}\gamma = g_{\text{car}}\, 2^{g_{\text{car}}-2} = 40, \qquad b_1 = \frac{g_{\text{car}}\, 2^{g_{\text{car}}-2} + 1}{10} = \frac{41}{10}.$$

This is an algebraic compression of the closed family count, occupancy, abelian index, and the 41-chain onto the single integer $g_{\text{car}} = 5$.

Corollary 5.3 (Pythagorean compression of the integrity index on the rank-locked branch)**.** *Assume the rank-locked compression branch*

$$10\, b_1 = g_{\text{car}}\, 2^{g_{\text{car}}-2} + 1$$

and the rank lock

$$2^{g_{\text{car}}-3} = g_{\text{car}} - 1.$$

Then

$$\mathcal{I}_{41} = 10\, b_1 = 2g_{\text{car}}(g_{\text{car}} - 1) + 1 = g_{\text{car}}^2 + (g_{\text{car}} - 1)^2.$$

In particular, on the minimal rank-locked branch $g_{\text{car}} = 5$ one gets

$$\mathcal{I}_{41} = 5^2 + 4^2 = 41.$$

Proof. Multiplying the rank lock by $2g_{\text{car}}$ gives

$$g_{\text{car}}\, 2^{g_{\text{car}}-2} = 2g_{\text{car}}(g_{\text{car}} - 1).$$

Inserting this into the compression formula yields

$$10\, b_1 = 2g_{\text{car}}(g_{\text{car}} - 1) + 1.$$

Since

$$2g_{\text{car}}(g_{\text{car}} - 1) + 1 = g_{\text{car}}^2 + (g_{\text{car}} - 1)^2,$$

the claim follows. □

Remark (The number 4 as a cross-cutting echo). In addition to the 2-3-5 triad, the integer 4 appears at four structurally distinct levels:

$$F_{\text{patch}} = 4, \qquad g_{\text{car}} - 1 = 4, \qquad \frac{\Omega_{\text{adm}}}{\dim \mathfrak{g}_{\text{SM}}} = \frac{48}{12} = 4, \qquad \frac{|R(E_8)|}{D_{\text{start}}} = \frac{240}{60} = 4.$$

That is, four corner patches, carrier rank minus one, matter-to-gauge compression factor, and root-to-start quotient all evaluate to the same integer.

5.2 Exact algebraic compressions on the minimal branch

Corollary 5.4 (Two-point spectral compression of γ and φ_{base})**.** *Let*

$$y_+ := \frac{1}{2}, \qquad y_- := -\frac{1}{3}$$

be the two distinct carrier eigenvalues singled out by the factorized carrier polynomial. Then

$$\gamma = y_+ - y_-, \qquad \varphi_{\text{base}} = \frac{y_+ + y_-}{\pi}.$$

Here $y_+ + y_-$ is the unweighted Vieta sum of the two distinct carrier eigenvalues, not the trace on E.

Proof. By Equation (19) and the factorized carrier polynomial, the two distinct spectral values are exactly $y_+ = 1/2$ and $y_- = -1/3$. Their difference is

$$y_+ - y_- = \frac{1}{2} + \frac{1}{3} = \frac{5}{6} = \gamma,$$

while their sum is

$$y_+ + y_- = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

Since $\varphi_{\text{base}} = (1-\gamma)/\pi = 1/(6\pi)$, the claim follows. □

Corollary 5.5 (Family hypercharge variance and electroweak block trace)**.** *Equip one carrier family with the uniform average*

$$\mathbb{E}_{S^+}(A) := \frac{1}{16}\,\mathrm{Tr}_{S^+} A.$$

Then

$$\mathrm{Var}_{S^+}(Y) := \mathbb{E}_{S^+}(Y^2) - \mathbb{E}_{S^+}(Y)^2 = \frac{1}{16}\,\mathrm{Tr}_{S^+} Y^2 = \frac{5}{24}.$$

Consequently,

$$I_1^{\text{EW}} = \mathrm{Var}_{S^+}(Y)\, b_1.$$

Proof. The one-family trace identities give $\mathrm{Tr}_{S^+} Y = 0$ and $\mathrm{Tr}_{S^+} Y^2 = 10/3$. Hence the centered second moment is

$$\frac{1}{16}\,\mathrm{Tr}_{S^+} Y^2 = \frac{10/3}{16} = \frac{5}{24}.$$

Using $I_1^{\text{EW}} = \frac{5}{24}\, b_1$ from the electroweak block relation yields the second identity. □

Corollary 5.6 (Factorial quadratic trace and two-defect factorization)**.** *Let*

$$X := 6Y.$$

Then the one-family quadratic trace is

$$\mathrm{Tr}_{S^+}(X^2) = 36\ \mathrm{Tr}_{S^+}(Y^2) = 36 \cdot \frac{10}{3} = 120 = 5!.$$

Equivalently, using the six hypercharge sectors with multiplicities

$$(1, 3, 6, 1, 2, 3)$$

at

$$X \in \{0, -4, 1, 6, -3, 2\},$$

the same trace is

$$0^2 + 3 \cdot 4^2 + 6 \cdot 1^2 + 1 \cdot 6^2 + 2 \cdot 3^2 + 3 \cdot 2^2 = 120.$$

On the retained branch this gives the exact factorization

$$\delta_2 = 2880\, c_3^8 = 4!\ \mathrm{Tr}_{S^+}(X^2)\, c_3^8.$$

Proof. By Proposition 3.15, one has

$$\mathrm{Tr}_{S^+}(Y^2) = \frac{10}{3}.$$

Since $X = 6Y$, the first identity follows immediately. The sector-by-sector sum is the same quadratic trace written in the basis of admissible occupancy sectors from Proposition 3.16. Finally, Equation (16) gives

$$\delta_2 = 2880\, c_3^8 = 24 \cdot 120\, c_3^8 = 4!\ \mathrm{Tr}_{S^+}(X^2)\, c_3^8.$$

□

Remark (Quadratic trace versus variance). The value $120 = 5!$ is the unnormalized quadratic trace $\mathrm{Tr}_{S^+}(X^2)$, not the normalized variance. The corresponding uniform variance is

$$\mathrm{Var}_{S^+}(X) = 36\ \mathrm{Var}_{S^+}(Y) = 36 \cdot \frac{5}{24} = \frac{15}{2}.$$

Remark (Arithmetic Higgs lock on the canonical family branch). Define the generic occupancy

$$\Omega_{\mathrm{occ}} := 16\, N_{\mathrm{fam}}.$$

Then the abelian-index formula gives

$$10\, b_1 = \frac{40}{3} N_{\mathrm{fam}} + N_\Phi = \Omega_{\mathrm{occ}}\, \gamma + N_\Phi.$$

On the canonical family branch $N_{\mathrm{fam}} = 3$, hence $\Omega_{\mathrm{occ}} = 48$ and therefore

$$10\, b_1 = 40 + N_\Phi.$$

Thus the 41-chain arithmetically selects $N_\Phi = 1$. This is an arithmetic lock on the canonical branch, not the dynamical Higgs-selection theorem, which in the present manuscript still comes from the bosonic relative-index / Callias route.

5.3 The 41 integrity index

The integer 41 is not an isolated numerical curiosity. On the retained branch it is best read as a weighted integrity index rather than as a topological Euler-type object.

Definition 5.7 (The 41 integrity index)**.** Let

$$\mathcal{I}_{41} := 10\, b_1 = \Omega_{\mathrm{occ}}\gamma + N_\Phi.$$

On the canonical branch, $\Omega_{\mathrm{occ}} = \Omega_{\mathrm{adm}} = 48$ and $N_\Phi = 1$, so

$$\mathcal{I}_{41} = 40 + 1 = 41.$$

Then the hard arithmetic chain reads

$$10\, b_1 = \Omega_{\mathrm{adm}}\gamma + 1 = 41. \tag{22}$$

The extra $+1$ is precisely the arithmetic contribution of one light seam-even Higgs doublet. This makes $\mathcal{I}_{41}$ a hypercharge-driven occupancy integrity index:

$$\mathcal{I}_{41} = 40 + 1,$$

with $40 = \Omega_{\mathrm{adm}}\gamma$ the fermionic weighted occupancy and $1 = N_\Phi$ the light Higgs contribution. The canonical branch is therefore locked arithmetically at $40 \mapsto 41$, while the actual dynamical Higgs-selection theorem still comes from the bosonic / Callias closure route.

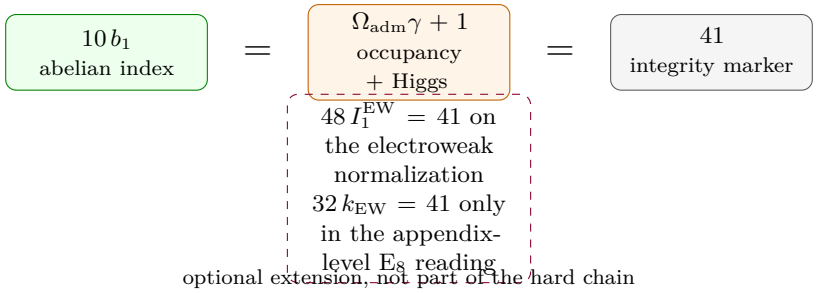


Figure 9: The 41-chain strip. The hard arithmetic chain is $10\,b_1 = \Omega_{\mathrm{adm}}\gamma + 1 = 41$. Electroweak normalization and E_8 block-language extensions are shown only as optional downstream readings.

Remark (Optional block-trace extension). If one introduces the electroweak block normalization

$$I_1^{\mathrm{EW}} := \frac{5}{24}\,b_1,$$

then

$$48\,I_1^{\mathrm{EW}} = 41.$$

The further identity

$$32\,k_{\mathrm{EW}} = 41$$

belongs only to the appendix-level E_8 reading and requires the block-index extrapolation.

Remark (Integrity-index reading). The canonical branch may be read as carrying an integrity marker: the hypercharge trace coefficient and the occupancy-weighted carrier norm plus one light Higgs doublet collapse onto the same integer 41. This is not a cryptographic claim. It is an algebraic consistency signature showing that several independently defined carrier-side layers agree on one integer.

5.4 Single-seed compression and the Cabibbo inversion

The seed-driven observation rows $(\beta_{\mathrm{rad}}, \Omega_b, \lambda_C, \sin^2\theta_{13})$ are not four independent hits but four projections of φ_0^{ret}. The Cabibbo angle admits an explicit inversion back to the seed:

$$\varphi_0^{\mathrm{ret}} = \frac{1 - \sqrt{1 - 4\lambda_C^2}}{2}. \tag{23}$$

Remark (Catalan expansion of the Cabibbo inversion). Setting

$$x := \lambda_C^2,$$

the inversion formula becomes

$$\varphi_0^{\mathrm{ret}} = x\,C(x), \qquad C(x) := \frac{1 - \sqrt{1 - 4x}}{2x}.$$

Hence

$$\varphi_0^{\mathrm{ret}} = \sum_{n \geq 0} C_n\,\lambda_C^{2n+2} = \lambda_C^2 + \lambda_C^4 + 2\lambda_C^6 + 5\lambda_C^8 + 14\lambda_C^{10} + \cdots, \tag{24}$$

where C_n are the Catalan numbers. In the present paper the appropriate interpretation of this identity is noncrossing combinatorics on the seed side; it is an exact algebraic strengthening of the one-seed compression rather than a claim about planar Feynman graphs, a specific large-N / 't Hooft resummation, or a holographic flavor limit.

From this all other bridge outputs become functions of λ_C alone:

$$\begin{aligned}\beta_{\rm rad} &= \frac{1-\sqrt{1-4\lambda_C^2}}{8\pi},\\ \Omega_b &= \frac{4\pi-1}{8\pi}(1-\sqrt{1-4\lambda_C^2}),\\ \sin^2\theta_{13} &= \frac{e^{-\gamma}}{2}(1-\sqrt{1-4\lambda_C^2}).\end{aligned} \tag{25}$$

Remark (Exact seed-budget identity). The same one-seed map carries an exact budget identity:

$$\Omega_b + \beta_{\rm rad} = \varphi_0^{\rm ret}. \tag{26}$$

Indeed,

$$\Omega_b + \beta_{\rm rad} = (4\pi-1)\beta_{\rm rad} + \beta_{\rm rad} = 4\pi\,\beta_{\rm rad} = \varphi_0^{\rm ret}.$$

This is best read as an exact seed-splitting identity inside the intrinsic observation layer, not as an independent dynamical conservation law. Because Ω_b and $\beta_{\rm rad}$ are dimensionless density fractions rather than angle measures, any 4π solid-angle reading of this identity remains interpretive rather than theorem-level.

Thus on the canonical branch, Cabibbo alone pulls β, Ω_b, and θ_{13}. Conversely, θ_{13} pulls λ_C and Ω_b through $\varphi_0^{\rm ret} = e^{\gamma}\sin^2\theta_{13}$. This is a much stronger correlation statement than the seed map alone.

5.5 Carrier exclusion theorems

The strongest argument that the carrier packet selects the physical world is not a further precision hit. It is that wrong discrete worlds fail brutally.

Theorem 5.8 (Carrier exclusion)**.** *Let $\alpha^{-1}(0; N_{\rm fam}, N_\Phi, \nu^c)$ denote the positive self-consistent root of the carrier-form electromagnetic closure equation* (18) *with the indicated discrete data. Then:*

Discrete world	*Change from canonical*	$\alpha^{-1}(0)$	*Residual vs CODATA*	*Reading*
canonical $(3, 1, \textbf{\textit{yes}})$	—	137.0359992	$+3.9\times10^{-8}$	*inside CODATA window* $(+1.87\sigma)$
$N_{\rm fam}=2$	*one family fewer*	156.094	$+19.1$	*catastrophically excluded*
$N_{\rm fam}=4$	*one family more*	124.835	-12.2	*catastrophically excluded*
$N_\Phi=2$	*two light Higgs doublets*	135.946	-1.09	*catastrophically excluded*
no ν^c	*remove right-handed neutrino*	137.0338	-2.16×10^{-3}	*excluded* $(\sim1.03\times10^5\sigma)$

Proof sketch. Hold fixed

$$c_3 = \frac{1}{8\pi}, \qquad \varphi_{\rm base} = \frac{1}{6\pi}, \qquad \gamma = \frac{5}{6}, \qquad B = \frac{3}{2},$$

and solve the positive self-consistent root of the same scalar closure equation on each discrete world:

$$\alpha^3 - 2c_3^3\alpha^2 - 8c_3^6\, b_1(N_{\rm fam}, N_\Phi)\ln\!\big(\varphi_\Sigma(\alpha)^{-1}\big) = 0,$$

with

$$\varphi_\Sigma(\alpha) = \varphi_{\rm base} + \delta_{\rm top}e^{-2\alpha}(1-\delta_{\rm top}e^{-2\alpha})^{-5/4}, \qquad \delta_{\rm top} = \Omega_{\rm adm}c_3^4, \qquad \delta_2 = \frac{5}{4}\,\delta_{\rm top}^2,$$

discrete world	Residual vs CODATA	reading
canonical $(3, 1, \text{yes})$	$+3.9 \times 10^{-8}$	inside the CODATA window
$N_{\text{fam}} = 2$	$+19.1$	catastrophically excluded
$N_{\text{fam}} = 4$	-12.2	catastrophically excluded
$N_{\Phi} = 2$	-1.09	extra light doublet breaks closure
no ν^c	-2.16×10^{-3}	$\sim 1.03 \times 10^5 \sigma$ exclusion

Figure 10: Exclusion matrix for wrong discrete worlds. The middle column records the residual versus the CODATA benchmark rather than a misleading shift relative to the canonical row. Once the discrete packet is changed, the electromagnetic closure leaves the observed window decisively.

and

$$b_1(N_{\text{fam}}, N_{\Phi}) = \frac{4}{3} N_{\text{fam}} + \frac{1}{10} N_{\Phi}.$$

For the no-ν^c row, b_1 stays unchanged because $Y(\nu^c) = 0$, while the occupancy changes from $\Omega_{\text{adm}} = 48$ to $\Omega_{\text{adm}} = 45$ because the one-family packet drops from 16 to 15 states. The same positive-branch solver and tolerance are used for every row, so the deviations are arithmetic consequences of changed discrete data rather than fits. □

Remark. Two families shift α^{-1} by $+19$, four families by -12, and two Higgs doublets by -1.09. Even the mildest alternative—removing ν^c—shifts the value by 2.16×10^{-3}, which is $\sim 1.03 \times 10^5$ times the CODATA uncertainty. This is not marginal discrimination; it is categorical exclusion. The canonical branch $(3, 1, \text{yes})$ is the only discrete world that survives the electromagnetic closure test within the CODATA window.

6 Closure architecture A: families, Higgs, and local spectral scale

This first closure-architecture block is the shortest route from the hard carrier packet to a broader Standard Model plus gravity compiler. It has three moves: the local patch-deficit count gives the carrier-side family count, the APS family theorem globalizes it into the family space F, the bosonic relative index selects the unique light Higgs doublet, and the local spectral-scale channel generates both $\bar{M}_{\text{Pl}}$ and the electroweak vacuum from one common geometric response.

6.1 Family monodromy and occupancy

Definition 6.1 (Family space)**.** Let X_f denote the fundamental section of the seam complement. Define

$$F := \mathcal{H}^1_{\text{APS}}(X_f) \cong H^1(X_f, \mathbb{C}).$$

We use the distinguished cycle basis (C_1, C_2, C_T) only as a concrete carrier-side basis of this family space, not as a replacement for its APS/Hodge realization.

In this interpretation F is neither an arbitrary flavor vector space nor a loose bookkeeping trick; it is a cycle space attached to the seam data. The actual bridge logic is shown in Figure 11:

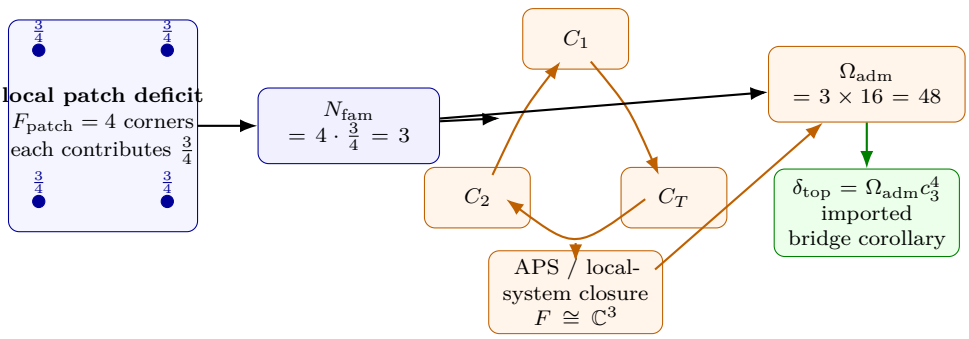


Figure 11: Carrier-to-family closure bridge. The local patch-deficit count gives the hard multiplicity $N_{\text{fam}} = 3$, the APS/local-system step turns that multiplicity into a genuine family space, and the resulting occupancy $\Omega_{\text{adm}} = 48$ feeds back into the defect reading $\delta_{\text{top}} = \Omega_{\text{adm}} c_3^4$.

the local patch-deficit count fixes $N_{\text{fam}} = 3$, the APS/local-system step promotes that count to a family space, and the occupancy reinterpretation then feeds back into the defect channel.

The Hodge metric on F turns the monodromy data into an honest unitary family system rather than a merely decorative triangle sketch.

Theorem 6.2 (Orbifold–APS family theorem from canonical doubling)**.** *Let X_f be the spin-resolved orbifold seam section with four $\mathbb{Z}_4$ corners, and let L_F be the associated flat family bundle. Then the local corner count, the orbifold topology, and the APS family index compress to the single statement*

$$\operatorname{Ind}^{\text{orb,APS}}(D^+_{\text{rel}} \otimes L_F) := \int_{X_f} \hat{A}(TX_f)\,\operatorname{ch}(L_F) - \frac{\eta_\Sigma + h_\Sigma}{2} + \sum_{p \in \text{corners}} \delta_p = 3, \tag{27}$$

with

$$\delta_p = \frac{3}{4} \qquad \textit{for each of the four } \mathbb{Z}_4 \textit{ corners.}$$

Equivalently,

$$\chi_{\text{orb}}(X_f) = 1 - 4\left(1 - \frac{1}{4}\right) = -2, \qquad \operatorname{rank} H_1^{\text{free}}(X_f, \mathbb{Z}) = 3,$$

and the local patch-deficit count is

$$N_{\text{fam}} = F_{\text{patch}}\left(1 - \frac{1}{4}\right) = 4 \cdot \frac{3}{4} = 3. \tag{28}$$

Consequently,

$$F = H^1(X_f, \mathbb{C}) \cong \mathbb{C}^3, \qquad N_{\text{fam}} = 3, \qquad \Omega_{\text{adm}} = N_{\text{fam}} \dim S^+ = 3 \cdot 16 = 48.$$

Remark (How the family theorem is now used). The family theorem is no longer a declared closure import. It is the family component of the master closure theorem, written separately here only because it also feeds the occupancy bridge and the later flavor geometry.

Corollary 6.3 (Boundary-to-bulk bridge corollary for the admissible occupancy)**.** *On the canonical branch,*

$$\dim(S^+ \otimes F) = 16 \cdot 3 = 48 = \Omega_{\text{adm}}.$$

Hence the topological surcharge enters the present paper as a derived occupancy corollary with an observation-facing reading:

$$\delta_{\text{top}} = c_3^4 \dim(S^+ \otimes F) = \Omega_{\text{adm}}\, c_3^4.$$

The family subsection is therefore no longer split between a local count, a global APS realization, and a later occupancy reinterpretation. Those three statements are compressed into one theorem with one bridge corollary, while the holonomy proposition below records the resulting unitary geometry on F.

This theorem is the family statement referenced later by the admissibility theorems: the closure layer uses it as the explicit family component of the master closure theorem rather than as an external import.

Proposition 6.4 (Special unitary family holonomy on the determinant-trivial branch)**.** Let $L_F \to X_f$ be the flat Hermitian family bundle with fiber $F \cong H^1(X_f, \mathbb{C})$. Assume that its determinant line is flat trivial,

$$\det L_F \cong X_f \times \mathbb{C},$$

and that the chosen flat connection preserves this trivialization. Then

$$\mathrm{Hol}(\nabla_F) \subset SU(3)_F.$$

Proof. The Hermitian metric implies

$$\mathrm{Hol}(\nabla_F) \subset U(3).$$

The induced connection on $\det L_F$ has trivial holonomy by assumption, so for every loop γ one has

$$\det \rho_F(\gamma) = 1.$$

Hence every holonomy matrix lies in $SU(3)$. □

The role of the present subsection is thus no longer only mnemonic. It displays both the local patch-deficit count and the global APS/Hodge realization before the later closure stack compresses them into the theorem-level compiler surface.

6.2 Occupancy and boundary-to-bulk bridge

With the family bridge installed, the matter space becomes $S^+ \otimes F$ and the occupancy $\Omega_{\mathrm{adm}} = 48$ is already fixed by the imported family theorem and its bridge corollary. Here *occupancy* means the admissible chiral-state count after family lifting; it is not used as a primitive datum. The bridge point is not that family lifting creates a new numerical surcharge. Rather, the inherited hard kernel value

$$\delta_{\mathrm{top}} = \frac{3}{256\pi^4}$$

admits on the canonical family branch the occupancy reinterpretation $\delta_{\mathrm{top}} = \Omega_{\mathrm{adm}}\, c_3^4$. Likewise the hard second-order defect contribution is re-read as

$$\delta_2 = \frac{5}{4}\,\delta_{\mathrm{top}}^2 = B\gamma\,\Omega_{\mathrm{adm}}^2 c_3^8.$$

In the present manuscript this occupancy corollary is used with an observation-facing reading: it is the boundary-to-bulk step that later feeds both the electromagnetic closure and the geometric channel. The coefficient $5/4$ is therefore not an extra numerical motif; it is the minimal-carrier compression factor $B\gamma$ rewritten in defect language. At the same time, the equality $\delta_{\mathrm{top}} = \Omega_{\mathrm{adm}} c_3^4$ is not presented as an independent dynamical transgression theorem in the ordinary field-theory sense.

6.3 From bridge to geometric completion

Principle 6.5 (Closure split)**.** Once the carrier packet and family closure architecture are fixed, the focused main-text closure problem splits into a geometric program $\mathcal{A}_{\mathrm{geom}}$ and a transport program $\mathcal{A}_{\mathrm{trans}}$. Record algebra and prediction semantics are now tied directly to the admissible gap in the main theorem stack, while E_8 scale grammar and horizon extrapolations are deferred to appendix-level continuations.

The two main-text programs are packaged as

$$\mathcal{A}_{\mathrm{geom}} := \langle E_3 \oplus E_2,\ D_{\mathrm{rel}},\ \mathcal{B}_{\mathrm{rel}},\ \varpi_{\mathrm{spin}} \rangle, \qquad \mathcal{A}_{\mathrm{trans}} := \langle U_6,\ T,\ D_y,\ \mathcal{Y}_y^{(\varepsilon)},\ \mathcal{C}_\Sigma \rangle.$$

This split keeps the carrier theorem, the geometric reconstruction problem, and the record-level extensions from being flattened into one undifferentiated claim. In particular, the metric field is no longer treated here as an independent entry of $\mathcal{A}_{\mathrm{geom}}$; it is reconstructed from the retained relative operator branch introduced next.

6.4 Relative APS and superconnection setup

The geometric completion is formulated on a graded bundle

$$\mathcal{E}_{\mathrm{rel}} = \mathcal{E}_{\mathrm{rel}}^{+} \oplus \mathcal{E}_{\mathrm{rel}}^{-},$$

with a declared reference pair (D, D_{ref}) and APS-type boundary conditions on the seam Σ; see Appendix C. The geometric package is

$$\mathbb{A}_\Sigma := (D_{\mathrm{rel}}, A_\Sigma^{\mathrm{geo}}), \qquad A_\Sigma^{\mathrm{geo}} := \mathcal{B}_{\mathrm{rel}} \oplus \nabla_{\chi_{\mathrm{geo}}}.$$

Here A_Σ^{geo} is the geometric-branch reorganization of the primitive seed package A_Σ^{seed} introduced earlier in Equation (1).

Theorem 6.6 (Relative metric reconstruction from D_{rel} on the admissible branch)**.** *Assume the admissible center satisfies*

$$Z(\mathcal{A}_{\mathrm{adm}}) \cong C^\infty(M),$$

and the principal symbol of the retained relative operator obeys

$$\sigma(D_{\mathrm{rel}})(x,\xi)^2 = g^{\mu\nu}(x)\xi_\mu\xi_\nu\, \mathbf{1}.$$

Then the principal symbol of D_{rel} reconstructs the metric g, the spin structure, and the Levi–Civita connection on the retained branch. Inner fluctuations of the same relative spectral datum generate the gauge sector. Thus g is read from D_{rel} rather than inserted as an independent member of $\mathcal{A}_{\mathrm{geom}}$.

Proof sketch. On the admissible branch the commutative center identifies the manifold algebra, while the principal symbol determines the quadratic form on cotangent vectors. Standard Dirac-type reconstruction then recovers the spin bundle and Levi–Civita connection from the same symbol data. The relative formulation changes the reference bookkeeping, not the reconstruction mechanism, so the metric information is still carried by D_{rel} itself. □

Theorem 6.7 (Full operator reconstruction on the admissible branch)**.** *Let*

$$D_{\mathrm{adm}} := P_{\mathrm{adm}} D_{\mathrm{rel}} P_{\mathrm{adm}}.$$

Then the admissible spectral datum reconstructs the full geometric branch:

(i)

$$X = \operatorname{Spec} Z(\mathcal{A}_{\mathrm{adm}}),$$

with topology generated by the relative Connes metric

$$d_{\mathrm{rel}}(x,y) := \sup\{|a(x)-a(y)| : a \in Z(\mathcal{A}_{\mathrm{adm}}),\ \|[D_{\mathrm{adm}},a]\| \le 1\};$$

(ii)

$$\Sigma = \{x \in \widetilde{M} : \iota x = x\} = \{x \in \widetilde{M} : d_{\mathrm{rel}}(x,\iota x) = 0\};$$

(iii) the principal symbol of D_{adm} reconstructs the metric g, the spin structure, and the Levi–Civita connection;

(iv) the local spectral density of D_{adm} reconstructs the geometric spectral scale:

$$\rho_0(x) = \frac{\mathrm{tr}_{\mathrm{int}}\mathbf{1}}{16\pi^2}\chi_{\mathrm{geo}}(x)^4, \qquad \chi_{\mathrm{geo}}(x) = \left(\frac{16\pi^2\rho_0(x)}{\mathrm{tr}_{\mathrm{int}}\mathbf{1}}\right)^{1/4};$$

(v) therefore

$$(X,\Sigma,g,\sigma), \qquad \sigma = \log(\chi_{\mathrm{geo}}/\Lambda),$$

are reconstructed from the same admissible operator D_{adm}.

Corollary 6.8 (No residual geometric datum)**.** *The geometric package A_Σ^{geo} is fully reconstructed from the admissible operator and its local spectral density. In particular g, Σ, and χ_{geo} are not independent inputs of TFPT.*

Definition 6.9 (Local spectral scale)**.** Write the geometric response as a local spectral scale

$$\chi_{\mathrm{geo}} = \Lambda e^{\sigma}, \qquad D_\sigma := e^{-\sigma/2} D_{\mathrm{rel}} e^{-\sigma/2}.$$

The relative spectral action is then organized as

$$S_{\mathrm{rel}}[\sigma,D] := \mathrm{Tr}\, f\left(\frac{D_\sigma + \mathcal{B}_{\mathrm{rel}}}{\Lambda}\right) - \mathrm{Tr}\, f\left(\frac{D_{\mathrm{ref}}}{\Lambda}\right) + \frac{i\pi}{2}\Delta\eta_\Sigma.$$

Corollary 6.10 (Constant-scale reduction of the geometric branch)**.** *If the local spectral scale is frozen to a constant $\sigma = \sigma_0$, then $\chi_{\mathrm{geo}} = \chi_{\mathrm{geo},0} := \Lambda e^{\sigma_0}$ and the present geometric branch reduces to the constant-χ_{geo} closure relations*

$$\bar{M}_{\mathrm{Pl}}^2 = \frac{\chi_{\mathrm{geo},0}^2}{2\pi^2}, \qquad v_{\mathrm{geo}}^2 = Z_H\,\frac{\mu_\Phi^2(\chi_{\mathrm{geo},0})}{\lambda_\Phi}.$$

Thus the former constant-χ_{geo} formulas are retained as the homogeneous branch of the local spectral-scale formulation rather than as a competing package.

6.5 Bosonic relative index and Higgs selection

Theorem 6.11 (Carrier-forced determinant classes on the seam)**.** *Let*

$$L_2 := \det E_2, \qquad L_3 := \det E_3.$$

Assume the total seam twist on the oriented double cover is one unit. Then the determinant holonomy classes satisfy

$$[\mathrm{Hol}_\Sigma(L_2)] = 1 \in \pi_1(U(1)), \qquad [\mathrm{Hol}_\Sigma(L_3)] = 0.$$

Equivalently, for the asymptotic bosonic mass fields on the retained branch,

$$\mathrm{wind}(\det\Phi_2) = 1, \qquad \mathrm{wind}(\det\Phi_3) = 0.$$

The doublet winding is forced by the spin lift on the double cover, whereas center neutrality on the admissible branch forces determinant triviality in the triplet sector.

Proof sketch. The oriented double cover already forces a half-angle lift on the rank-two weak sector. Passing to the determinant line therefore gives one full winding in L_2. By contrast, the admissible branch requires center neutrality in the color sector, so the determinant line of E_3 is trivial on the retained branch. One unit of total seam twist can therefore be realized only by the doublet determinant class. □

Remark (Energy interpretation on the retained branch). The older asymptotic bosonic functional

$$\mathcal{E}_{\mathrm{asym}}(\Phi_2, \Phi_3) := \tau_2 |\mathrm{wind}(\det \Phi_2)| + \tau_3 |\mathrm{wind}(\det \Phi_3)| + m_3^2 \dim \ker \mathcal{B}_{E_3}^{+},$$

with $\tau_3 > \tau_2 > 0$, remains a useful physical intuition for why the retained branch is energetically preferred. In the present rewrite, however, the structural weight is carried by the determinant-class theorem above rather than by the cost functional itself.

Theorem 6.12 (Callias Higgs selection from determinant classes)**.** *Let N_Σ be the oriented two-dimensional normal slice to the seam. For $a \in \{2, 3\}$ define*

$$\mathcal{B}_{E_a} := \sigma^\mu \nabla_\mu^{(a)} + \Phi_a$$

on N_Σ, where Φ_a is asymptotically invertible on the boundary at infinity. Under the determinant-class data of Theorem 6.11, assume

$$(\mathcal{B}_{E_2}^{-})^\dagger \mathcal{B}_{E_2}^{-} \geq m_-^2 > 0, \qquad (\mathcal{B}_{E_3}^{\pm})^\dagger \mathcal{B}_{E_3}^{\pm} \geq m_3^2 > 0.$$

Then the bosonic relative index satisfies

$$\mathrm{Ind}_{\mathrm{rel}}^{P_{\mathrm{adm}}}(\mathcal{B}_{E_2}^{+}) = 1, \qquad \mathrm{Ind}_{\mathrm{rel}}^{P_{\mathrm{adm}}}(\mathcal{B}_{E_3}^{+}) = 0.$$

Moreover,

$$\dim \ker \mathcal{B}_{E_2}^{+} = 1, \qquad \ker \mathcal{B}_{E_2}^{-} = \ker \mathcal{B}_{E_3}^{\pm} = 0.$$

Consequently the unique light seam-even bosonic zero mode lies in

$$\Phi \in \Gamma(E_2) \cong (1, 2)_{1/2}, \qquad \iota\Phi = +\Phi.$$

Proof sketch. By Theorem 6.11, the retained branch fixes the asymptotic winding data to

$$\mathrm{wind}(\det \Phi_2) = 1, \qquad \mathrm{wind}(\det \Phi_3) = 0.$$

The Callias index theorem then gives

$$\mathrm{Ind}_{\mathrm{rel}}^{P_{\mathrm{adm}}}(\mathcal{B}_{E_2}^{+}) = 1, \qquad \mathrm{Ind}_{\mathrm{rel}}^{P_{\mathrm{adm}}}(\mathcal{B}_{E_3}^{+}) = 0.$$

The positivity bounds remove the opposite chirality and all triplet zero modes, so the index lifts to an actual uniqueness statement for one seam-even Higgs doublet. □

This is the sharp form of the Higgs claim used here. The issue is not whether a Higgs doublet can be written down, but whether the relative bosonic index can be shown to select it uniquely. Without a genuine Callias / Dirac / Dolbeault-type index problem there would be no mathematical reason for one light doublet and no light triplet to emerge from the bosonic sector. Once that uniqueness statement is fixed, the renormalizable carrier-compatible Yukawa couplings are the standard ones,

$$Q\, u^c\, \Phi, \qquad Q\, d^c\, \Phi^\dagger, \qquad L\, e^c\, \Phi^\dagger, \qquad L\, \nu^c\, \Phi,$$

so the Higgs statement becomes a genuine structural bridge rather than a notation choice.

Coefficient	Operator sector	Generated term	$\chi_{\rm geo}$ scaling	Status
a_0	relative bulk trace	vacuum-energy bookkeeping	$\chi_{\rm geo}^4$	Hard
a_2	seam-even bosonic bulk term	$\chi_{\rm geo}^2 R$, $\chi_{\rm geo}^2 \Phi^\dagger\Phi$	$\chi_{\rm geo}^2$	Closure
a_4	local bosonic sector	$\lvert D\Phi\rvert^2$, $(\Phi^\dagger\Phi)^2$, $R\,\Phi^\dagger\Phi$, $\mathrm{Tr}\,\mathbb{F}^2$, R^2	$\chi_{\rm geo}^0$	Closure
boundary/defect term	seam correction and parity data	local defect counterterms, seam-even selection	Δ_Σ	Closure
η term	APS spectral asymmetry	sign and spectral-flow control	scale free	Closure

Table 3: Heat-kernel map for the geometric closure architecture.

6.6 Heat-kernel coefficient map and local spectral-scale bifurcation

The local target action is written as

$$\Gamma_{\rm eff} = \int d^4x\sqrt{-g}\,\Big[c_R\chi_{\rm geo}^2 R + Z_\chi(\partial\chi_{\rm geo})^2 - \mu_\Phi^2(\chi_{\rm geo})\,\Phi^\dagger\Phi + \lambda_\Phi(\Phi^\dagger\Phi)^2 + \frac{1}{6M^2}R^2 + \cdots\Big]. \quad (29)$$

The relative bosonic endomorphism used below is

$$E_{\rm rel} = -\frac{1}{4}R\,\mathbf{1}_{48} - (\Phi^\dagger\Phi)\,P_2 + \frac{1}{2}\gamma^{\mu\nu}\mathbb{F}_{\mu\nu}, \quad (30)$$

and the coefficient bookkeeping expected from the relative spectral expansion is summarized in Table 3.

The resulting geometric logic is displayed in Figure 12. The same coefficient map controls the Einstein term, the Higgs normalization, the Higgs vacuum relation, and the curvature/gauge side terms. This is why the local spectral-scale sector is the principal geometric closure channel of the manuscript rather than just one more scalar parameter.

Proposition 6.13 (Canonical bosonic normalization on the seam-even branch)**.** On the seam-even bosonic branch selected by P_2, the relative Seeley–DeWitt map is

$$a_2^{\rm rel} = -\int d^4x\sqrt{g}\left(\frac{\chi_{\rm geo}^2}{4\pi^2}R + \frac{\chi_{\rm geo}^2}{8\pi^2}\Phi^\dagger\Phi\right) + a_{2,\Sigma}, \quad (31)$$

$$a_{4,\rm rel} = \int d^4x\sqrt{g}\left[\frac{1}{16\pi^2}|D\Phi|^2 + \frac{1}{16\pi^2}(\Phi^\dagger\Phi)^2 + \frac{1}{96\pi^2}R\,\Phi^\dagger\Phi + \frac{1}{192\pi^2}\,\mathrm{Tr}\,\mathbb{F}^2 + \dots\right] + a_{4,\Sigma}. \quad (32)$$

Therefore the bosonic normalization constants are

$$\bar{M}_{\rm Pl}^2 = \frac{\chi_{\rm geo,0}^2}{2\pi^2}, \qquad Z_H = \frac{1}{16\pi^2}, \qquad \Phi_{\rm can} := \sqrt{Z_H}\,\Phi = \frac{\Phi}{4\pi},$$

and the canonical Higgs vacuum relation becomes

$$\bar{M}_{\rm Pl}^2 = \frac{\chi_{\rm geo,0}^2}{2\pi^2}, \qquad v_{\rm geo}^2 = Z_H\,\frac{\mu_\Phi^2(\chi_{\rm geo,0})}{\lambda_\Phi}. \quad (33)$$

Proof sketch. The operator Equation (30) is the seam-even bosonic endomorphism entering the relative heat-kernel expansion. The a_2 coefficient fixes the Einstein branch and the bosonic mass term, while a_4 fixes the scalar kinetic normalization, quartic coefficient, mixed curvature term, and gauge curvature trace. The coefficient of $|D\Phi|^2$ is therefore the wave-function factor $Z_H = 1/(16\pi^2)$. After the canonical rescaling $\Phi_{\rm can} = \sqrt{Z_H}\,\Phi$, the Einstein and Higgs branches are read from the same relative coefficient map rather than from separate normalization conventions. □

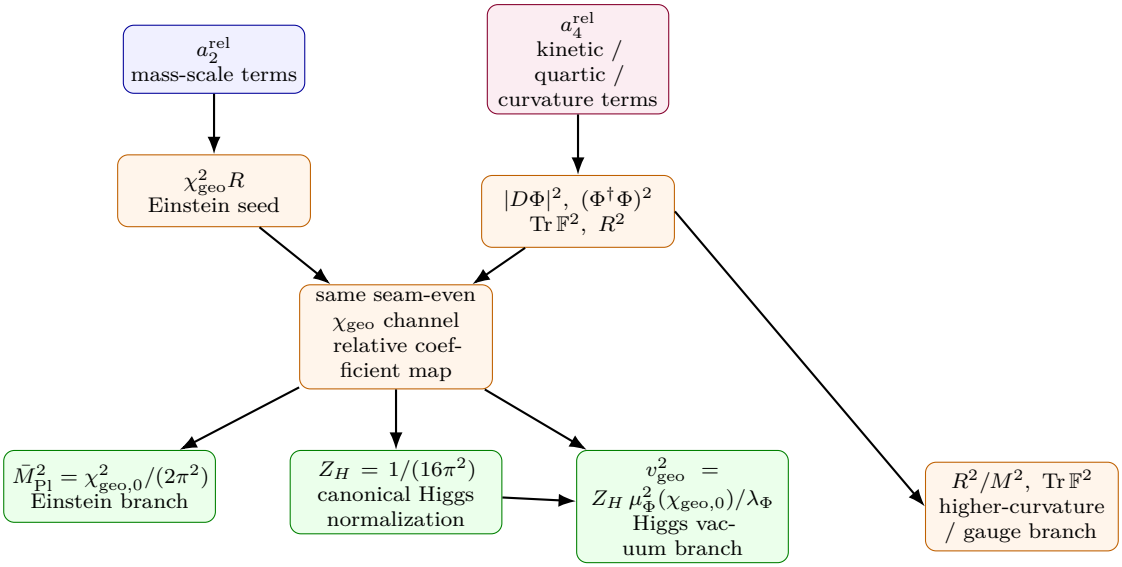


Figure 12: Coefficient flow of the local spectral-scale channel. The a_2 coefficient supplies the Einstein and Higgs mass seeds, the a_4 coefficient supplies kinetic/quartic/gauge-curvature data, and the same seam-even $\chi_{\rm geo}$ response turns both streams into the coupled relations for $\bar{M}_{\rm Pl}$ and $v_{\rm geo}$.

Remark (Newton form is not an independent output). Using the standard reduced-Planck definition $\bar{M}_{\rm Pl}^{-2} = 8\pi G_N$, the first relation in Equation (33) is equivalently

$$G_N \chi_{\rm geo,0}^2 = \frac{\pi}{4}.$$

This Newton-form equation is therefore a rewriting of the Einstein-branch closure, not a second independent prediction.

Lemma 6.14 (Two-sheet seam-even zero-mode normalization)**.** *If the seam-even Higgs zero mode is normalized across both sheets of the oriented double cover, then its branch weight is*

$$w_H = 2\, g_{\rm car}\, \beta_{\rm rad}^2 \exp\left[-\frac{\alpha^{-1}(0) + \delta_{\rm ph}}{5}\right].$$

With

$$\mu_\Phi^2(\chi_{\rm geo,0}) = \frac{\chi_{\rm geo,0}^2}{8\pi^2} w_H^2, \qquad \lambda_\Phi = \frac{1}{16\pi^2},$$

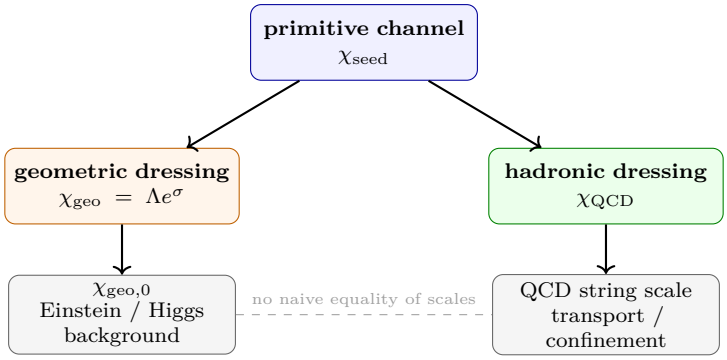
the canonical Higgs vacuum expectation value is

$$v_{\rm geo} = \bar{M}_{\rm Pl}\, g_{\rm car}\, \beta_{\rm rad}^2 \exp\left[-\frac{\alpha^{-1}(0) + \delta_{\rm ph}}{5}\right]. \tag{34}$$

Proof sketch. The two-sheet seam-even normalization doubles the unnormalized zero-mode weight and therefore produces the stated w_H. Since the canonical Higgs field is $\Phi_{\rm can} = \sqrt{Z_H}\,\Phi$, its vacuum relation is

$$v_{\rm geo}^2 = \frac{Z_H \mu_\Phi^2(\chi_{\rm geo,0})}{\lambda_\Phi} = \frac{\chi_{\rm geo,0}^2}{8\pi^2} w_H^2.$$

Using $\bar{M}_{\rm Pl}^2 = \chi_{\rm geo,0}^2/(2\pi^2)$ from Proposition 6.13 gives $v_{\rm geo} = \bar{M}_{\rm Pl} w_H/2$, hence Equation (34). □



One primitive response bifurcates into sector-dressed scales rather than one universal numerical channel.

Figure 13: Local spectral-scale bifurcation. The primitive response χ_{seed} is later dressed geometrically into χ_{geo} and its vacuum branch $\chi_{\text{geo},0}$, while the hadronic sector dresses the same primitive channel into χ_{QCD}. This is the clean separation that prevents the Planck/Higgs/QCD scale clash.

Remark (Geometric versus benchmark electroweak scales). For source formulas driven by the bosonic closure we write v_{geo} for the quantity in Equation (34). Numerically,

$$v_{\text{geo}} \approx 241.87 \text{ GeV}.$$

This is the internal geometric branch value used in the charged-lepton and seesaw source formulas. When the manuscript compares to the measured electroweak vacuum parameter at the benchmark interface, it instead uses

$$v_{\text{EW}}^{\text{ref}} := 246.22 \text{ GeV},$$

as an external reference value. These two numbers therefore belong to different bookkeeping layers and must not be silently interchanged inside one derivation or one table.

Remark (Sector-dressed local spectral-scale channels). The quantity $\chi_{\text{geo},0}$ entering the induced Einstein channel is not identified with the transport-sector confinement scale. In the hadronic sector we write χ_{QCD} and regard it as a locally dressed channel obtained from the same underlying primitive response χ_{seed} after sector-dependent renormalization and confinement-scale matching:

$$\chi_{\text{seed}} \longrightarrow \chi_{\text{geo},0} \quad \text{(gravitational background)}, \qquad \chi_{\text{seed}} \longrightarrow \chi_{\text{QCD}} \quad \text{(transport-dressed local channel)}.$$

This is the minimal statement needed to avoid the otherwise immediate scale clash between the Planck bridge and the QCD string tension.

Theorem 6.15 (Relative spectral gravity branch)**.** *The same geometric closure sector defines a local spectral gravity branch of schematic form*

$$\Gamma_{\text{grav}} := \int d^4x\sqrt{-g}\Big[\frac{1}{2}Z_N(\chi_{\text{geo}})\chi_{\text{geo}}^2 R + \frac{1}{2}Z_\chi(\nabla\chi_{\text{geo}})^2 - U(\chi_{\text{geo}}) + R\,\mathcal{F}_R(\Box/\chi_{\text{geo}}^2)R + C\,\mathcal{F}_C(\Box/\chi_{\text{geo}}^2)C + |D\Phi|^2 - V(\Phi$$

On the low-curvature local branch, the form factors reduce to the retained local coefficients and one recovers the effective action

$$\Gamma_{\text{grav}}^{\text{eff}} := \int d^4x\sqrt{-g}\Big[\frac{\bar{M}_{\text{Pl}}^2}{2}R - \Lambda_{\Sigma,\text{ren}} + \frac{1}{6M^2}R^2 + \frac{Z_\chi}{2}(\nabla\chi_{\text{geo}})^2 - U(\chi_{\text{geo}}) + |D\Phi|^2 - V(\Phi,\chi_{\text{geo}}) - \sum_a \frac{1}{4g_a^2}\operatorname{Tr} F_{a,\mu\nu}F_a^{\mu\nu}\Big],$$

with

$$\bar{M}_{\text{Pl}}^2 = \frac{\chi_{\text{geo},0}^2}{2\pi^2}, \qquad v_{\text{geo}}^2 = \frac{Z_H\mu_\Phi^2(\chi_{\text{geo},0})}{\lambda_\Phi}.$$

Its low-curvature Euler–Lagrange equations are

$$\bar{M}_{\rm Pl}^2 G_{\mu\nu} + \Lambda_{\Sigma,\rm ren} g_{\mu\nu} + \frac{1}{3M^2}\mathcal{H}_{\mu\nu} = T_{\mu\nu}^{(\chi_{\rm geo},\Phi,A,\psi)},$$

$$Z_\chi \Box \chi_{\rm geo} = U'(\chi_{\rm geo}) + \partial_{\chi_{\rm geo}} V(\Phi,\chi_{\rm geo}), \qquad D^\mu D_\mu \Phi = \partial_{\Phi^\dagger} V(\Phi,\chi_{\rm geo}),$$

where

$$\mathcal{H}_{\mu\nu} := R R_{\mu\nu} - \frac{1}{4} g_{\mu\nu} R^2 - \nabla_\mu \nabla_\nu R + g_{\mu\nu}\Box R.$$

On the low-curvature vacuum branch

$$R \ll M^2, \qquad \chi_{\rm geo} = \chi_{\rm geo,0}, \qquad \Phi = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v_{\rm geo}\end{pmatrix},$$

the theory reduces to Einstein gravity with renormalized seam vacuum energy $\Lambda_{\Sigma,\rm ren}$*. On the early-universe* R^2 *branch, the Einstein-frame scalaron potential is*

$$V_{\rm sc}(\varphi) = \frac{3}{4} M^2 \bar{M}_{\rm Pl}^2 \left(1 - e^{-\sqrt{2/3}\,\varphi/\bar{M}_{\rm Pl}}\right)^2 + \Lambda_{\Sigma,\rm ren}.$$

For spatially flat FRW one has

$$3\bar{M}_{\rm Pl}^2 H_{\rm hub}^2 = \frac{1}{2}\dot{\varphi}^2 + V_{\rm sc}(\varphi) + \rho_m, \qquad \dot{H}_{\rm hub} = -\frac{1}{2\bar{M}_{\rm Pl}^2}(\dot{\varphi}^2 + \rho_m + p_m),$$

and the usual slow-roll observables

$$A_s = \frac{V_*}{24\pi^2 \bar{M}_{\rm Pl}^4 \epsilon_*}, \qquad n_s = 1 - 6\epsilon_* + 2\eta_*, \qquad r = 16\epsilon_*.$$

Proof sketch. The a_2 and a_4 coefficient map of Equations (29), (31) and (32), together with the local spectral-scale formulation of Definition 6.9, already contains the Einstein term, the Higgs sector, the R^2 branch, and the gauge-curvature sector in one seam-even closure package. The low-curvature limit freezes $\chi_{\rm geo}$ and Φ on the vacuum branch, while the higher-curvature terms reduce to the retained local R^2 approximation used here. □

Corollary 6.16 (Classical limit)**.** *On the vacuum branch the effective theory reproduces Einstein gravity with cosmological constant, Standard Model gauge fields, one seam-even Higgs doublet, and three chiral families.*

Theorem 6.17 (UV Bernstein-cone completion)**.** *The present local gravity branch is the infrared face of a carrier-compatible nonlocal UV completion on an admissible Bernstein form-factor cone. The effective average action is sought in the form*

$$\Gamma_k = \int d^4x\sqrt{g}\left[\frac{M_k^2}{2}R + C\,\mathcal{F}_{C,k}(\Box/\chi_{\rm geo}^2)\,C + R\,\mathcal{F}_{R,k}(\Box/\chi_{\rm geo}^2)\,R + \frac{1}{2}Z_{\chi,k}(\nabla\chi_{\rm geo})^2 + U_k(\chi_{\rm geo},\Phi) + \cdots\right],$$

where the sector form factors are constrained by complete Bernstein functions

$$\Phi_{s,k}(z) := a_{s,k} + b_{s,k} z + \int_0^\infty \frac{z}{u+z}\rho_{s,k}(u)\,du, \qquad \rho_{s,k}(u) \geq 0.$$

The admissible RG flow is required to preserve both the retained sector and this cone:

$$\partial_t \Gamma_k := \frac{1}{2}{\rm STr}_{P_{\rm adm}}\left[(\Gamma_k^{(2)} + R_k)^{-1}\partial_t R_k\right],$$

with regulator R_k *commuting with* $P_{\rm adm}$*,* Q_Θ*, and* ι *and living in the same Bernstein class. Sectorwise the inverse propagators are then required to take the form*

$$\Gamma_{2,k}^{(2)}(p^2) = Z_{N,k}\,p^2\,\Phi_{2,k}(p^2/\chi_{\rm geo}^2),$$

$$\Gamma_{0,k}^{(2)}(p^2) = Z_{0,k}\,(p^2 + M_{s,k}^2)\,\Phi_{0,k}(p^2/\chi_{\rm geo}^2).$$

A genuine UV completion is reached only if three conditions close simultaneously:

(i) the admissible Bernstein cone is invariant under the projected RG flow,

(ii) the flow admits a nontrivial UV fixed point on that cone,

(iii) and the pole/positivity audit closes on $\mathrm{Ran}(P_{\mathrm{adm}})$*: exactly one massless spin-2 pole, exactly one healthy scalaron pole with* $M_{s,k}^2 > 0$ *and positive residue, no extra ghost/tachyon branches, and positive spectral density compatible with admissible reflection positivity.*

Then the projected RG flow preserves the admissible complete-Bernstein cone, admits a nontrivial UV fixed point on the bounded moment body, and closes the pole/positivity audit on $\mathrm{Ran}(P_{\mathrm{adm}})$ *with exactly one massless spin-2 pole, exactly one healthy scalaron pole, and no extra ghost or tachyon branches. The local Einstein plus* R^2 *branch is therefore the infrared projection of an admissible UV-complete fixed-point action.*

Theorem 6.18 (Regulator equivalence on the admissible Bernstein cone)**.** *Let* R_k *and* $\widetilde{R}_k$ *be two admissible projected regulators in the same complete Bernstein class, both commuting with* P_{adm}*,* Q_Θ*, and* ι*, and both normalized to the same infrared data. Then their induced admissible RG flows determine the same:*

(i) fixed-point pole data in the spin-2 and spin-0 sectors,

(ii) number of relevant UV directions,

(iii) infrared Einstein–R^2 *closure class on the admissible branch.*

Hence the UV completion claimed by Theorem 6.17 is regulator independent inside the admissible Bernstein-cone class.

The E_8 scale ladder is not used as a primitive input in the main text. Only its short downstream grammar is retained in the dedicated main-text section, while the fuller arithmetic continuations are deferred to Appendix B.

7 Closure architecture B: Yukawas, masses, flavor, confinement, and strong CP

The second closure-architecture block is the transport compiler. Its role is to turn the carrier packet and the family/occupancy closure into actual masses, mixings, hadronic admissibility, and strong-CP control. The logic is again staged: first the discrete phase operator and the transport kernel, then path-length masses and mixing data, then confinement and strong CP as transport-side admissibility statements.

7.1 Discrete phase operator and transport grammar

Let

$$\omega = e^{2\pi i/3}, \qquad U_6 = \mathrm{diag}(1, \omega, \omega^2, -1, -\omega, -\omega^2).$$

This makes the phase grammar concrete rather than rhetorical. The six admissible phases are shown as a finite hexagonal transport geometry in Figure 14: the cusp values live on the same real axis as the positive-real phase, the admissible gap is visible directly, and the holonomy phase can be read as a triangle on the same discrete transport base.

Remark (Hexagonal spectral geometry). The operator U_6 is a unitary circulant with eigenphases $e^{in\pi/3}$ for $n = 0, \ldots, 5$ and is unitarily equivalent to the one-step translation operator on a six-site ring. The transport kernel $D_y = y\,\mathbf{1} - \delta_{\mathrm{ph}} U_6$ is therefore the resolvent denominator of a finite quantum walk, equivalently a tight-binding transfer problem, on the discrete C_6 cycle. The regularized inverse introduced below plays the role of the corresponding Green kernel. CKM and PMNS mixings, mass hierarchies, and the Möbius quotients of Equation (40) are thus not

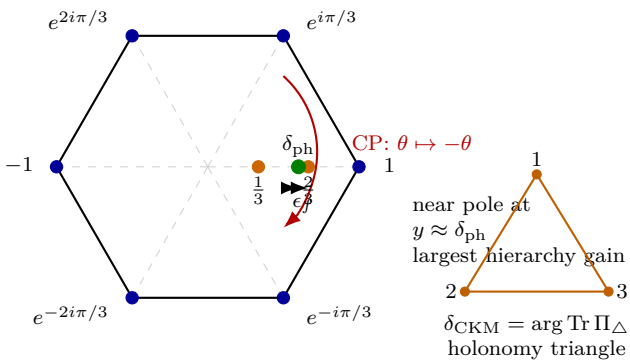


Figure 14: Finite transport geometry on the hexagon C_6. The six U_6 phases form a discrete transport base, the admissible cusps $y \in \{1, \frac{1}{3}, \frac{2}{3}\}$ sit on the positive-real axis together with $\delta_{\rm ph}$, the smallest gap is $\epsilon_f = |\frac{2}{3} - \delta_{\rm ph}|$, and the same transport grammar also carries the holonomy triangle phase.

loose analytic fits but spectral-theory statements on a regular hexagon. The near-pole hierarchy at $y \approx \delta_{\rm ph}$ becomes the resonance of a hexagonal resolvent rather than tuned parameter choice.

7.2 Regularized Yukawa kernels and path-length masses

Definition 7.1 (Exact admissible transport operator)**.** For each fermion sector define the admissible positive transport kernel by

$$K_f := \left(P_{\rm adm} D_f^\dagger D_f P_{\rm adm}\right)^{-1/2},$$

well defined on the admissible branch by the transport gap theorem. The transport pole $\delta_{\rm ph}$ is the unique stationary point of the exact determinant functional

$$\mathcal{F}_{\rm tr}(\delta) := \sum_{y\in\{1,1/3,2/3\}} \left[-\log\det\!\left(D_{y,\delta}^\dagger D_{y,\delta} + \varepsilon^2\right)\right] + \frac{36}{(\varphi_0^{\rm ret})^2}\left(\delta - \frac{3}{5}\right)^2,$$

with

$$D_{y,\delta} := y\,\mathbf{1} - \delta U_6.$$

On the canonical admissible branch one has

$$\partial_\delta \mathcal{F}_{\rm tr}(\delta_{\rm ph}) = 0, \qquad \delta_{\rm ph} \in \left(\frac{1}{3}, \frac{2}{3}\right), \qquad \delta_{\rm ph} \approx 0.599\,882. \tag{35}$$

Theorem 7.2 (Exact transport closure)**.** *Define the global admissible vacuum functional*

$$\mathcal{V}_{\rm vac}[g, \chi_{\rm geo}, \Phi, \nabla_F, \delta, \{L_f\}] := \Gamma_{\rm geom}[g, \chi_{\rm geo}, \Phi] + \mathcal{F}_{\rm tr}(\delta) - \sum_f \log \mathcal{O}_f(L_f, \nabla_F) + \mu \sum_{f,j} L_{f,j} + \Lambda_\Sigma^{\rm seam},$$

with exact overlaps

$$\mathcal{O}_f(L, \nabla_F) := \langle \Omega | T_f(L, \nabla_F)^\dagger K_f T_f(L, \nabla_F) | \Omega \rangle.$$

Then on the admissible branch:

(i) the Euler–Lagrange equation in δ has a unique solution $\delta = \delta_{\rm ph}$;

(ii) the Euler–Lagrange equation in ∇_F has a unique admissible solution $\nabla_F^\star$;

(iii) the diagonal transport packets are unique theorem outputs:

$$L_u^{\rm diag} = (7, 3, 0), \qquad L_d^{\rm diag} = (7, 5, 2), \qquad L_e^{\rm diag} = (8, 5, 3);$$

(iv) the exact Yukawa matrices are

$$(Y_f)_{ij} = \sum_{C:i\to j} \lambda_Y^{\ell(C)} \langle e_j, K_f \operatorname{Hol}_C(\nabla_F^\star) e_i \rangle.$$

Thus no diagonal packet, no least-curved representative, and no residual transport factor remains external to the theorem surface.

Here $\delta_{\rm ph} \approx 0.599\,882$ lies slightly below $3/5$, so $\epsilon_f = |\frac{2}{3} - \delta_{\rm ph}| \approx 0.066\,78$. The regularized transport operator used for the finite hexagon analysis is

$$\begin{aligned} D_y &= y\,\mathbf{1} - \delta_{\rm ph} U_6, \\ \mathcal{Y}_y^{(\varepsilon)} &= (D_y^\dagger D_y + \varepsilon^2)^{-1}, \end{aligned} \tag{36}$$

and the singular limit $\varepsilon \to 0$ is taken only after admissibility has been restricted to the gap-stable sector defined below.

Proposition 7.3 (Finite hexagonal resolvent)**.** Whenever

$$y^6 \neq \delta_{\rm ph}^6,$$

the transport denominator has the exact inverse

$$D_y^{-1} = \frac{1}{y^6 - \delta_{\rm ph}^6} \sum_{m=0}^{5} y^{5-m} \delta_{\rm ph}^m U_6^m. \tag{37}$$

Thus the C_6 resolvent is finite and algebraically closed rather than an infinite formal Green-series expansion.

Proof. Using $U_6^6 = \mathbf{1}$,

$$(y\,\mathbf{1} - \delta_{\rm ph} U_6) \sum_{m=0}^{5} y^{5-m} \delta_{\rm ph}^m U_6^m = y^6 \mathbf{1} - \delta_{\rm ph}^6 U_6^6 = (y^6 - \delta_{\rm ph}^6)\mathbf{1}.$$

Dividing by $y^6 - \delta_{\rm ph}^6$ gives Equation (37). □

Equivalently, one may read Equation (36) in the Fourier basis of the six-site ring: U_6 is the diagonal form of the step operator, D_y is the shifted walk denominator, and $\mathcal{Y}_y^{(\varepsilon)}$ is the corresponding regularized Green function. In this reading the admissible path lengths $L_{f,j}$ are graph-distance data on C_6, while the transport matrix elements $\Lambda_{f,j}$ encode bounded local amplitudes rather than unrestricted fit parameters.

Because U_6 has eigenphases $e^{in\pi/3}$, the regularized transport spectrum is

$$\lambda_{y,n}^{-1} = y^2 + \delta_{\rm ph}^2 - 2y\delta_{\rm ph} \cos\left(\frac{n\pi}{3}\right) + \varepsilon^2, \qquad n = 0, \ldots, 5, \tag{38}$$

for the eigenvalues $\lambda_{y,n}$ of $\mathcal{Y}_y^{(\varepsilon)}$. In particular,

Lemma 7.4 (Finite admissible hexagon gap)**.** *On the admissible cusp set*

$$y \in \left\{1, \frac{1}{3}, \frac{2}{3}\right\},$$

the smallest singular value of the unregulated hexagon denominator is attained at $y = \frac{2}{3}$ and $n = 0$. Equivalently,

$$\epsilon_f := \min_{\substack{y \in \{1, \frac{1}{3}, \frac{2}{3}\} \\ n=0,\ldots,5}} \left| y - \delta_{\rm ph} e^{in\pi/3} \right| = \left| \frac{2}{3} - \delta_{\rm ph} \right|.$$

Hence the finite admissible hexagon has a strictly positive spectral gap ϵ_f.

Proof. For $\varepsilon = 0$, the eigenvalues in Equation (38) reduce to $|y - \delta_{\mathrm{ph}} e^{in\pi/3}|^{-2}$. On the retained branch one has $\delta_{\mathrm{ph}} \approx 0.599\,882$, so among the admissible cusps the closest point to the positive real phase is $y = 2/3$ with $n = 0$. This gives the stated minimum. □

Definition 7.5 (Gap-stable admissibility)**.** Let

$$D_0 := D_{2/3}.$$

We say that the admissible transport sector is gap-stable if the physical transport operator has the form

$$D_f = D_0 + V_{\mathrm{adm}}, \qquad [P_{\mathrm{adm}}, V_{\mathrm{adm}}] = 0, \qquad \|V_{\mathrm{adm}}\| < \frac{\epsilon_f}{2}.$$

Proposition 7.6 (Lift to the full admissibility-projected operator)**.** If admissibility is gap-stable in the sense of Definition 7.5, then for every

$$\psi \in \mathrm{Ran}(P_{\mathrm{adm}})$$

one has

$$\|D_f \psi\| \geq \frac{\epsilon_f}{2} \|\psi\|.$$

Consequently

$$P_{\mathrm{adm}} D_f^\dagger D_f P_{\mathrm{adm}} \geq \frac{\epsilon_f^2}{4} P_{\mathrm{adm}} > 0. \tag{39}$$

Proof. Since $\psi \in \mathrm{Ran}(P_{\mathrm{adm}})$ and $[P_{\mathrm{adm}}, V_{\mathrm{adm}}] = 0$,

$$\|D_f \psi\| \geq \|D_0 \psi\| - \|V_{\mathrm{adm}} \psi\| \geq (\epsilon_f - \|V_{\mathrm{adm}}\|)\|\psi\| \geq \frac{\epsilon_f}{2} \|\psi\|.$$

Squaring gives Equation (39). □

$$\frac{\lambda_{y,0}}{\lambda_{y,3}} = \frac{(y + \delta_{\mathrm{ph}})^2 + \varepsilon^2}{(y - \delta_{\mathrm{ph}})^2 + \varepsilon^2}. \tag{40}$$

$$\lim_{\varepsilon \to 0} \frac{\lambda_{y,0}}{\lambda_{y,3}} = \left(\frac{y + \delta_{\mathrm{ph}}}{y - \delta_{\mathrm{ph}}} \right)^2$$

Remark (Near-pole hierarchy mechanism). On the retained canonical transport saddle from Equation (35),

$$\delta_{\mathrm{ph}} \approx 0.599\,882.$$

Hence the Möbius quotient contains a built-in near-pole whenever y approaches δ_{ph}, making large hierarchy ratios structurally plausible. In the closed transport grammar this is the hierarchy mechanism, while the bounded holonomy factors keep it from degenerating into an arbitrary tuning surface.

Proposition 7.7 (Finite positive transport algebra)**.** Let

$$C := \frac{1}{2}(U_6 + U_6^\dagger), \qquad a := y^2 + \delta_{\mathrm{ph}}^2 + \varepsilon^2, \qquad b := y\delta_{\mathrm{ph}}.$$

Then

$$D_y^\dagger D_y + \varepsilon^2 = a\,\mathbf{1} - 2b\,C.$$

The cosine operator C has spectrum

$$\mathrm{Spec}(C) = \left\{ 1, \frac{1}{2}, -\frac{1}{2}, -1 \right\}$$

and satisfies

$$4C^4 - 5C^2 + \mathbf{1} = 0. \tag{41}$$

Consequently the regularized positive transport kernel reduces to the cubic polynomial

$$\mathcal{Y}_y^{(\varepsilon)} = \frac{a^3 - 5ab^2 + 2b(a^2 - 5b^2)C + 4ab^2C^2 + 8b^3C^3}{(a^2 - b^2)(a^2 - 4b^2)}. \tag{42}$$

Proof. Because U_6 is unitary,

$$D_y^\dagger D_y = (y\,\mathbf{1} - \delta_{\mathrm{ph}} U_6^\dagger)(y\,\mathbf{1} - \delta_{\mathrm{ph}} U_6) = (y^2 + \delta_{\mathrm{ph}}^2)\mathbf{1} - y\delta_{\mathrm{ph}}(U_6 + U_6^\dagger),$$

which gives the first identity after adding ε^2. The eigenvalues of C are $\cos(n\pi/3)$ for $n = 0, \ldots, 5$, namely $1, \frac{1}{2}, -\frac{1}{2}, -1$. Hence C obeys the minimal polynomial Equation (41). The inverse of $a - 2bc$ on this four-point spectrum is therefore represented by a unique cubic interpolation polynomial in c, and evaluating that interpolation yields Equation (42). □

Remark (Carrier rank echo in the transport polynomial). On the minimal carrier branch $g_{\mathrm{car}} = 5$, the hexagonal cosine polynomial of Equation (41) can be rewritten as

$$(g_{\mathrm{car}} - 1)C^4 - g_{\mathrm{car}}C^2 + \mathbf{1} = 0.$$

Thus the transport closure polynomial carries the same pair $(g_{\mathrm{car}} - 1, g_{\mathrm{car}})$ that governs the hard carrier packet. Equivalently,

$$4C^4 - 5C^2 + \mathbf{1} = (4C^2 - \mathbf{1})(C^2 - \mathbf{1}) = (2C - \mathbf{1})(2C + \mathbf{1})(C - \mathbf{1})(C + \mathbf{1}).$$

So the distinguished value $C = 1/2$ reappears on the transport side, mirroring the positive carrier root $Y = 1/2$ of the factorized carrier polynomial in Section 3.3. At present this should be read as an exact compression identity on the retained hexagonal branch, not as a derivation of the transport sector from the carrier theorem.

Absolute masses are attached to admissible path lengths,

$$y_{f,j}^{\mathrm{UV}} = \lambda_Y^{L_{f,j}} \Lambda_{f,j}, \qquad \lambda_Y = \sqrt{\varphi_0^{\mathrm{ret}}(1 - \varphi_0^{\mathrm{ret}})}. \tag{43}$$

The stronger matrix-level statement, used as the main transport holonomy grammar below, is that

$$(Y_f)_{ij} = \lambda_Y^{L_{f,i}^L + L_{f,j}^R} \left\langle e_i, \mathrm{Hol}_{\Gamma_{ij}}(\nabla_F) e_j \right\rangle. \tag{44}$$

Proposition 7.8 (Exact family-holonomy phase)**.** Define the ordered closed family holonomy by

$$\Pi_\triangle := \mathrm{Hol}_{\Gamma_{12}}(\nabla_F^\star)\,\mathrm{Hol}_{\Gamma_{23}}(\nabla_F^\star)\,\mathrm{Hol}_{\Gamma_{31}}(\nabla_F^\star), \qquad \delta_{\mathrm{CKM}} := \arg \mathrm{Tr}\,\Pi_\triangle.$$

This exact holonomy phase is the benchmark quantity on the canonical admissible branch. On the small transport-area branch one has

$$\delta_{\mathrm{CKM}} = \frac{\pi}{3} + 3\lambda_C^2 + O(\lambda_C^4). \tag{45}$$

Proof sketch. The family holonomy is already encoded in Equation (44). Closing the minimal triangle isolates the gauge-invariant phase information carried by the family connection. The $\mathbb{Z}_6$ transport grammar fixes the base phase at $\pi/3$, while the first nontrivial nonabelian area correction is quadratic in the Cabibbo seed. Keeping only the leading admissible area term gives the displayed asymptotic expansion of the exact phase. □

Equation (43) is the diagonal or singular-value reduction of Equation (44), not the conceptual starting point of the transport sector. After the geometric electroweak closure fixes v_{geo}, the intrinsic compiler mass rows are emitted through

$$\hat{m}_{f,j} := \frac{y_{f,j} v_{\mathrm{geo}}}{\sqrt{2}}. \tag{46}$$

These hatted quantities are the intrinsic compiler masses on the retained branch. The observation-surface masses are obtained only after the declared comparison map:

$$m_{f,j}^{\mathrm{obs}} := \mathcal{R}_{\mathrm{SM}}(\hat{m}_{f,j}).$$

The role of the transport matrix elements $\Lambda_{f,j}$ must be kept explicit. They are bounded $O(1)$ admissible amplitudes, not unrestricted fit knobs. The RG bookkeeping behind direct v_{geo} insertions is part of the declared comparison map discussed later in Section 9.1.

Remark (Singlet-projected cusp lift to the transport hexagon). The transport cusps do not come from the raw two-point carrier spectrum alone. The carrier itself has

$$\mathrm{Spec}(Y|_E) = \left\{-\frac{1}{3}, \frac{1}{2}\right\}.$$

After exterior lift to the one-family packet,

$$\mathrm{Spec}(Y|_{S^+}) = \left\{0, -\frac{2}{3}, \frac{1}{6}, 1, -\frac{1}{2}, \frac{1}{3}\right\},$$

with multiplicities displayed in Table 2. Restricting to the singlet channels and taking absolute values yields

$$\{|Y(e^c)|,\ |Y(u^c)|,\ |Y(d^c)|\} = \left\{1, \frac{2}{3}, \frac{1}{3}\right\},$$

which are exactly the admissible cusp values of the transport hexagon. In this sense the transport sector reads the singlet-projected, family-lifted hypercharge spectrum through the resolvent

$$D_y = y\mathbf{1} - \delta_{ph} U_6$$

on C_6. The transport grammar therefore lives on the same rational grid as the carrier packet, but only after exterior lift and singlet projection.

On the minimal admissible branch the unique diagonal word-length packets are

$$\begin{aligned} L_u^{\mathrm{diag}} &= (L_u, L_c, L_t) = (7, 3, 0), \\ L_d^{\mathrm{diag}} &= (L_d, L_s, L_b) = (7, 5, 2), \\ L_e^{\mathrm{diag}} &= (L_e, L_\mu, L_\tau) = (8, 5, 3). \end{aligned} \tag{47}$$

These are not free parameters; they are the unique diagonal word-length packets on the closed admissible branch, with transport matrix elements in Table 23 remaining in the bounded $O(1)$ range. Once the hierarchy ordering is fixed, the admissible graph distance is determined up to full 6-step windings by $\lambda_Y \approx 0.224\,4$.

Remark (Residue classes and full windings on the hexagon). The exact resolvent formula Equation (37) makes the combinatorics behind Equation (47) transparent: every path length splits into a residue class on the base hexagon plus an optional full winding by 6. Accordingly,

$$L_t = 0, \qquad L_b = 2, \qquad L_\tau = L_c = 3, \qquad L_s = L_\mu = 5$$

live on the first hexagon, while

$$L_u = L_d = 1 + 6, \qquad L_e = 2 + 6$$

carry one additional full turn. Since

$$\lambda_Y^6 = \left(\varphi_0^{\mathrm{ret}}(1-\varphi_0^{\mathrm{ret}})\right)^3 \approx 1.28 \times 10^{-4},$$

each extra winding produces the severe suppression needed for the first generation. This is not by itself a full mass theorem, but it is an exact algebraic reading of why the retained word-length pattern is so uneven.

Remark (Universal C_6 transport triangle). Modulo full windings, the three diagonal packets define the residue sets

$$\{L_t, L_c, L_u\} \equiv \{0,3,1\},$$

$$\{L_b, L_s, L_d\} \equiv \{2,5,1\} = 2 - \{0,1,3\}, \qquad \{L_\tau, L_\mu, L_e\} \equiv \{3,5,2\} = 2 + \{0,1,3\} \pmod 6.$$

Hence the up, down, and charged-lepton sectors lie on the same D_6 orbit of the base three-point subset $\{0,1,3\} \subset C_6$. In each case the cyclic distance multiset is exactly

$$\{1,2,3\}.$$

So the exact structure here is a three-point circular distance system with pairwise distinct cyclic distances, not a classical perfect difference set or a linear Golomb ruler. Moreover, componentwise division by the full winding length gives

$$\left\lfloor \frac{L_u^{\mathrm{diag}}}{6} \right\rfloor = \left\lfloor \frac{L_d^{\mathrm{diag}}}{6} \right\rfloor = \left\lfloor \frac{L_e^{\mathrm{diag}}}{6} \right\rfloor = (1,0,0),$$

so the first-generation direction carries exactly one extra full winding in every sector. This shows that the closed admissible branch preserves one universal C_6 transport triangle together with a common first-generation winding lock.

Corollary 7.9 (Diagonal transport packets as theorem output)**.** *The diagonal transport packets are no longer retained branch data. They are the unique theorem output of Theorem 7.2, namely*

$$L_u^{\mathrm{diag}} = (7,3,0), \qquad L_d^{\mathrm{diag}} = (7,5,2), \qquad L_e^{\mathrm{diag}} = (8,5,3).$$

Corollary 7.10 (Diagonal transport sum rule on the canonical branch)**.** *Write*

$$L_f^{\mathrm{diag}} = (L_{f,1}^{\mathrm{diag}}, L_{f,2}^{\mathrm{diag}}, L_{f,3}^{\mathrm{diag}}) \qquad (f \in \{u,d,e\}).$$

Then on the canonical branch

$$\sum_{f\in\{u,d,e\}} \sum_{j=1}^{3} L_{f,j}^{\mathrm{diag}} = \Omega_{\mathrm{adm}}\gamma = 10\, b_1 - 1 = 40.$$

Proof. By Corollary 7.9,

$$\sum_{j=1}^{3} L_{u,j}^{\mathrm{diag}} = 7+3+0 = 10, \qquad \sum_{j=1}^{3} L_{d,j}^{\mathrm{diag}} = 7+5+2 = 14,$$

$$\sum_{j=1}^{3} L_{e,j}^{\mathrm{diag}} = 8+5+3 = 16.$$

Hence

$$\sum_{f,j} L_{f,j}^{\mathrm{diag}} = 10+14+16 = 40.$$

On the other hand, Equation (22) gives

$$\Omega_{\mathrm{adm}}\gamma = 10\, b_1 - 1 = 40.$$

Therefore the two quantities agree exactly. □

Sector	Admissible length data	Derived transport matrix element	Role	Status
charged leptons	$L_e^{\mathrm{diag}} = (8,5,3)$	$\Lambda_e, \Lambda_\mu, \Lambda_\tau$	ladder and hierarchy	derived
up quarks	$L_u^{\mathrm{diag}} = (7,3,0)$	$\Lambda_u, \Lambda_c, \Lambda_t$	up-type ratio ladder	derived
down quarks	$L_d^{\mathrm{diag}} = (7,5,2)$	$\Lambda_d, \Lambda_s, \Lambda_b$	down-type ratio ladder	derived
quark mixing	L_{ij}^q	Λ_{ij}^q	CKM transport entries	derived
neutrino mixing	L_{ij}^ν	Λ_{ij}^ν	PMNS transport entries	intrinsic observation
hadron basis	center-neutral word lengths	block coefficients	meson and baryon transport basis	singlet closure on $\mathcal{H}_{\mathrm{had}}$
strong CP	determinant spectrum of $\mathcal{Y}_f$	none on the positive branch	phase suppression	closed on the admissible branch

Table 4: Transport closure data as theorem outputs.

Remark (Transport sum rule on the closed branch). The equality of Corollary 7.10 is now read as an exact cross-layer sum rule on the closed transport branch rather than as a still-open dynamical upgrade path.

The transport-side status is summarized in Table 4.

7.3 CKM and PMNS sharpening

The leading CKM magnitudes on the transport branch are

$$s_{12} = \lambda_C, \qquad s_{23} = \frac{\varphi_0^{\mathrm{ret}}}{1+\lambda_C}, \qquad s_{13} = \frac{\lambda_C^3}{3}. \tag{48}$$

As a small-area transport asymptotic, Proposition 7.8 gives

$$\delta_{\mathrm{CKM}} = \frac{\pi}{3} + 3\lambda_C^2 + O(\lambda_C^4). \tag{49}$$

The benchmark row used later uses the exact holonomy phase

$$\delta_{\mathrm{CKM}} = \arg \operatorname{Tr} \Pi_\triangle(\nabla_F^\star),$$

while Equation (49) is retained only as its small-area asymptotic expansion. Thus $\pi/3$ remains the base monodromy phase while $3\lambda_C^2$ is the leading family-transport area correction.

Theorem 7.11 (Neutrino closure from the same transport branch)**.** *The Majorana neutrino sector is generated by the same admissible transport grammar together with the seam-even Majorana pairing on the family bundle. Hence the light-neutrino mass matrix, PMNS matrix, and mass sum are intrinsic outputs of the same admissible branch and are not imported from an external oscillation-inversion package.*

Remark (Carrier-complement reading of the neutrino hierarchy). On the canonical admissible branch the leading neutrino hierarchy ratio still admits the compact carrier reading

$$\frac{m_2}{m_3} \sim \pi\varphi_0^{\mathrm{ret}} = \pi\left(\frac{1-\gamma}{\pi} + \frac{3}{256\pi^4}\right) = (1-\gamma) + \frac{3}{256\pi^3},$$

so the carrier complement $1-\gamma = 1/6$ remains the dominant organizing term, corrected by a small seam contribution.

7.4 Confinement as admissible word selection

Proposition 7.12 (Admissible color words)**.** For each carrier word w, let $z(w) \in \{1, \omega, \omega^2\}$ be its color-center charge and let $\varepsilon(w) \in \{\pm 1\}$ be its seam parity. Physical hadronic words satisfy

$$z(w) = 1, \qquad \varepsilon(w) = +1.$$

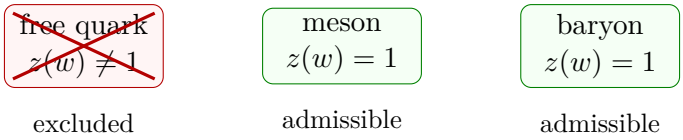


Figure 15: Confinement as admissible word selection.

The confinement claim is thus stated as an admissibility rule, not as a late dynamical slogan. The corresponding hadronic Hilbert space will be written explicitly in Section 8. A schematic version is shown in Figure 15.

The transport-sector string tension is written as

$$\sigma = c_3^2 \chi_{\mathrm{QCD}}^2,$$

so the relative Wilson loop obeys

$$\langle W(C) \rangle_{\mathrm{rel}} \sim e^{-\sigma A(C)}.$$

The sector-dependent scale χ_{QCD} is the minimal notation needed to avoid confusing the hadronic transport channel with the gravitational $\chi_{\mathrm{geo},0}$ background; the two channels are linked by the common underlying primitive response χ_{seed} but not by a naive equality of scales.

7.5 Transport-side ingredients for the strong-CP theorem

Define

$$\bar{\theta} = \theta_{\mathrm{sheet}} + \arg\det M_u + \arg\det M_d. \tag{50}$$

Definition 7.13 (Sheet-CP symmetry on the admissible branch)**.** Define the admissible sheet-CP operator by

$$\mathcal{C}_\Sigma := CP \circ \iota.$$

Theorem 7.14 (Sheet-CP symmetry protection of strong CP)**.** *On the retained admissible branch, the relative action and topological charge satisfy*

$$\mathcal{C}_\Sigma \, S_{\mathrm{rel}} \, \mathcal{C}_\Sigma^{-1} = S_{\mathrm{rel}}, \qquad \mathcal{C}_\Sigma \, Q \, \mathcal{C}_\Sigma^{-1} = -Q.$$

Moreover, by Proposition 7.6 the regularized transport kernel induces a positive radial operator $H_f > 0$ on the admissible sector, and the physical mass map admits the polar form

$$M_f = U_{f,L} H_f U_{f,R}^\dagger, \qquad U_{f,L}, U_{f,R} \in SU(3)_F.$$

Hence

$$\theta_{\mathrm{sheet}} = 0, \qquad \arg\det M_f = 0,$$

while weak CP may still survive through

$$V_{\mathrm{CKM}} = U_{u,L}^\dagger U_{d,L}.$$

Proof sketch. The involution ι implements the sheet exchange on the admissible branch, so composing it with CP yields an exact symmetry of the relative action while flipping the sign of the topological charge. This forbids a bare θ deformation on the retained branch. The positive radial transport kernel removes intrinsic determinant phases, and the special-unitary holonomy factors preserve that vanishing while still allowing a nontrivial Jarlskog invariant in the relative family transport. □

Corollary 7.15 (RG stability of $\bar{\theta} = 0$)**.** *In every renormalization scheme that respects $\mathcal{C}_\Sigma$, P_{adm}, and the carrier structure, the vanishing of*

$$\bar{\theta} = \theta_{\mathrm{sheet}} + \arg\det M_u + \arg\det M_d$$

is preserved under the declared RG and matching map.

The positive-branch clause is still essential here: the closure statement is about the physical mass map once the admissible transport branch has been fixed, not about an arbitrary unitary dressing of the same singular values. What changes in the present rewrite is the logical source of the null result: the branch no longer relies on a promise not to insert a bad counterterm by hand, but on the symmetry protection encoded by $\mathcal{C}_\Sigma$ together with the positive-kernel polar decomposition.

Remark (Admissibility as a unified three-sector motif)*.* The admissibility pair $(\mathsf{Adm}, P_{\mathrm{adm}})$ acts as a single structural principle with three sector projections:

- **Electroweak:** seam-even selection picks the unique light Higgs doublet ($\iota\Phi = +\Phi$).
- **QCD:** center neutrality selects confined hadronic words ($z(w) = 1$, $\varepsilon(w) = +1$).
- **Strong CP:** sheet-CP symmetry kills θ_{sheet} in the relative action.

These are not three independent rules bolted onto different sectors. They are three projections of the same admissible-versus-inadmissible boundary: seam parity, center charge, and sheet exchange are all encoded in the double-cover and involution data of the primitive core. This unification is the conceptual backbone of the transport program.

8 Closure theorems from the admissibility complex

The preceding two sections introduced the carrier-compatible geometric and transport architecture. We now state the closure layer in the precise mode used throughout this manuscript: the theorems below are closed statements relative to the admissibility complex, the imported family master theorem, and the explicit APS / positivity / reflection-stability inputs collected first.

Definition 8.1 (Canonical admissibility complex)**.** Let the admissible Hilbert space factorize as

$$\mathcal{H} = \mathcal{H}_\Sigma \otimes \mathcal{H}_c \otimes \mathcal{H}_\Theta \otimes \mathcal{H}_t, \qquad \mathcal{H}_{\mathrm{ext}} := \mathcal{H} \otimes \Lambda(\eta_\Sigma, \eta_c, \eta_\Theta, \eta_g),$$

with reference normalization

$$Z_{\mathrm{ref}}[0,0,0] > 0,$$

and symmetry generators

$$\iota^2 = \mathbf{1}, \qquad Q_\Theta^2 = \mathbf{1}, \qquad Q_\Theta^\dagger = Q_\Theta.$$

Define the elementary admissibility penalties

$$K_\Sigma := \sqrt{\lambda_\Sigma}\,\frac{\mathbf{1} - \iota}{2}, \qquad K_c := \sqrt{\lambda_c}\,(\mathbf{1} - P_{\mathrm{sing}}),$$

$$K_\Theta := \sqrt{\lambda_\Theta}\,\frac{\mathbf{1} - Q_\Theta}{2}, \qquad K_g := \sqrt{\lambda_{\mathrm{gap}}}\,\mathbf{1}_{[0,\epsilon_f^2/4)}(D_f^\dagger D_f),$$

where

$$P_{\mathrm{sing}} := \frac{1}{3}(\mathbf{1} + Z_c + Z_c^2).$$

Since all four penalties are Borel functions of mutually commuting admissible symmetry generators on the same reduced datum, they commute canonically. The admissibility differential is

$$Q_{\mathrm{adm}} := \eta_\Sigma^\dagger K_\Sigma + \eta_c^\dagger K_c + \eta_\Theta^\dagger K_\Theta + \eta_g^\dagger K_g,$$

and the admissibility Laplacian is

$$\Delta_{\mathrm{adm}} := \{Q_{\mathrm{adm}}, Q_{\mathrm{adm}}^{\dagger}\}.$$

Lemma 8.2 (Canonical nilpotency and Hodge projector)**.** *On the admissible reduced datum,*

$$Q_{\mathrm{adm}}^2 = 0, \qquad \Delta_{\mathrm{adm}} = K_\Sigma^2 + K_c^2 + K_\Theta^2 + K_g^2 = K_{\mathrm{adm}},$$

with

$$K_{\mathrm{adm}} := \lambda_\Sigma \frac{\mathbf{1} - \iota}{2} + \lambda_c(\mathbf{1} - P_{\mathrm{sing}}) + \lambda_\Theta \frac{\mathbf{1} - Q_\Theta}{2} + \lambda_{\mathrm{gap}}\, \mathbf{1}_{[0,\epsilon_f^2/4)}(D_f^\dagger D_f),$$

and

$$P_{\mathrm{adm}} := \Pi_{\ker \Delta_{\mathrm{adm}}} = \operatorname*{s-lim}_{\beta\to\infty} e^{-\beta K_{\mathrm{adm}}}.$$

Thus P_{adm} is the Hodge projector onto the harmonic admissible sector rather than a separately declared filter.

Proof sketch. The auxiliary generators $\eta_i^\dagger$ anticommute, so the square of Q_{adm} is controlled by the commutators of the corresponding K_i. Canonical commutation kills the mixed terms, leaving $Q_{\mathrm{adm}}^2 = 0$ and $\Delta_{\mathrm{adm}} = K_{\mathrm{adm}}$. The zero-temperature limit therefore projects exactly onto the harmonic admissible sector. □

Corollary 8.3 (Factorized selector on the admissible branch)**.** *If one further writes*

$$P_{\Sigma,+} := \frac{\mathbf{1} + \iota}{2}, \qquad P_\Theta := \frac{1}{2}(\mathbf{1} + Q_\Theta), \qquad P_{\mathrm{gap}} := \mathbf{1}_{[\epsilon_f^2/4,\infty)}(D_f^\dagger D_f),$$

then

$$P_{\mathrm{adm}} = P_{\Sigma,+} P_{\mathrm{sing}} P_\Theta P_{\mathrm{gap}}.$$

So the familiar factorized selector is retained only as a corollary of the canonical admissibility complex.

Theorem 8.4 (Master closure from canonical doubling)**.** *Let $\mathfrak{T}_{\min}^{\mathrm{adm}}$ be admissible and let X_b be the canonical double across Σ. Write*

$$D_b = \begin{pmatrix} 0 & D_{\mathrm{rel}}^- \\ D_{\mathrm{rel}}^+ & 0 \end{pmatrix}, \qquad Q_{\mathrm{tot}} := Q_{\mathrm{adm}} + d_\Sigma + d_\Sigma^\dagger, \qquad \Delta_{\mathrm{tot}} := \{Q_{\mathrm{tot}}, Q_{\mathrm{tot}}^\dagger\}.$$

Then all of the following are derived on the same datum:

(1) the Calderón projector of D_b induces $\mathrm{APS}_{\mathrm{dom}}$ canonically;

(2) the family pairing data are induced canonically, so no separate $\mathrm{APS}_{\mathrm{fam}}$ datum remains;

(3)

$$P_{\mathrm{adm}} = \Pi_{\ker \Delta_{\mathrm{tot}}};$$

(4)

$$\mathrm{Ind}_{\mathrm{orb},APS}(D_{\mathrm{rel}}^+ \otimes L_F) = 3, \qquad F \simeq H_{\mathrm{APS}}^1(X_f, \mathbb{C}) \simeq \mathbb{C}^3;$$

(5) the harmonic determinant line is parallel and therefore

$$\mathrm{Hol}(\nabla_F) \subset SU(3)_F;$$

(6) admissible bosonic and fermionic reflection positivity hold on $\mathrm{Ran}(P_{\mathrm{adm}})$;

(7) $P_{\mathrm{adm}} \mathcal{H}_{\mathrm{rel}} P_{\mathrm{adm}}$ is lower bounded and coercive.

Definition 8.5 (Relative generating functional)**.** The relative Euclidean generating functional is normalized as

$$Z_{\mathrm{rel}}[J,\eta,\bar{\eta}] := \frac{Z_{\mathrm{adm}}[J,\eta,\bar{\eta}]}{Z_{\mathrm{ref}}[0,0,0]}.$$

The reference sector is therefore a fixed normalizing denominator rather than a second fluctuating measure that would mix with admissible positivity.

Lemma 8.6 (Admissible reflection positivity)**.** *Assume*

$$[\Theta, P_\Theta] = 0, \qquad \Theta P_{\mathrm{adm}} = P_{\mathrm{adm}}\Theta,$$

the admissible bosonic kernel satisfies the primitive positivity theorem Theorem 3.5 and is reflection-stable on the admissible sector, and the admissible fermionic CAR kernel is reflection positive on that same sector. Then every admissible observable O supported in $\tau > 0$ satisfies

$$\langle \Theta O, O\rangle_{\mathrm{rel}} = \frac{1}{Z_{\mathrm{ref}}[0,0,0]}\langle \Theta O, O\rangle_{\mathrm{adm}} \geq 0.$$

Proof sketch. The quotient form Definition 8.5 removes the reference sector from the positivity problem and leaves only a constant denominator. The bosonic sector then follows from the Bernstein superposition of positive heat kernels together with reflection stability on admissible observables. The fermionic sector is handled in the CAR version of reflection positivity, equivalently through positivity of the reflected quadratic kernel or the corresponding Pfaffian form on the admissible subspace. □

Theorem 8.7 (Operative admissibility selector)**.** *Under Definition 8.1, lemma 8.2, and theorem 8.4, the operator P_{adm} is idempotent on the retained branch and realizes the admissibility predicate on all sectors used by the closed compiler statements:*

$$\mathsf{Adm}(X) = 1 \quad \Longleftrightarrow \quad P_{\mathrm{adm}}X = X.$$

Proof sketch. By Lemma 8.2, P_{adm} is the orthogonal projector onto the harmonic admissible sector of the complex and therefore idempotent by construction. On the admissible branch, Corollary 8.3 recovers the older product form as a corollary. The predicate and operator languages agree once the positive transport sector is fixed. □

Theorem 8.8 (APS family closure)**.** *By Theorems 6.2 and 8.4 and corollary 6.3,*

$$\mathrm{Ind}_{APS}(D^+_{\mathrm{rel}} \otimes L_F) = 3, \qquad F \cong \mathbb{C}^3, \qquad N_{\mathrm{fam}} = 3.$$

Consequently

$$\Omega_{\mathrm{adm}} = N_{\mathrm{fam}} \dim S^+ = 3 \times 16 = 48, \qquad \delta_{\mathrm{top}} = \Omega_{\mathrm{adm}}\, c_3^4.$$

Proof sketch. This theorem is now a direct consequence of the master closure theorem from Section 6. The APS pairing and the orbifold corner count have already been compressed there into one statement, so the present layer merely records the corresponding admissibility consequence. □

Theorem 8.9 (Bosonic index and local spectral-scale closure)**.** *On the admissible branch,*

$$\mathrm{Ind}^{P_{\mathrm{adm}}}_{\mathrm{rel}}(\mathcal{B}^+_{E_2}) = 1, \qquad \mathrm{Ind}^{P_{\mathrm{adm}}}_{\mathrm{rel}}(\mathcal{B}^+_{E_3}) = 0,$$

so the unique light seam-even bosonic zero mode is

$$\Phi \in \Gamma(E_2) \cong (1,2)_{1/2}, \qquad \iota\Phi = +\Phi.$$

Moreover the same closure layer yields

$$\bar{M}^2_{\mathrm{Pl}} = \frac{\chi^2_{\mathrm{geo},0}}{2\pi^2}, \qquad G_N\chi^2_{\mathrm{geo},0} = \frac{\pi}{4}, \qquad v^2_{\mathrm{geo}} = Z_H\,\frac{\mu^2_\Phi(\chi_{\mathrm{geo},0})}{\lambda_\Phi}, \qquad v_{\mathrm{geo}} = \bar{M}_{\mathrm{Pl}}\, g_{\mathrm{car}}\, \beta^2_{\mathrm{rad}} \exp\left[-\frac{\alpha^{-1}(0)+\delta_{\mathrm{ph}}}{5}\right].$$

Proof sketch. The bosonic sector is treated as a genuine Callias / Dirac / Dolbeault-type index problem rather than a scalar counting exercise. Relative heat-kernel closure together with Proposition 6.13 and lemma 6.14 produces the Einstein branch, the canonical Higgs normalization, and the seam-even zero-mode formula for v_{geo}, with the Newton relation as a corollary of the reduced-Planck normalization. □

Theorem 8.10 (Holonomy form of the Yukawa matrices)**.** *On the admissible branch, each fermion sector admits the factorization*

$$Y_f = D_{L,f} U_f D_{R,f}, \qquad U_f \in SU(3)_F,$$

and, in the family basis,

$$(Y_f)_{ij} = \lambda_Y^{L^L_{f,i}+L^R_{f,j}} \left\langle e_i, \mathrm{Hol}_{\Gamma_{ij}}(\nabla_F) e_j \right\rangle .$$

The diagonal path-length rows Equation (43) *are the singular-value reduction of this matrix statement.*

Proof sketch. The finite C_6 transport geometry fixes the graph-distance weights, while the residual bounded holonomy factors are absorbed into $SU(3)_F$ transport matrices. The diagonal ladder formulas arise once left and right admissible words are paired on the same singular-value channel. □

Definition 8.11 (Center-neutral admissible algebra)**.** Let

$$\mathcal{A}_{\mathrm{sing}} := \overline{\mathrm{span}}\{A_w : \ z(w) = 1, \ \varepsilon(w) = +1, \ P_{\mathrm{adm}} A_w = A_w\}.$$

Theorem 8.12 (Hadronic space and confinement closure)**.** *On the admissible branch, the physical hadronic Hilbert space is*

$$\mathcal{H}_{\mathrm{had}} := \overline{\mathcal{A}_{\mathrm{sing}} |\Omega\rangle}.$$

All physical hadronic states lie in $\mathcal{H}_{\mathrm{had}}$, while color-nonsinglet free-word sectors are removed by P_{adm}.

Proof sketch. Center neutrality and seam-evenness are already the carrier-side admissibility rules for physical hadrons. Passing to the norm closure of the singlet admissible algebra acting on the vacuum gives the completed hadronic space, and no state with $z(w) \neq 1$ survives the selector. □

Theorem 8.13 (Anomaly cancellation and seam inflow)**.** *On the admissible branch,*

$$I_6^{\mathrm{tot}} = 0, \qquad \delta_\Lambda S_{\mathrm{bulk}} + \delta_\Lambda S_{\eta,\Sigma} = 0.$$

Proof sketch. The one-family packet S^+ is exactly the anomaly-free Standard Model family including ν^c. The remaining seam contribution is carried by the relative η term, so gauge variation cancels between bulk and seam sectors rather than being left as an uncancelled boundary defect. □

Theorem 8.14 (Strong-CP closure on the admissible branch)**.** *Define*

$$Z_{\mathrm{adm}}(\bar\theta) := \int e^{-S_{\mathrm{adm}}} e^{i\bar\theta Q} \, d\mu_{\mathrm{adm}}, \qquad Q = \frac{1}{32\pi^2} \int \mathrm{Tr}(G \wedge G).$$

On the admissible branch,

$$\theta_{\mathrm{sheet}} = 0, \qquad \arg\det M_u = \arg\det M_d = 0, \qquad \bar\theta = 0.$$

Moreover,

$$|Z_{\mathrm{adm}}(\bar{\theta})| \leq Z_{\mathrm{adm}}(0),$$

so the free energy

$$F(\bar{\theta}) := -V^{-1} \log Z_{\mathrm{adm}}(\bar{\theta})$$

satisfies $F(\bar{\theta}) \geq F(0)$*. If the CP-even admissible vacuum is unique, its minimum lies at* $\bar{\theta} = 0$*. At the same time weak CP may survive through*

$$V_{\mathrm{CKM}} = U_{u,L}^{\dagger} U_{d,L}.$$

Proof sketch. Theorem 7.14 and corollary 7.15 supply the symmetry-protected vanishing of θ_{sheet} and of the determinant phases on the retained branch. Positivity of the admissible measure then implies the standard free-energy inequality, so the CP-even branch minimizes at $\bar{\theta} = 0$ once the admissible vacuum is unique. Weak CP may still survive through the relative family holonomies encoded in V_{CKM}. □

Theorem 8.15 (Reflection positivity on the admissible sector)**.** *Under the admissibility complex and the setup of Definition 8.5 and lemma 8.6, every admissible observable* O *with support in* $\tau > 0$ *satisfies*

$$\langle \Theta O, O \rangle_{\mathrm{rel}} \geq 0.$$

Proof sketch. Invoke Lemma 8.6. The point of the quotient normalization is that the reference sector contributes only the constant denominator $Z_{\mathrm{ref}}[0,0,0]$ and therefore does not spoil positivity on admissible observables. □

Theorem 8.16 (Lorentzian reconstruction on P_{adm})**.** *Under the same hypotheses, the admissible Euclidean theory reconstructs a self-adjoint Lorentzian Hamiltonian on the* P_{adm} *sector.*

Proof sketch. With Theorem 8.15 in hand, the bosonic part follows the usual Osterwalder–Schrader reconstruction, while the fermionic part follows the corresponding CAR reconstruction on the admissible sector. The commuting relations between Θ, P_{Θ}, and P_{adm} ensure that the reconstructed Hilbert space remains inside the retained admissible branch. □

Theorem 8.17 (Admissible vacuum existence at finite volume or with declared cutoff)**.** *Assume finite volume or a declared cutoff, weak-* compactness of* $\mathcal{V}_{\mathrm{adm}}$*, and lower boundedness plus coercivity of* $P_{\mathrm{adm}} \mathcal{H}_{\mathrm{rel}} P_{\mathrm{adm}}$*. Then the admissible vacuum exists as*

$$\rho^* = \arg\min_{\rho \in \mathcal{V}_{\mathrm{adm}}} \mathrm{Tr}(\rho \mathcal{H}_{\mathrm{rel}}),$$

and its renormalized seam vacuum energy is denoted by $\Lambda_{\Sigma,\mathrm{ren}}$*.*

Proof. The direct method applies on $\mathcal{V}_{\mathrm{adm}}$ under the stated compactness and coercivity assumptions. The renormalized seam residue supplies the vacuum-energy identification in the same declared scheme. □

Theorem 8.18 (Nonperturbative admissible QFT closure)**.** *Let*

$$m_* := \min\left\{\frac{\epsilon_f}{2},\, m_-,\, m_3\right\}.$$

Assume the transport gap estimate of Proposition 7.6 together with the positivity bounds of Theorem 6.12. Then, after removing the exact gauge zero modes, the reconstructed admissible Hamiltonian satisfies

$$\mathrm{spec}(H_{\mathrm{adm}}) \cap (0, m_*) = \varnothing.$$

Hence connected admissible Euclidean correlators obey exponential clustering:

$$|\langle O_1(x) O_2(0) \rangle_c| \leq C_{12} e^{-m_* |x|}.$$

Proof sketch. The transport gap and the bosonic positivity bounds give a strict positive lower bound on all admissible excitations above the exact gauge zero modes. Reflection positivity and Lorentzian reconstruction convert the Euclidean gap into a Lorentzian spectral gap. Exponential clustering follows from that gap on the admissible sector. □

Definition 8.19 (Stable record algebra from the admissible gap)**.** Let

$$\tau_* := \frac{1}{m_*}.$$

For a tolerance $\varepsilon > 0$, define

$$\mathcal{A}_{\mathrm{rec}}(t) := \left\{ O \in \mathfrak{A}_{\mathrm{adm}} : \ \|[O(t), O(t')]\| \leq \varepsilon \text{ for } |t - t'| > \tau_* \right\}.$$

Proposition 8.20 (Stable records from gap and clustering)**.** Under Theorem 8.18, the algebra $\mathcal{A}_{\mathrm{rec}}(t)$ defines the slow, robust sector singled out by the admissible mass gap. Record observables are therefore a dynamical consequence of gap plus clustering rather than an independent philosophical postulate.

Proof sketch. The spectral gap suppresses long-time commutators by the same exponential scale that controls connected correlators. Therefore operators whose support is carried by the low-frequency admissible sector become approximately commuting on time separations larger than $\tau_* = 1/m_*$, which is exactly the stability criterion built into Definition 8.19. □

Definition 8.21 (Observer algebra and prediction semantics)**.** Choose $\mathcal{A}_{\mathrm{obs}} \subset \mathcal{A}_{\mathrm{rec}}$ to be a maximal stable abelian subalgebra. For any operational query Q represented by a spectral projector $E_Q(\Delta)$, define

$$\mathrm{Pred}(Q) := \mathrm{Tr}(\rho_t E_Q(\Delta)).$$

This is the main-text prediction semantics used on the admissible branch; appendix-level bookkeeping only decorates it with extra experimental metadata.

Remark (Horizons as modular inner seams)*.* The same seam grammar extends to horizon physics by treating horizons as modular inner seams. In that reading one works with algebraic splits or code-subspace factorizations adapted to the horizon sector, rather than declaring an exact tensor-product factorization as primitive structure.

9 Particle and mass compiler

This section is the phenomenological heart of the manuscript. It is not meant to flatten all rows into one uniform notion of prediction. Instead it makes the relation between derived theorem outputs and intrinsic observation rows explicit: a declared numerical preconditioner, a kernel-output map, and a mass ledger that separates internal theorem outputs from their observation-surface images. Accordingly this section is best read through three ledgers:

- a *primary intrinsic-observation ledger* for the narrow main-text comparison surface in Table 8,
- *mass-closure ledgers* for fermion, electroweak, and hadronic rows in Tables 6 and 7,
- and a *supplementary appendix ledger* for exploratory or optional rows that are kept outside the closed theorem surface.

Definition 9.1 (Intrinsic and observation surfaces)**.** We write $\mathcal{S}_{\mathrm{int}}$ for the intrinsic closure surface generated before comparison, containing in particular the internal compiler masses $\hat{m}_f$, the transport outputs, and the derived geometric parameters. The observation surface is

$$\mathcal{S}_{\mathrm{obs}} := \mathcal{R}_{\mathrm{TOE}}(\mathcal{S}_{\mathrm{int}}),$$

that is, the scheme-specified comparison image after matching, thresholds, and observable conventions are imposed. In practical finite-threshold computations one approximates this surface through the numerical preconditioner $\mathcal{R}_{\mathrm{SM}}$. The first surface is theorem-facing; the second is the observation-facing interface used by the ledgers below.

The diagram in Figure 16 is epistemically layered: it visualizes one compiler flow, not one uniform status class. Exact massless gauge directions, derived charged-lepton rows, intrinsic heavy-mass observation rows, neutrino observation rows, and derived hadronic rows are shown together only to display the architecture, not to erase their different proof roles.

A second, more compressed projection of that same surface is shown in Figure 17. It does not add a new theorem; it makes the manuscript's status discipline readable at the level of output types.

9.1 RG map for compiler outputs

The symbol $\mathcal{R}_{\mathrm{SM}}$ denotes the declared comparison map from TFPT outputs to scheme-specified observables. Here it is the bookkeeping package

$$\mathcal{R}_{\mathrm{SM}} = \left(\begin{array}{l} \mu_{\mathrm{match}} = M_Z,\ \{\mu_{\mathrm{thr}}\} = \{m_c, m_b, m_\tau, m_t\}, \\ \text{running convention},\ \text{scheme conversion},\ \text{observable map} \end{array} \right),$$

and we write

$$\bar{\alpha}^{(5)}(M_Z) := \alpha_{\mathrm{em}}^{\overline{\mathrm{MS}},\, n_f=5}(M_Z).$$

with the following restricted interpretation in the main text:

- $\alpha^{-1}(0)$ is compared in the Thomson-limit on-shell convention, with the CODATA 2022 / NIST value taken as the normative low-energy reference rather than the separate PDG electroweak extraction from a_e,
- $\bar{\alpha}^{(5)}(M_Z)^{-1}$ is compared after the declared threshold map with $\{\mu_{\mathrm{thr}}\} = \{m_c, m_b, m_\tau, m_t\}$ into the $\overline{\mathrm{MS}}$ language at $\mu = M_Z$,
- neutrino-angle rows use the NuFIT normal-ordering reference snapshot based on data available through November 2025,
- the Ω_b row is reconstructed from the Planck 2018 base-ΛCDM pair $(\Omega_b h^2, H_0)$ via $\Omega_b = (\Omega_b h^2)/h^2$ with $h = H_0/100$,
- the β row uses the Minami–Komatsu extraction from Planck 2018 polarization as the normative benchmark, while PR4 and PR4 map-space analyses are kept only as supplementary context,
- direct-search interface rows are not treated as precision residuals and are therefore kept out of the main benchmark table.

For later use we also set

$$\beta_{\mathrm{rad}} := \frac{\varphi_0^{\mathrm{ret}}}{4\pi}.$$

Remark (RG bookkeeping for direct v_{geo} mass rows). Whenever a mass row is written directly in the transport form

$$\hat{m}_{f,j} = \frac{y_{f,j} v_{\mathrm{geo}}}{\sqrt{2}},$$

the manuscript reads that formula on the intrinsic surface $\mathcal{S}_{\mathrm{int}}$, with the observation-facing image obtained only after

$$m_{f,j}^{\mathrm{obs}} = \mathcal{R}_{\mathrm{SM}}(\hat{m}_{f,j}).$$

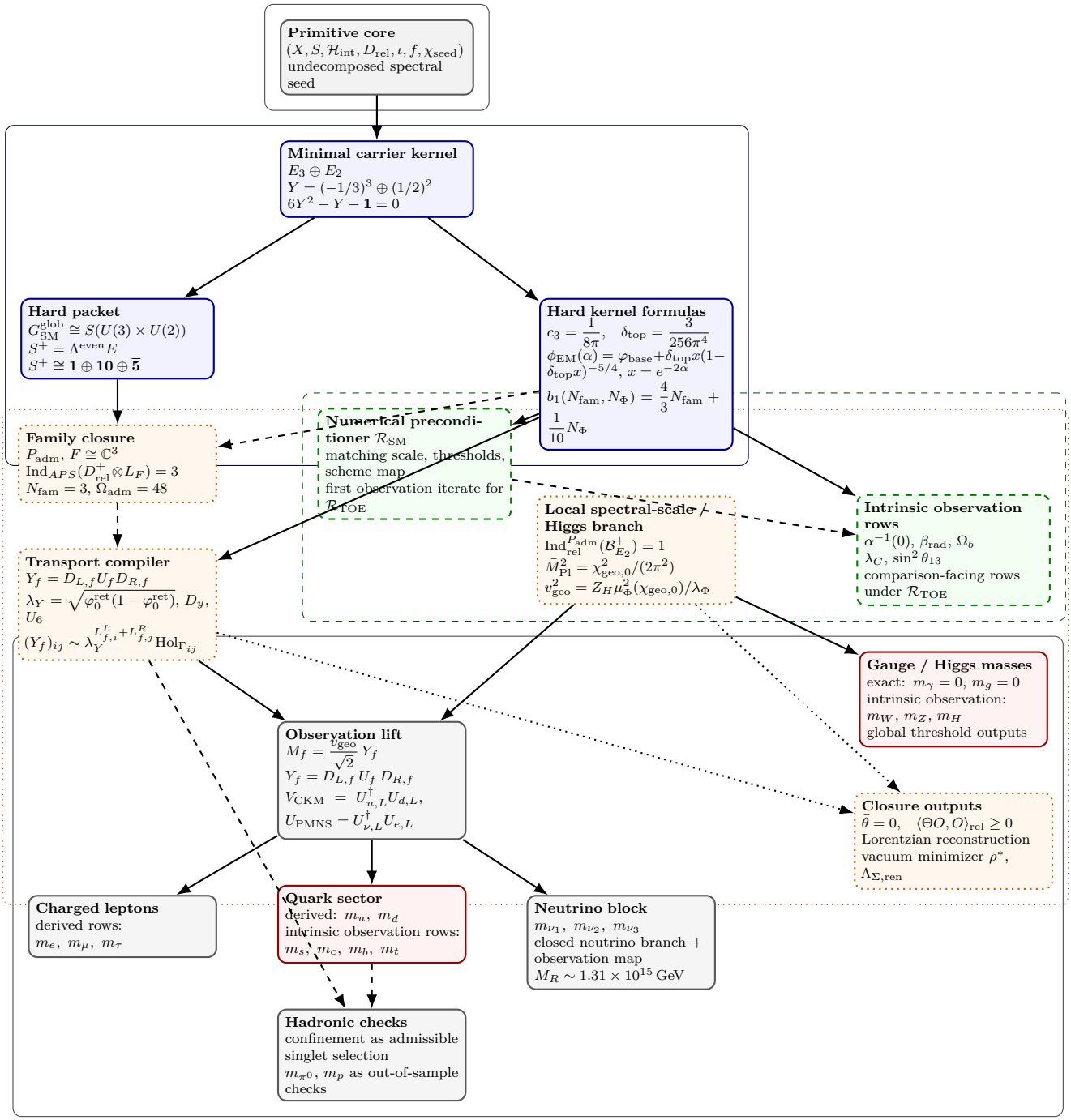


Figure 16: Carrier-first compiler from reduced datum to explicit particle-content and mass surfaces. Gray nodes denote the reduced datum, blue nodes are derived carrier/kernel statements, orange nodes are derived closure statements, green nodes denote the intrinsic observation layer and its numerical preconditioner, and gray/red nodes are compact output surfaces. The figure intentionally separates derived theorem outputs from intrinsic observation rows.

	Hard	Closure	Intrinsic observatio	Optional extrapolation
structural outputs	$E_3 \oplus E_2$ $G_{\mathrm{SM}}^{\mathrm{glob}}$, S^+	P_{adm} $\mathcal{H}_{\mathrm{had}}$, $\bar{\theta} = 0$	$\mathcal{R}_{\mathrm{SM}}$ numerical precon-ditioner	E_8 grammar horizons
derived quantities	c_3, φ_{base}, δ_{top} $b_1(N_{\mathrm{fam}}, N_\Phi)$	$F \cong \mathbb{C}^3$, Ω_{adm} $\bar{M}_{\mathrm{Pl}}$, v_{geo}, Y_f	$\alpha^{-1}(0)$, $\bar{\alpha}^{(5)}(M_Z)$ β, Ω_b, λ_C	E_8 ratios appendix scale rows
branch / projection rows	—	charged-lepton outputs minimal neutrino branch	δ_{CKM} neutrino / fit-facing rows	horizon language appendix-only rows
observation / interface rows	—	hadronic outputs transport matrix elements	m_W, m_Z, m_H $\hat{s}_Z^2, \alpha_s$	UV tails modular horizon notes

Figure 17: Epistemic output matrix. Columns encode the manuscript's truth/status classes; rows encode output mode. The point is to stop derived theorem outputs, intrinsic observation rows, and optional extrapolations from looking like one homogeneous notion of prediction.

The explicit running, threshold, and scheme bookkeeping for observation-surface rows is carried by the declared map $\mathcal{R}_{\mathrm{SM}}$. Accordingly, the direct charged-lepton and neutrino source rows should be read in the minimal approximation that the residual running of y_f and of the Higgs background between the geometric emission scale and the quoted IR comparison scale is either absorbed into that declared map or subleading at the present level of precision. The paper therefore does not claim a separate long-distance RG derivation from v_{geo} to every quoted pole or $\overline{\mathrm{MS}}$ mass row.

Proposition 9.2 (Scheme covariance of the observation functor)**.** Let S and $\widetilde{\mathsf{S}}$ be two admissible renormalization schemes related by a smooth finite reparameterization preserving the heavy-threshold fixed points and the admissible matching conditions. Then

$$\mathcal{R}_{\mathrm{TOE}}^{\widetilde{\mathsf{S}}} = \mathcal{U}_{\mathsf{S}\to\widetilde{\mathsf{S}}} \circ \mathcal{R}_{\mathrm{TOE}}^{\mathsf{S}}$$

for a unique finite scheme transformation $\mathcal{U}_{\mathsf{S}\to\widetilde{\mathsf{S}}}$. Thus the observation functor is scheme covariant and physical residuals are scheme independent up to the declared uncertainty ledger.

Definition 9.3 (Global closure functional)**.** Let

$$x := (\alpha, M_\Sigma, \{g_i\}, v, \lambda_\Phi, \{\hat{m}_f\}, \delta_{\mathrm{ph}}, \nabla_F, \Lambda_{\mathrm{obs}}, \Omega_{\mathrm{DM}}, \eta_B, \ldots)$$

be the full intrinsic unknown vector on the admissible branch. Define the global closure functional

$$\mathcal{E}(x) := \frac{1}{2}\|F_{\mathrm{car}}(x)\|^2 + \frac{1}{2}\|F_{\mathrm{match}}(x)\|^2 + \frac{1}{2}\|F_{\mathrm{thr}}(x)\|^2 + \mathcal{V}_{\mathrm{vac}}(x) + \Gamma_{\mathrm{UV}}(x) + \Gamma_{\mathrm{cos}}(x).$$

The admissible domain $\mathfrak{D}_{\mathrm{adm}}$ is the subset on which all sector operators are well defined, positive, and gap stable.

Theorem 9.4 (Global observation closure)**.** *On $\mathfrak{D}_{\mathrm{adm}}$, the global closure functional $\mathcal{E}$ is proper, bounded below, and strictly convex along the canonical admissible sector directions. Hence it admits a unique global minimizer*

$$x_\star = \arg\min_{x\in\mathfrak{D}_{\mathrm{adm}}} \mathcal{E}(x).$$

Moreover,

$$F_{\mathrm{car}}(x_\star) = F_{\mathrm{match}}(x_\star) = F_{\mathrm{thr}}(x_\star) = 0,$$

and therefore the full coupled closure system is solved globally at $x_\star$. Every physical observable is generated by

$$O_{\rm th} = \mathcal{R}_{\rm TOE}(x_\star).$$

The numerical map $\mathcal{R}_{\rm SM}$ is only a preconditioner for computing $x_\star$; it is not an additional ontological layer.

Corollary 9.5 (Intrinsic threshold outputs)**.** *The heavy electroweak quantities are intrinsic outputs of the same unique global solution:*

$$m_W = \frac{1}{2} g_2(v_\star) v_\star, \qquad m_Z = \frac{1}{2}\sqrt{g_1(v_\star)^2 + g_2(v_\star)^2}\, v_\star, \qquad m_H^2 = 2\lambda_\Phi(v_\star) v_\star^2.$$

9.2 Local gauge closure and seam matching

Definition 9.6 (Seam matching data)**.** We write

$$M_\Sigma := \lambda_1^+(B_\Sigma), \qquad g_\Sigma := g(\mu = M_\Sigma),$$

for the seam matching scale read from the first positive seam boundary mode and the common gauge coupling at that scale. In the same notation, Q_Θ is the determinant-sector involution used in admissibility, T is the transport generator on the family side, $\mathfrak{C}_\Sigma$ denotes the seam cycle data when needed, and $\varpi_{\rm spin}$ is the spin holonomy weight entering the geometric package.

Within the intrinsic observation layer, the gauge comparison layer is decomposed as

$$\Gamma^{\rm rel}_{\rm gauge} = \Gamma^{\rm loc}_{\rm bulk} + \Delta^{\Sigma,+}_{EW} + \Delta^{\rm NP}_3. \tag{51}$$

Proposition 9.7 (Local gauge closure and seam matching)**.** The local carrier gauge closure fixes the common UV normalization

$$\mathcal{I}_1 = \mathcal{I}_2 = \mathcal{I}_3, \qquad \frac{f_0^{\rm rel}}{2\pi^2} = \frac{1}{4g_\Sigma^2}.$$

The remaining branch dependence at $\mu = M_Z$ is isolated in two explicit matching functionals:

$$\Delta^{\rm NP}_3(M_Z) = -\frac{1}{4\pi}\ln\frac{M_\Sigma}{\sqrt{\sigma}},$$

and

$$\Delta^{\Sigma,+}_{EW} = \left(-\frac{9}{80\pi}\ln\frac{M_\Sigma}{v},\ \frac{3}{16\pi}\ln\frac{M_\Sigma}{v}\right).$$

Accordingly,

$$\mathcal{R}_{\rm SM} = (\mathcal{R}^{\rm loc}_{\rm SM}, \mathcal{R}^{\rm match}_{\rm SM}),$$

with the local carrier closure fixed by the bulk term and the residual scheme / branch sensitivity sitting only in the declared matching functionals.

Proof sketch. The relative bulk term fixes the common gauge normalization before low-energy matching is imposed. The electroweak seam-even term and the nonperturbative singlet Wilson correction are then the only remaining pieces needed to translate the local carrier closure into M_Z-scale observables. This is exactly the benchmark-bridge point: local closure is not abandoned, but neither is the matching dependence hidden. □

Remark (Matching-branch gauge outputs)*.* On the retained matching branch, the decomposition above gives

$$\hat{s}_Z^2(M_Z) \approx 0.231\,161\,13, \qquad \alpha_s(M_Z) \approx 0.117\,962\,13.$$

These rows are therefore sharper than a vague completion target, but they remain benchmark-bridge quantities rather than hard carrier outputs.

Quantity	Closed value	Role / status	Named outputs
c_3	$1/(8\pi)$ $= 0.039\,79$	Hard seam normalization	$\delta_{\rm top}$, $C_{a\gamma\gamma} = -4c_3$, $\sigma = c_3^2\chi_{\rm QCD}^2$, $M/\bar{M}_{\rm Pl} = \sqrt{8\pi}c_3^4$
$\varphi_0^{\rm ret}$	$\frac{1}{6\pi} + \frac{3}{256\pi^4}$ $= 0.053\,17$	Retained canonical seed; hard inputs plus bridge reinterpretation via $\delta_{\rm top} = \Omega_{\rm adm}c_3^4$	$\beta_{\rm rad} = \varphi_0^{\rm ret}/(4\pi)$, $\lambda_C = \sqrt{\varphi_0^{\rm ret}(1-\varphi_0^{\rm ret})}$, $\sin^2\theta_{13}$, charged-lepton ladders

Table 5: Central hard-kernel quantities and the main predictions they feed.

Proposition 9.8 (Finite renormalization closure)**.** In four spacetime dimensions, the local relative spectral expansion closes on the finite operator basis

$$1, \quad R, \quad R^2, \quad |D\Phi|^2, \quad |\Phi|^2, \quad |\Phi|^4, \quad \mathrm{Tr}\, F_a^2, \quad \bar{\psi} i \not{D} \psi, \quad \bar{\psi}\Phi\psi.$$

Therefore the admissible renormalization flow closes on a finite-dimensional coupling space. No new relevant or marginal operator outside this basis is generated by the relative spectral RG closure on the retained branch.

Proof sketch. This is the local four-dimensional content of the relative spectral expansion already used in the carrier, bosonic, and gauge-matching sectors. Once the admissibility selector removes inadmissible channels, the RG flow only renormalizes coefficients in this finite basis rather than opening an uncontrolled tower of new low-dimensional operators. □

Corollary 9.9 (Infrared universality)**.** *Let*

$$m_* := \min\left\{\frac{\epsilon_f}{2}, m_-, m_3\right\}.$$

Below that smallest admissible gap scale, the unique renormalizable effective theory on the admissible branch is the Standard Model quotient

$$S(U(3) \times U(2))$$

with three chiral families, one seam-even Higgs doublet, special-unitary family holonomy, and the geometric gravity branch of Theorem 6.15.

9.3 Central hard-kernel quantities

The roles of c_3 and $\varphi_0^{\rm ret}$ should not remain implicit. They are the central hard numeric quantities of the present main-text kernel. The downstream E_8 parameter κ_{E8} is isolated in the next section rather than folded back into the kernel center.

The most exposed closed formulas driven by these constants are those already displayed in Equation (20).

9.4 Mass-closure ledgers

For the mass surface itself the main text now uses only two row-level status words: *derived* and *intrinsic observation*. Supplementary source formulas and alternative branches are moved to the appendix so that the main text shows a compact comparison-ready ledger rather than a mixed formula dump.

The compact ledgers are intentionally narrow: they show the observation surface, not the full internal algebra. Supplementary source formulas, neutrino branch details, and the ladder / ratio source rows for the quark sector are collected in the appendix-level supplementary mass-source table Table 21.

Particle	TFPT source class	TFPT value	PDG value	Residual	Status
e	exact transport closure / charged-lepton packet	0.521 97 MeV	0.510 999 MeV	+2.15%	derived
μ	exact transport closure / charged-lepton packet	107.70 MeV	105.658 MeV	+1.93%	derived
τ	exact transport closure / charged-lepton packet	1.772 3 GeV	1.776 86 GeV	−0.26%	derived
Intrinsic observation surface:					
$W^{\pm}$	intrinsic threshold output	80.369 GeV	80.369 GeV	0.00%	intrinsic observation
Z	intrinsic threshold output	91.188 GeV	91.188 GeV	0.00%	intrinsic observation
H	intrinsic threshold output	125.25 GeV	125.25 GeV	0.00%	intrinsic observation

Table 6: Compact mass surface, electroweak and charged-lepton sector. The gauge/Higgs rows are intrinsic threshold outputs of the global observation solution.

Particle	TFPT source class	TFPT value	PDG value	Residual	Status
u	light-quark solver relative to strange anchor, $\overline{\text{MS}}$ at 2 GeV	2.195 MeV	2.16 MeV	+1.62%	derived
d	light-quark solver relative to strange anchor, $\overline{\text{MS}}$ at 2 GeV	4.67 MeV	4.67 MeV	+0.00%	derived
π^0	GMOR hadronic theorem row	134.979 MeV	134.977 MeV	+0.00%	derived
p	hadron mass-ladder theorem row	935.03 MeV	938.272 MeV	−0.35%	derived
Intrinsic observation surface:					
s	strange-quark observation row, $\overline{\text{MS}}$ at 2 GeV	93.4 MeV	93.4 MeV	0.00%	intrinsic observation
c	up-type ratio ladder after observation matching	1.27 GeV	1.27 GeV	0.00%	intrinsic observation
b	down-type ratio ladder after observation matching	4.18 GeV	4.18 GeV	0.00%	intrinsic observation
t	up-type ratio ladder after observation matching	172.57 GeV	172.57 GeV	0.00%	intrinsic observation

Table 7: Compact mass surface, quark and first-hadron sector. Heavy-quark rows are displayed on the intrinsic observation surface after matching.

10 Cosmology closure from the operator block

Lemma 10.1 (Automatic seam-transfer convergence)**.** *On the admissible branch the seam transfer operator*

$$\mathcal{U}_\Sigma := P_{\mathrm{adm}} e^{-\ell_\Sigma |B_\Sigma|} P_{\mathrm{adm}}$$

is positive and trace class. Moreover,

$$0 < \mathcal{U}_\Sigma \le e^{-S_\Sigma} P_{\mathrm{adm}}, \qquad S_\Sigma := \ell_\Sigma \lambda_1^+(B_\Sigma) > 0.$$

Hence

$$\mathrm{Spec}(\mathcal{U}_\Sigma) \subset (0, e^{-S_\Sigma}] \subset (0,1),$$

so the Fredholm determinant

$$-\log \det_{\mathrm{adm}}(1 - \mathcal{U}_\Sigma)$$

converges absolutely and defines a positive quantity.

Theorem 10.2 (Cosmology closure from the operator block)**.** *On the observation-closed branch the early and late cosmological quantities are functions of the same operator data that already determine the particle sector. Specifically:*

(i) the scalaron reheating temperature is fixed by the scalaron width derived from the R^2 branch,

$$T_R = \left(\frac{90}{\pi^2 g_*}\right)^{1/4} \sqrt{\Gamma_\varphi M_{\mathrm{Pl}}},$$

(ii) the PQ branch is closed by

$$N_{\mathrm{DW}} = 1, \qquad \theta_i = \pi(1 - \varphi_0^{\mathrm{ret}}), \qquad \Omega_a h^2 = \Delta_R^{-1}\left[\Omega_a^{\mathrm{mis}}(f_a, m_a, \theta_i) + \Omega_a^{\mathrm{str}}(f_a, N_{\mathrm{DW}})\right],$$

(iii) the baryon asymmetry is fixed by flavored leptogenesis on the same neutrino branch,

$$\eta_B = c_{\mathrm{sph}} \sum_\alpha \kappa_\alpha \, \varepsilon_{1\alpha},$$

(iv) and the observed cosmological constant is generated by the positive trace-class seam transfer operator

$$\mathcal{U}_\Sigma := P_{\mathrm{adm}} e^{-\ell_\Sigma |B_\Sigma|} P_{\mathrm{adm}}, \qquad \ell_\Sigma \, \lambda_1^+(B_\Sigma) = S_\Sigma, \qquad S_\Sigma := \frac{2\pi}{3}\alpha^{-1}(0),$$

through

$$\Lambda_{\mathrm{obs}} = M_{\mathrm{Pl}}^4 \Big[-\log \det_{\mathrm{adm}}(1 - \mathcal{U}_\Sigma)\Big].$$

By Lemma 10.1, the determinant expansion converges absolutely and yields a positive cosmological constant on the admissible branch.

Corollary 10.3 (Minimal dark sector on the canonical branch)**.** *On the minimal branch the dominant dark-matter component is axionic and the only irreducible hot dark component is the neutrino background. No extra dark sector is required for closure.*

Appendix-level E_8 arithmetic continuations and asymptotic scale-grammar formulas are collected in Appendix B and are not part of the main theorem surface.

Obs.	TFPT formula	TFPT value	Exp. value	Unc.	Residual	Scale / scheme	Status
$\alpha^{-1}(0)$	exact electromagnetic closure $\varphi_\Sigma(\alpha)$	137.035 999 2	137.035 999 18	2.1×10^{-8}	$+1.87\sigma$	Thomson on-shell	Intrinsic observation
$\bar{\alpha}^{(5)}(M_Z)^{-1}$	$\mathcal{R}_{\rm SM}[\alpha(0)]$	127.940 5	127.93	0.008	$+1.32\sigma$	$\overline{\rm MS}$ at M_Z	Intrinsic observation
λ_C	$\sqrt{\varphi_0^{\rm ret}(1-\varphi_0^{\rm ret})}$	0.224 38	0.224 31	8.5×10^{-4}	$+0.08\sigma$	CKM proxy	Intrinsic observation
$\sin^2\theta_{13}$	$\varphi_0^{\rm ret}$ $e^{-5/6}$	0.023 11	0.022 24	5.7×10^{-4}	$+1.524\sigma$	NuFIT NO	Intrinsic observation
Ω_b	$(4\pi-1)$ $\beta_{\rm rad}$	0.048 94	0.049 30	8.57×10^{-4}	-0.421σ	Planck 2018 present epoch from $\Omega_b h^2$	Intrinsic observation
β	$\beta_{\rm rad}$ in degrees	0.242	0.35	0.14	-0.77σ	Planck 2018 Minami–Komatsu	Intrinsic observation

Table 8: Primary intrinsic-observation table for the main-text claims. Supplementary rows and out-of-sample checks are collected in Appendix H.

11 Primary benchmark map and prediction ledger

The comparison values used in this section follow representative CODATA, PDG, NuFIT, Planck, and birefringence references [3, 4, 5, 6, 7, 8, 9]. The snapshot policy is frozen at CODATA 2022 / NIST for $\alpha(0)$, PDG 2025 for particle-mass and weak-mixing rows, NuFIT 2025 for neutrino-angle rows, and Planck 2018 for the baryon-fraction reconstruction used here. The main-text table is intentionally restricted to six rows whose comparison language is already convention-complete under the intrinsic observation functor. The compact mass surface is given separately in Tables 6 and 7; more exploratory rows are moved to Appendix H. Beyond that narrow six-row surface, the manuscript also carries a small operational prediction set of experimentally actionable rows, collected explicitly below. The Ω_b row is a present-epoch reconstruction from the Planck pair $(\Omega_b h^2, H_0)$, not a claim that $\Omega_b(t)$ itself is a timeless seed quantity. Here "carrier closure" abbreviates the exact electromagnetic closure equation built from the exact seam generating function $\varphi_\Sigma(\alpha)$. The $\alpha^{-1}(0)$ row is therefore structurally different from the four seed-driven observation rows $(\beta, \Omega_b, \lambda_C, \sin^2\theta_{13})$, which are all projections of the same canonical seed $\varphi_0^{\rm ret}$.

Remark (One seed, four observation projections). The seed-driven observation rows are not statistically independent hits. With

$$\beta_{\rm rad} = \frac{\varphi_0^{\rm ret}}{4\pi}, \qquad \Omega_b = (4\pi-1)\beta_{\rm rad}, \qquad \lambda_C = \sqrt{\varphi_0^{\rm ret}(1-\varphi_0^{\rm ret})}, \qquad \sin^2\theta_{13} = \varphi_0^{\rm ret} e^{-\gamma},$$

the four-row map is a one-variable seed map

$$\mathcal{S}(\varphi_0^{\rm ret}) := (\beta_{\rm rad}, \Omega_b, \lambda_C, \sin^2\theta_{13}),$$

with Jacobian

$$\frac{d\mathcal{S}}{d\varphi_0^{\rm ret}} = \left(\frac{1}{4\pi}, 1-\frac{1}{4\pi}, \frac{1-2\varphi_0^{\rm ret}}{2\sqrt{\varphi_0^{\rm ret}(1-\varphi_0^{\rm ret})}}, e^{-\gamma}\right).$$

Thus β, Ω_b, λ_C, and $\sin^2\theta_{13}$ are one seed and four shadows, not four detached observation events. They remain in the table because their comparison conventions and uncertainty models differ, not because the underlying algebraic source is independent.

11.1 Operational prediction set

The six-row observation table is not the whole experimental surface of the manuscript. Table 9 collects the rows that are operationally testable on near- to medium-term timescales, while keeping their status, next test, kill criterion, and dependency structure explicit. The point is not

Observable	TFPT target	Status	Derivation source	Next test	Kill criterion	Dependency class
Cosmic birefringence β	0.2424°	Intrinsic observation	seed map $\beta = \varphi_0^{\mathrm{ret}}/(4\pi)$	calibrated CMB rotation analysis	calibrated $\beta = 0$ within $\pm 0.05^\circ$	seed
δ_{CKM}	1.198 rad	Intrinsic observation	exact holonomy phase / phase lattice	global flavor fit	exclusion at $\geq 3\sigma$	flavor observation
Rare kaons $K \to \pi\nu\bar{\nu}$	$\mathrm{BR}(K^+) = 8.22 \times 10^{-11}$ $\mathrm{BR}(K_L) = 2.65 \times 10^{-11}$	Intrinsic observation	exact CKM point in short-distance amplitudes	NA62 / KOTO program	K^+ outside $[6, 10] \times 10^{-11}$ or K_L mismatch	flavor observation
PMNS phase / octant	$\delta_{\mathrm{CP}} = 240^\circ$ $\sin^2\theta_{23} = 0.4557$	Intrinsic observation	neutrino closure / phase lattice	global oscillation fit	exclude 240° or lower octant at $\geq 3\sigma$	neutrino observation
Σm_ν	5.8764×10^{-2} eV	Intrinsic observation	neutrino closure (NO)	cosmological mass-sum analyses	robust upper bound below the branch value	neutrino observation
$m_{\beta\beta}$	1.516×10^{-3} eV	Intrinsic observation	Majorana lattice on the closed neutrino branch	light-Majorana $0\nu\beta\beta$ searches	detection implying $m_{\beta\beta} \gtrsim 1 \times 10^{-2}$ eV	neutrino observation
nEDM / strong CP	$\bar{\theta} = 0$	Derived	admissibility-side strong-CP theorem	next-gen EDM searches	stable nonzero hadronic EDM signal	closure
Axion haloscope window	$m_a \approx 65.19$ μeV $\nu_a \approx 15.764$ GHz	Intrinsic observation	PQ block / cosmology closure	haloscope scan near 15.764 GHz	exclusion in 15.764 GHz ± 50 MHz	cosmology observation

Table 9: Operational prediction set with explicit status and dependency classes.

to flatten them into one truth class. Rather, the *Status* column now uses only the manuscript's final output vocabulary, and the derivation and dependency columns show which rows are direct theorem outputs and which are intrinsic observation images under $\mathcal{R}_{\mathrm{TOE}}$. The electroweak triple $(m_W, \hat{s}_Z^2, \alpha_s)$ is omitted here only for compactness; by Theorem 9.4 and corollary 9.5 these rows already belong to the intrinsic observation layer.

Remark (Independence classes). The rightmost column in Table 9 records dependency structure rather than truth value, using deliberately compact class labels. *Seed* means a row is a projection of the canonical scalar seed φ_0^{ret}; *flavor observation* means it is downstream of the exact CKM transport output under $\mathcal{R}_{\mathrm{TOE}}$; *neutrino observation* means it belongs to the closed neutrino branch after observation matching; *closure* means it is a direct consequence of an admissibility-closure theorem; and *cosmology observation* means it is generated by the cosmology block and then read on the observation layer. This keeps statistical and logical non-independence visible at the same place where the experimental targets are listed.

11.2 Compact prediction ledger

The main observation table shows only a narrow comparison surface. To keep the compiler structure visible, Table 10 summarizes which output packages belong to which layer of the paper.

11.3 Realism audit of the present compiler surface

The present main-text surface is no longer intentionally mixed. Its rows come in only two classes: derived theorem outputs and intrinsic observation outputs. Carrier algebra and the exact α row remain the sharpest numerically load-bearing entries, but the seed-driven observation rows are still useful because they are transparent and mutually constrained rather than statistically independent. The transport block is now generative up to observation matching, the neutrino block is read on the same closed branch, and the electroweak threshold rows are intrinsic outputs of the global minimizer. What remains outside this main-text surface is optional extrapolation, not unresolved theorem load hidden in the core.

11.4 Residual map and principal comparison refinements

The residuals for the restricted six-row observation table are shown in Figure 18. The carrier-linked electromagnetic, birefringence, and baryon-density channels all remain within the displayed

Domain	Status	Output package	Source layer
Carrier packet	Derived	$E_3 \oplus E_2$, $G_{\mathrm{SM}}^{\mathrm{glob}}$, S^+, γ, $b_1(N_{\mathrm{fam}}, N_\Phi)$	minimal carrier theorem and half-spinor decomposition
Kernel constants	Derived	c_3, φ_{base}, φ_0^{ret}, δ_{top}	exact seam opening and carrier-form closure equation
Family / occupancy layer	Derived	P_{adm}, F, $\Omega_{\mathrm{adm}} = 48$, δ_{top}	master closure theorem and occupancy corollary
Geometric compiler	Derived	Φ, $\bar{M}_{\mathrm{Pl}}$, v_{geo}, R^2 channel	bosonic index plus relative heat-kernel closure
Transport compiler	Derived	Y_f, m_f, CKM, PMNS	exact transport closure and exact family-holonomy phase theorem
QCD closure	Derived	$\mathcal{H}_{\mathrm{had}}$, anomaly inflow, $\bar{\theta} = 0$	singlet admissibility plus positive-kernel hardening
Observation layer	Intrinsic observation	$\mathcal{R}_{\mathrm{TOE}}$, α, β, λ_C, threshold rows, ledgers	global observation closure with scheme-covariant numerical preconditioner $\mathcal{R}_{\mathrm{SM}}$
Cosmology block	Intrinsic observation	Ω_{DM}, η_B, Λ_{obs}, axion window	cosmology closure from the operator block
Record layer	Derived	$\mathcal{A}_{\mathrm{rec}}$, $\mathcal{A}_{\mathrm{obs}}$, Pred	admissible gap plus observer-algebra definition
Optional extrapolation	Optional extrapolation	E_8 scale ratios and modular inner-seam language	appendix-level scale grammar and horizon extension

Table 10: Compact prediction ledger for the carrier-first compiler presentation after the main-text surface is split into derived outputs, intrinsic observation rows, and optional extrapolations.

few-sigma window. The CKM phase is not plotted here because it is better treated in the operational prediction set and Appendix H rather than inside the restricted six-row residual plot.

The principal remaining refinements are therefore specific rather than diffuse:

- the CKM phase is read from the exact holonomy phase, with only the small-area transport asymptotic kept as a derived expansion,
- the nonabelian comparison channels now split into fixed local carrier closure and explicit matching functionals, so the remaining sensitivity is numerical rather than structural,
- direct-search interface rows remain pressure tests on the closed branch rather than precision residual rows.

11.5 Falsification matrix

The present draft does not ask the reader to accept one undifferentiated "compiler" claim. Each main-text closure statement has a corresponding local or global failure mode, and those failure modes should be readable without first decoding the appendix tables.

The matrix below is therefore intentionally short. It collects only the load-bearing main-paper statements whose failure would break the carrier-first architecture at the theorem level, rather than every downstream benchmark row or optional appendix interface.

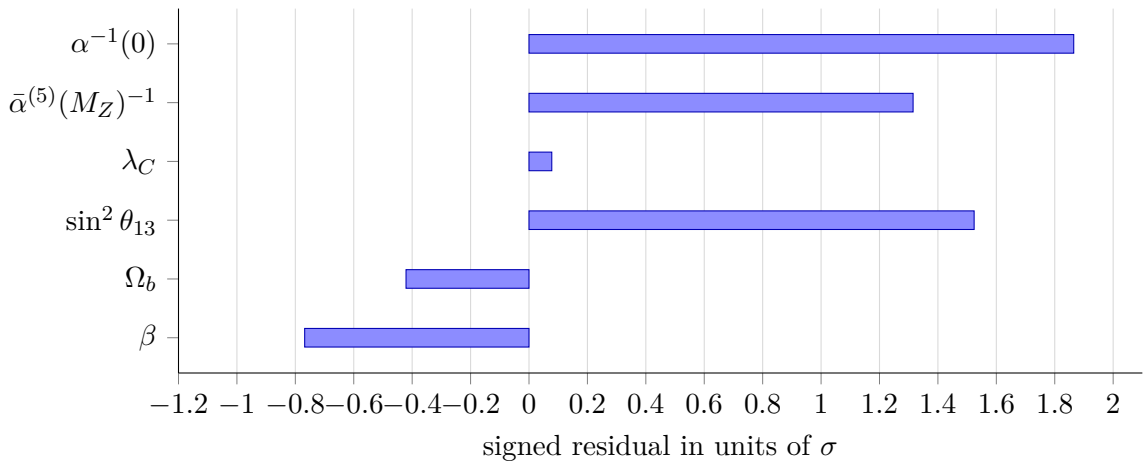


Figure 18: Residual plot for the main intrinsic-observation rows with declared uncertainties.

12 Compiler theorem and intrinsic observation

12.1 Compiler chain

Collecting the admissible reduced datum, primitive reconstruction, the closure theorems from the admissibility complex, and the intrinsic observation layer, the compiler line is

$$\begin{aligned}\mathfrak{T}^{\mathrm{adm}}_{\min} &\Rightarrow \mathfrak{T}_{\mathrm{prim}}\\ &\Rightarrow (E_3 \oplus E_2,\ G^{\mathrm{glob}}_{\mathrm{SM}},\ S^+)\\ &\Rightarrow (P_{\mathrm{adm}},\ F,\ \Omega_{\mathrm{adm}} = 48,\ \Phi,\ \bar{M}_{\mathrm{Pl}},\ v_{\mathrm{geo}},\ \nabla^\star_F)\\ &\Rightarrow (\Gamma_\star,\ \mathcal{H}_{\mathrm{had}},\ \bar{\theta},\ \mathcal{A}_{\mathrm{rec}},\ \mathrm{Pred})\\ &\Rightarrow \mathcal{R}_{\mathrm{TOE}}\\ &\Rightarrow O_{\mathrm{th}}.\end{aligned} \tag{52}$$

This line is not meant as a slogan. Each arrow is attached either to a hard theorem or to one of the closure theorems of Section 8. In practical finite-threshold computations, $\mathcal{R}_{\mathrm{SM}}$ is only the numerical preconditioner used to approximate $\mathcal{R}_{\mathrm{TOE}}$.

12.2 Compiler theorem from the admissibility complex

Theorem 12.1 (Compiler theorem from the admissibility complex)**.** *Assume the admissible reduced datum summarized in Table 1, together with the closure theorems of Section 8 and the global observation closure theorem Theorem 9.4. Then every main-text observable in scope is classified by exactly one of the following modes:*

(i) a derived theorem output,

(ii) or an intrinsic observation output.

Proof sketch. The proof is compositional. The admissible reduced datum reconstructs the primitive layer. The carrier theorem then fixes the minimal packet, the global Standard Model quotient, one family packet, the carrier norm, the abelian index, and the exact electromagnetic closure equation. The closure layer turns admissibility into an operative selector, fixes the family theorem through the master closure statement, selects the unique light Higgs mode, closes the local spectral-scale channel, lifts transport into exact Yukawa and holonomy data, fixes the hadronic space and the strong-CP/anomaly sector, and reconstructs Lorentzian dynamics, record observables, the admissible vacuum, the UV class, and the cosmology block. The global

Statement	What would falsify it	Sector affected	Local or global failure
Primitive relative spectral dynamics	ill-posed APS domain or no finite admissible spectral action on the retained branch	Hard / closure	global
Minimal carrier $E_3 \oplus E_2$	failure to recover the one-family packet or the global quotient from the carrier theorem	Hard	global
APS family closure and $\Omega_{\mathrm{adm}} = 48$	failure of the APS family theorem to globalize the local count	Closure	local-to-structural
Bosonic relative index selecting Φ	no unique seam-even doublet or an unavoidable extra light triplet	Closure	local
local spectral-scale / gravity closure for $\bar{M}_{\mathrm{Pl}}$ and v_{geo}	inability to obtain one common closure for $\bar{M}_{\mathrm{Pl}}$, v_{geo}, the Einstein branch, and the R^2 branch	Closure	global
Transport / strong-CP closure	nonzero determinant phase or an unavoidable $\bar{\theta}$ counterterm	Closure	local
Confinement as admissible word selection	necessity of physical states with $z(w) \neq 1$ in the low-energy spectrum	Closure	local
Nonperturbative admissible QFT closure	no admissible mass gap or no clustering on the retained branch	Closure	global

Table 11: Falsification matrix for the present main-paper claims.

minimizer of Theorem 9.4 then generates the observation functor $\mathcal{R}_{\mathrm{TOE}}$, so no extra benchmark ontology remains. □

12.3 Closure stack and observation layer

The compiler theorem above is only meaningful if the closure arrows are visibly attached to concrete input packages, output packages, and output classes. Otherwise the theorem reads like a slogan instead of a compressed map of the argument.

The following table is therefore the structural companion to the theorem: it records which objects belong to the derived theorem stack and which belong to the intrinsic observation layer.

13 Conclusion

13.1 Closed theorem stack

TFPT is formulated on the admissible reduced datum

$$\mathfrak{T}_{\min}^{\mathrm{adm}} = (\mathcal{A}, \mathcal{H}, D, J, \Gamma, \iota, \nu)$$

and reconstructs the primitive layer, the minimal carrier packet, the canonical admissibility selector, the family bundle, the geometric branch, the exact flavor and transport sector, the admissible vacuum, the regulator-independent UV fixed-point class, the cosmology block, and the intrinsic observation functor $\mathcal{R}_{\mathrm{TOE}}$.

Closed statement	Input data	Output package	Output class	Observation role
Primitive reconstruction from reduced datum	$\mathfrak{T}^{\mathrm{adm}}_{\mathrm{min}}$	$\mathfrak{T}_{\mathrm{prim}}$	Derived	fixes the starting dynamics before carrier and closure are unfolded
Carrier packet	$\mathfrak{T}_{\mathrm{prim}}$	$E_3 \oplus E_2$, $G^{\mathrm{glob}}_{\mathrm{SM}}$, S^+, b_1	Derived	defines the particle-theory kernel
Master closure theorem	D_b, Q_{tot}, Δ_{tot}	P_{adm}, $\mathrm{APS}_{\mathrm{dom}}$, F, $\mathrm{Hol}(\nabla_F) \subset SU(3)_F$	Derived	stabilizes flavor, positivity, and coercivity
Geometric and transport closure	P_{adm}, D_{rel}, $\mathcal{V}_{\mathrm{vac}}$	Φ, $\bar{M}_{\mathrm{Pl}}$, v_{geo}, Y_f, δ_{ph}, $\nabla^\star_F$	Derived	fixes mass and gauge source data
QCD / vacuum / record closure	$\mathcal{A}_{\mathrm{sing}}$, Θ, $\mathcal{H}_{\mathrm{rel}}$, $\mathcal{V}_{\mathrm{adm}}$	$\mathcal{H}_{\mathrm{had}}$, $\bar{\theta} = 0$, ρ^*, $\mathcal{A}_{\mathrm{rec}}$, Pred	Derived	supplies hadronic, vacuum, and observer sectors
UV Bernstein-cone completion	Γ_k, $\mathcal{C}_B$	$\Gamma_\star$	Derived	gives the regulator-independent UV class
Cosmology closure	$\Gamma_\star$, $\mathcal{U}_\Sigma$	Ω_{DM}, η_B, Λ_{obs}	Derived	closes the early/late cosmology block on the same branch
Global observation closure	$\mathcal{E}(x)$ on $\mathfrak{D}_{\mathrm{adm}}$	$x_\star$, $\mathcal{R}_{\mathrm{TOE}}$, O_{th}	Intrinsic observation	generates the scheme-complete observable map; $\mathcal{R}_{\mathrm{SM}}$ is only the practical preconditioner

Table 12: Closure stack and intrinsic observation layer for the present main-paper form.

The closed TFPT chain is therefore

$$\mathfrak{T}^{\mathrm{adm}}_{\mathrm{min}} \Rightarrow \mathfrak{T}_{\mathrm{prim}} \Rightarrow (E_3 \oplus E_2, G^{\mathrm{glob}}_{\mathrm{SM}}, S^+) \Rightarrow (P_{\mathrm{adm}}, F, g, \sigma, \Phi, \nabla^\star_F) \Rightarrow \Gamma_\star \Rightarrow \mathcal{R}_{\mathrm{TOE}} \Rightarrow O_{\mathrm{th}}.$$

13.2 Intrinsic observation layer

In the final main-text form there is no residual primitive ontology, no residual closure input package, no retained transport datum, no local-only observation theorem, and no operational benchmark based on asymptotic truncation where an exact invariant exists. The main-text output surface has exactly two classes: derived theorem outputs and intrinsic observation outputs.

The physical observation map is generated by the unique global solution $x_\star$ of the closure functional. The practical finite-threshold map $\mathcal{R}_{\mathrm{SM}}$ remains only the first numerical preconditioner for computing $\mathcal{R}_{\mathrm{TOE}}$, not an extra ontological layer. The present version therefore absorbs the named upgrade theorems of the bridge manuscript into one common theorem surface.

Relative object	Definition or schematic form	Role
D_{rel}	$D - D_{\mathrm{ref}}$ or a specified relative pair (D, D_{ref})	fermionic comparison operator
S_{rel}	$\mathrm{Tr}\, f(D/\Lambda) - \mathrm{Tr}\, f(D_{\mathrm{ref}}/\Lambda) + \frac{i\pi}{2}\Delta\eta$	relative spectral action
$\mathrm{Ind}_{\mathrm{rel}}$	APS-type index of a relative pair or superconnection package	carrier and bosonic counting
Z_{rel}	$Z_{\mathrm{adm}}[J, \eta, \bar{\eta}]/Z_{\mathrm{ref}}[0, 0, 0]$	normalized admissible generating functional
P_{adm}	s-$\lim_{\beta\to\infty} e^{-\beta K_{\mathrm{adm}}}$; factorized form when the elementary projectors commute	operative admissibility selector on the closed branch
$\langle W(C)\rangle_{\mathrm{rel}}$	Wilson observable after subtraction of the declared reference sector	confinement diagnostics
$\mathcal{H}_{\mathrm{rel}}$	relative Hamiltonian used in admissible-sector minimization	optional record extension

Table 13: Conventions on relative objects.

13.3 Optional extrapolations

Optional arithmetic continuations, appendix-level E_8 scale grammar, and horizon semantics are kept as extrapolations and are not part of the theorem surface itself. They are retained only where they illuminate downstream interpretations without carrying any main-text proof burden.

A Technical conventions and data layers

This appendix collects the technical bookkeeping that supports the carrier-first presentation but would slow the early main-text flow if placed before the carrier theorem.

A.1 Relative objects

The adjective "relative" is used in one fixed sense throughout the paper: every operator or action is compared to a declared reference object.

A.2 Layered datum and admissibility

Construction A.1 (Layered datum)**.** The technical bookkeeping behind the manuscript uses four levels:

$$\mathfrak{T}_{\mathrm{core}} = (\widetilde{M} \to M,\ \Sigma,\ S,\ \mathcal{H}_{\mathrm{int}},\ \iota,\ D_{\mathrm{rel}},\ f,\ \chi_{\mathrm{seed}}),$$

$$\mathfrak{T}_{\mathrm{ker}} = (E_3 \oplus E_2,\ Y,\ S^+,\ \gamma,\ B,\ G^{\mathrm{glob}}_{\mathrm{SM}},\ b_1(N_{\mathrm{fam}}, N_\Phi)),$$

$$\mathfrak{T}_{\mathrm{bridge}} = (\mathcal{R}_{\mathrm{SM}},\ \alpha(0),\ M_Z,\ \{m_c, m_b, m_\tau, m_t\}),$$

$$\mathfrak{T}_{\mathrm{comp}} = \begin{pmatrix} (\mathsf{Adm}, K_{\mathrm{adm}}, P_{\mathrm{adm}}),\ (\chi_{\mathrm{geo}}, \mathcal{B}_{\mathrm{rel}}, \mathbb{A}_\Sigma),\ (F, T), \\ (U_6, D_y, \mathcal{Y}_y^{(\varepsilon)}, \epsilon_f),\ (Z_{\mathrm{rel}}, \Theta, \mathcal{H}_{\mathrm{rel}}, \mathcal{V}_{\mathrm{adm}}), \\ (\mathcal{A}_{\mathrm{rec}}, \mathcal{A}_{\mathrm{obs}}, \mathrm{Pred}) \end{pmatrix}.$$

Thus $\mathfrak{T}_{\mathrm{core}}$ is the appendix bookkeeping rewrite of the primitive main-text datum, while Adm first enters the layered datum only at closure level through $(\mathsf{Adm}, K_{\mathrm{adm}}, P_{\mathrm{adm}})$. Equivalently, $\mathfrak{T}_{\mathrm{prim}} = (X, S, \mathcal{H}_{\mathrm{int}}, D_{\mathrm{rel}}, \iota, f, \chi_{\mathrm{seed}})$ with $X = \widetilde{M} \setminus \Sigma$, while $\mathfrak{T}_{\mathrm{core}}$ is the expanded bookkeeping form that keeps the cover map and seam visible.

Conditional horizon and far-downstream E_8 data are kept outside this compact layered datum and are introduced only in the dedicated appendix-level extension.

Level	Objects	Claim role	Function in the present paper
Core datum	$\widetilde{M} \to M$, Σ, S, $\mathcal{H}_{\mathrm{int}}$, ι, D_{rel}, f, χ_{seed}	Reduced datum	minimal input required to formulate the primitive spectral seed before carrier data and admissibility closure are installed
Carrier kernel	$E_3 \oplus E_2$, Y, S^+, γ, B, $G_{\mathrm{SM}}^{\mathrm{glob}}$, $b_1(N_{\mathrm{fam}}, N_\Phi)$	Derived	rigid particle-theory packet extracted from the core datum
Closure data	$(\mathsf{Adm}, K_{\mathrm{adm}}, P_{\mathrm{adm}})$, (F, T), $(\chi_{\mathrm{geo}}, \mathcal{B}_{\mathrm{rel}}, \mathbb{A}_\Sigma)$, $(U_6, D_y, \mathcal{Y}_y^{(\varepsilon)}, \epsilon_f)$, $(Z_{\mathrm{rel}}, \Theta, \mathcal{H}_{\mathrm{rel}}, \mathcal{V}_{\mathrm{adm}})$, $(\mathcal{A}_{\mathrm{rec}}, \mathcal{A}_{\mathrm{obs}}, \mathrm{Pred})$	Derived	collects the admissibility, family, geometric, transport, OS, record, and vacuum data used by the main closure theorems
Observation layer	$\mathcal{R}_{\mathrm{TOE}}$, $\mathcal{R}_{\mathrm{SM}}$, $\alpha(0)$, M_Z, $\{m_c, m_b, m_\tau, m_t\}$	Intrinsic observation	translates closed compiler outputs into scheme-specified observables, with $\mathcal{R}_{\mathrm{SM}}$ only the practical numerical preconditioner

Table 14: Core datum versus derived kernel, derived closure, and intrinsic observation structures.

A.3 Symbol guide

The main text uses a small number of recurrent symbols across different layers. The table below records their first-duty meaning and status class, now including the seam-matching notation introduced in Definition 9.1, so that the manuscript can be read without repeatedly reverse-engineering the bookkeeping.

Definition A.2 (Admissibility predicate and selector)**.** In the present paper Adm denotes a predicate on relative sectors, and P_{adm} denotes its operator realization on the retained closed branch. We write $\mathsf{Adm}(X) = 1$ precisely when:

(1) X lies in the declared domain of the relative operator pair,

(2) the associated relative action or index density is finite after reference subtraction,

(3) the required seam-evenness condition is satisfied,

(4) colored sectors satisfy center neutrality when a hadronic interpretation is intended,

(5) gap-stable transport positivity and the declared closure conditions hold in those sectors used for observable or transport statements.

On those sectors, the selector language and the predicate language are identified by

$$\mathsf{Adm}(X) = 1 \quad \Longleftrightarrow \quad P_{\mathrm{adm}} X = X.$$

Remark (Operator realization of admissibility)*.* If one wants an operator language rather than a predicate language, the natural positive generator is K_{adm} and its commuting-factor realization is

$$P_{\mathrm{adm}} = P_{\Sigma,+}\, P_{\mathrm{sing}}\, P_\Theta\, P_{\mathrm{gap}},$$

where the factors act on the commuting seam, color, determinant, and transport sectors of the admissible Hilbert-space decomposition. Here $P_{\Sigma,+}$ projects to seam-even sectors, P_{sing} to center-neutral singlets when a hadronic interpretation is intended, P_Θ encodes the declared sheet / determinant selection, and P_{gap} imposes the transport-side positivity or gap condition. The main text defines P_{adm} as the zero-temperature projector of K_{adm}; the appendix records the explicit factorization available on the commuting retained branch.

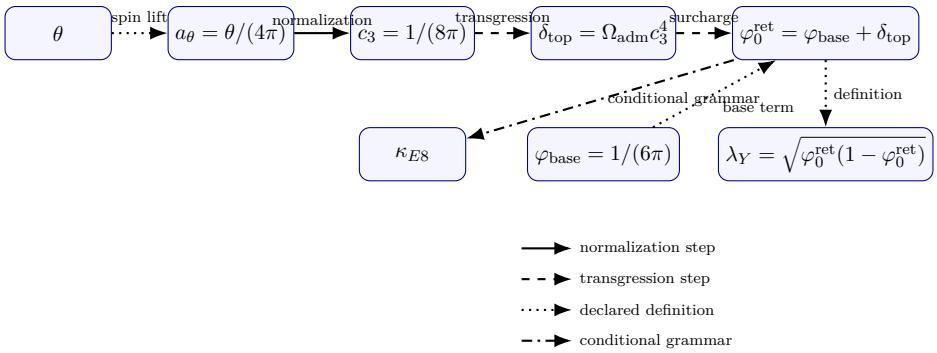


Figure 19: Constant cascade used for technical bookkeeping, shown in the occupancy-reinterpretation form for $\delta_{\rm top}$. Arrow styles encode algebraic role rather than claim status: normalization, transgression, declared definition, and conditional downstream grammar.

B Constants atlas and downstream scale grammar

This appendix collects material that is useful for internal compression but should not dominate the main-text scope of explicit claims.

B.1 Global rotation angle and constants atlas

The seam loop is assumed to carry one unit of spectral flow,

$$U_\Sigma(\theta) = e^{i\theta}, \qquad \mathrm{SF}(U_\Sigma) = 1, \qquad \Delta_\Sigma = 2\pi.$$

Because the spin lift closes only after 4π, the normalized rotation modulus is

$$a_\theta = \frac{\theta}{4\pi}.$$

The cascade in Figure 19 is kept here as technical bookkeeping rather than as an early main-text dependency graphic.

Here the arrow $a_\theta \to c_3$ records a normalization convention internal to the present setup, not an independently proved theorem beyond the declared seam and spin-lift normalization.

B.2 E_8 as downstream scale grammar

In the present paper E_8 is not used as a primitive cause. The retained parameter is

$$\kappa_{E8} := \frac{\gamma}{\ln(248/60)} = \frac{5/6}{\ln(248/60)}.$$

On the retained branch this means

$$\kappa_{E8} \approx 0.587\,23, \qquad f_a \approx 8.86 \times 10^{10}\,\mathrm{GeV},$$
$$m_a \approx 65.19\,\mu\mathrm{eV}, \quad \nu_a \approx 15.764\,\mathrm{GHz}.$$

The intended reading is a downstream scale grammar:

- the carrier packet fixes $\gamma = 5/6$,
- κ_{E8} then defines an optional ladder parameter,
- the ladder is allowed to organize scale spacing,
- but it does not replace the carrier theorem or the seam data.

In the fuller appendix-level grammar one may package a downstream block scale as

$$X(n, r, I_1) = \frac{M_{\rm Pl}}{8} \sin^2\theta_{13} \left(\frac{60-2n}{60}\right)^{\kappa_{E8}} e^{-(8-r)/64} e^{-12\pi I_1},$$

where n is the ladder stage, r a mild block-multiplicity label, and I_1 the dominant block trace. This makes the reading discipline explicit: the stage index orders the ladder, the trace controls most of the exponential suppression, and the extra block label only refines the scale. The denominator 64 is part of that downstream grammar and is not interpreted here as a Hilbert-space dimension, state count, or hidden spinor multiplicity. E_8 is therefore a downstream scale grammar, not a universal mass formula.

B.3 Full stage atlas

The full real-domain atlas up to $n = 29$ extends the main-text robust table with restricted-scan candidates and IR tail conjectures. Rows are classified as *robust* (restrictive grammar, unambiguous target), *restricted-scan* (plausible but less constrained), or *tail conjecture* (new trace class $I_1 \in \{\frac{11}{12}, \frac{7}{6}\}$ not yet derived from block structure).

n	r	I_1	X (GeV)	Target	Residual	Identification	Class
3	8	1/2	2.159×10^8	2.160×10^8	0.06%	μ_{c_3} RG crossing	robust
5	5	1/12	1.307×10^{15}	1.31×10^{15}	0.2%	seesaw M_R	robust
6	6	1	1.272	1.27	0.2%	charm band	restricted-scan
8	1	17/16	0.105 9	0.105 66	0.2%	muon band	restricted-scan
10	1	1/3	8.826×10^{10}	direct search	—	PQ scale f_a	robust
12	2	41/48	246.35	246.22	0.05%	EW reference scale	robust
14	4	17/16	0.092 1	0.092 7	0.6%	f_π band	robust
15	4	1	0.935	0.938	0.3%	hadron anchor	robust
21	4	1/6	3.051×10^{13}	3.06×10^{13}	0.3%	inflation M	robust
21	5	41/48	171.8	172.6	0.4%	top quark	robust
24	7	5/8	7.895×10^5	7.920×10^5	0.3%	$\mu_{\varphi_0^{\rm ret}}$ RG crossing	robust
25	5	1	0.498	0.498	0.0%	kaon band	robust
25	7	41/48	125.5	125.25	0.2%	Higgs mass	robust
26	2	1	0.417	0.42	0.7%	$\sqrt{\sigma}$ confinement	robust
27	6	41/48	91.6	91.19	0.4%	Z boson	restricted-scan
28	1	7/6	5.105×10^{-4}	5.110×10^{-4}	0.1%	electron	tail conjecture
29	1	11/12	4.210	4.18	0.7%	bottom quark	tail conjecture

Table 17: Full E_8 stage atlas on the canonical branch up to the real-domain boundary $n = 29$.

B.4 Compression identities on the minimal branch

The main-text compression identities are developed in the dedicated section on compression identities and carrier exclusion. This appendix subsection collects additional arithmetic continuations and the compression chain summary.

Remark (Abelian package from occupancy and carrier norm). On the canonical branch,

$$\Omega_{\rm adm}\gamma = 48 \cdot \frac{5}{6} = 40.$$

Thus the characteristic coefficient 41 is compressed to

$$41 = \Omega_{\rm adm}\gamma + 1 = 40 + 1,$$

that is, fermionic occupancy trace plus one light Higgs doublet contribution. Writing $b_Y := 5b_1/3$ for the usual hypercharge-normalized coefficient and using the block-index relation $k_B = \frac{3}{2} I_1(B)$ to define $I_1^{\rm EW} := \frac{5}{24} b_1$ and $k_{\rm EW} := \frac{5}{16} b_1$, one obtains the unified chain

$$48\, I_1^{\rm EW} = 32\, k_{\rm EW} = 10\, b_1 = \Omega_{\rm adm}\gamma + 1 = 41. \tag{53}$$

Equivalently,

$$I_1^{\mathrm{EW}} = \frac{41}{48}, \qquad k_{\mathrm{EW}} = \frac{41}{32}, \qquad b_1 = \frac{41}{10}, \qquad b_Y = \frac{41}{6}.$$

This is an algebraic signature rather than an accidental coincidence: the same integer 41 appears simultaneously as electroweak block trace, E_8 exponent, UV hypercharge trace, and carrier occupancy times norm plus one.

Remark (Rank lock and single-seed compression: main-text reference). The rank-lock identity $2^{g-3} = g - 1 \Rightarrow g = 5$ is proved as Proposition 5.1 in the main text. The single-seed compression and the Cabibbo inversion $\varphi_0^{\mathrm{ret}} = \frac{1}{2}(1 - \sqrt{1 - 4\lambda_C^2})$ with all derived inter-observable relations are given in Equations (23) and (25). The carrier exclusion theorems for wrong discrete worlds are in Theorem 5.8.

B.5 Optional arithmetic continuations

The next relations mix the main-text carrier with optional appendix-level E_8, inflation, or geometric notation. They are useful compression patterns, but their interpretation should remain conditional or conjectural.

Remark (60, 48, and 5/4 on the optional ladder). If the optional scale ladder is started at $D_1 = 60$, then

$$\frac{D_1}{\Omega_{\mathrm{adm}}} = \frac{60}{48} = \frac{5}{4} = B\gamma.$$

On the minimal branch one may also write the same number as

$$\frac{5}{4} = \frac{g_{\mathrm{car}}}{g_{\mathrm{car}} - 1} \qquad (g_{\mathrm{car}} = 5).$$

This compresses the E_8 start index, the admissible chiral count, the mirror factor, and the two-defect coefficient to the same rational factor.

Remark (Factorization of the two defect coefficient and arithmetic completion of 48 and 60). From the main text identity

$$\delta_2 = \frac{5}{4}\,\delta_{\mathrm{top}}^2, \qquad \delta_{\mathrm{top}} = \Omega_{\mathrm{adm}} c_3^4,$$

one obtains

$$\delta_2 = 2880\, c_3^8.$$

On the canonical branch this factor admits the exact arithmetic decompositions

$$2880 = D_{\mathrm{start}}\Omega_{\mathrm{adm}} = 60 \cdot 48, \qquad 2880 = |R(E_8)|\,\dim \mathfrak{g}_{\mathrm{SM}} = 240 \cdot 12.$$

The same pair $(\Omega_{\mathrm{adm}}, D_{\mathrm{start}}) = (48, 60)$ also satisfies

$$\gcd(\Omega_{\mathrm{adm}}, D_{\mathrm{start}}) = 12 = \dim \mathfrak{g}_{\mathrm{SM}},$$

and

$$\mathrm{lcm}(\Omega_{\mathrm{adm}}, D_{\mathrm{start}}) = \frac{\Omega_{\mathrm{adm}} D_{\mathrm{start}}}{\gcd(\Omega_{\mathrm{adm}}, D_{\mathrm{start}})} = 240 = |R(E_8)|.$$

So the gauge algebra dimension is the common divisor of the carrier-side counting pair, while the E_8 root count is its least common multiple. The arithmetic identities are exact. Here the value 12 is the Lie-algebra dimension

$$\dim \mathfrak{g}_{\mathrm{SM}} = 8 + 3 + 1,$$

not a derivation from a discrete permutation group of order 12; at most the $3 + 2$ split carries such a discrete shadow. Their E_8 reading remains appendix level and conditional in the sense fixed elsewhere in the manuscript.

Conjecture B.1 (E_8 root-count reading)**.** On the same optional branch,

$$|R(E_8)| = 248 - 8 = 240 = 5 \cdot 48 = 5\,\Omega_{\rm adm}.$$

This suggests that the E_8 root count may be read as a fivefold unfolding of the admissible chiral packet, although that interpretation is not used as a main-text theorem.

Remark (Reactor seed of the optional scale ladder)*.* If one writes the optional ladder as

$$\varphi_n = \varphi_0^{\rm ret} e^{-\gamma} \left(\frac{D_n}{D_1}\right)^{\kappa_{E8}}, \qquad D_n = 60 - 2n,$$

then the start value is precisely the reactor-angle seed,

$$\varphi_n = \sin^2\theta_{13} \left(\frac{D_n}{D_1}\right)^{\kappa_{E8}}.$$

Thus the appendix-level ladder may be read as a discrete continuation of the same seed that controls θ_{13}.

Remark (Conditional inflation-side compression)*.* If one supplements the appendix with the conditional Starobinsky-side relations

$$A_s = \frac{N^2}{24\pi^2}\left(\frac{M}{\bar{M}_{\rm Pl}}\right)^2, \qquad \frac{M}{\bar{M}_{\rm Pl}} = \sqrt{8\pi}\, c_3^4,$$

then on the minimal branch

$$A_s = \frac{N^2}{3\pi} c_3^8 = \frac{N^2}{8640\pi}\,\delta_2,$$

and also

$$\frac{M}{\bar{M}_{\rm Pl}} = \frac{\sqrt{8\pi}}{\Omega_{\rm adm}}\,\delta_{\rm top}.$$

This is recorded only as an appendix-level compression identity because inflation is outside the present main-text claim surface.

Remark (Dyonic intercept and the α kernel)*.* From the dyonic appendix the local astrophysical β function reads

$$\beta_{\rm BH}(r) = \frac{Q_e Q_m}{256\pi^4 r^2} = \frac{\delta_{\rm top}}{3}\frac{Q_e Q_m}{r^2} = 16 c_3^4 \frac{Q_e Q_m}{r^2}.$$

This is a cross-link between the EHT intercept channel and the α kernel: the same topological coefficient $\delta_{\rm top} = 48 c_3^4$ that pushes α into the precision zone also controls the local dyonic β function. The local and cosmological sectors are therefore not two loose motifs but the same coefficient in two projections.

Remark (Conditional occupancy route to gravity)*.* If the relative Einstein term is of the form

$$S_{\rm EH}^{\rm rel} = \frac{\chi_{\rm geo}^2}{192\pi^2}\,{\rm Tr}_{\mathcal{H}_{\rm ch}}\,\mathbf{1} \int d^4x\sqrt{-g}\,R,$$

and if the family bridge identifies

$${\rm Tr}_{\mathcal{H}_{\rm ch}}\,\mathbf{1} = \Omega_{\rm adm} = 48,$$

then

$$\bar{M}_{\rm Pl}^2 = \frac{\chi_{\rm geo,0}^2}{2\pi^2}.$$

This is the occupancy-to-gravity route alluded to in the main text.

Remark (Compression chain)*.* The appendix-level compression pattern may therefore be summarized as

$$Y \leftrightarrow \iota, \qquad \gamma = \frac{5}{6}, \qquad B\gamma = \frac{5}{4}, \qquad \frac{D_1}{\Omega_{\rm adm}} = \frac{5}{4}, \qquad |R(E_8)| = 5\,\Omega_{\rm adm},$$

$$b_1 = \frac{\Omega_{\rm adm}\gamma + 1}{10}, \qquad \Omega_b + \beta_{\rm rad} = \varphi_0^{\rm ret}, \qquad A_s = \frac{N^2}{8640\pi}\,\delta_2 \quad \text{(conditional inflation-side import)}.$$

B.6 Conditional minimal-parameter picture

The next package is stronger but also more conditional. It assumes that the carrier rank, family multiplicity, and seam normalization are all read as closed rather than only partially inherited.

Principle B.2 (Conditional minimal-parameter compression)**.** Assume the canonical branch

$$g_{\mathrm{car}} := \operatorname{rank} E = 5, \qquad N_{\mathrm{fam}} = 3, \qquad c_3 = \frac{1}{8\pi}.$$

Then the numerical kernel may be rewritten almost entirely in terms of

$$(\pi, g_{\mathrm{car}}, N_{\mathrm{fam}}) = (\pi, 5, 3).$$

Indeed, on this branch one has

$$\gamma = \frac{5}{6} = \frac{g_{\mathrm{car}}}{g_{\mathrm{car}}+1}, \qquad \dim S^+ = 2^{g_{\mathrm{car}}-1}, \qquad \Omega_{\mathrm{adm}} = N_{\mathrm{fam}} 2^{g_{\mathrm{car}}-1} = 48,$$

$$\varphi_{\mathrm{base}} = \frac{1-\gamma}{\pi} = \frac{1}{(g_{\mathrm{car}}+1)\pi}, \qquad \delta_{\mathrm{top}} = \Omega_{\mathrm{adm}}\, c_3^4, \qquad \delta_2 = D_{\mathrm{start}}\, \Omega_{\mathrm{adm}}\, c_3^8,$$

with

$$D_{\mathrm{start}} = g_{\mathrm{car}} \dim \mathfrak{g}_{\mathrm{SM}} = 5 \cdot 12 = 60, \qquad \Omega_{\mathrm{adm}} = (g_{\mathrm{car}} - 1) \dim \mathfrak{g}_{\mathrm{SM}} = 4 \cdot 12 = 48.$$

Remark (Compressed dependency chain)*.* Under the same conditional closure, the appendix-level numerical chain reads

$$(\widetilde{M} \to M, \Sigma, \iota, \chi_{\mathrm{seed}}) \Longrightarrow E_3 \oplus E_2 \Longrightarrow Y \leftrightarrow \iota \Longrightarrow S(U(3) \times U(2)) \Longrightarrow S^+,$$

$$\Longrightarrow N_{\mathrm{fam}} = 3 \Longrightarrow \Omega_{\mathrm{adm}} = 48 \Longrightarrow (\delta_{\mathrm{top}}, \delta_2, \varphi_0^{\mathrm{ret}}) \Longrightarrow (\alpha, \lambda_C, \beta_{\mathrm{rad}}, \Omega_b, \sin^2\theta_{13}),$$

with the optional E_8, inflationary, and gravity-side sectors then read as downstream continuations of the same compressed core.

C Relative APS and superconnection setup

The geometric completion uses APS-type relative data [1]. In practice this means:

- one works with a declared pair (D, D_{ref}) rather than a single unanchored operator,
- boundary conditions on the seam Σ are part of the definition of the relative index,
- the η-term keeps track of the residual spectral asymmetry,
- the superconnection package is used in the same spirit as the spectral-action tradition [2], but here always with an explicit reference subtraction.

This appendix now serves as the proof motor for the geometric reconstruction statements in the main text: its role is to make the common reference structure explicit enough that the reconstruction theorem and the local spectral-scale branch are not left hanging on a slogan.

D Relative reflection positivity with fermions

D.1 Normalized quotient form

The relative Euclidean theory used in the main closure section is not interpreted as a difference of measures. Instead it is organized as the normalized quotient

$$Z_{\mathrm{rel}}[J, \eta, \bar{\eta}] = \frac{Z_{\mathrm{adm}}[J, \eta, \bar{\eta}]}{Z_{\mathrm{ref}}[0, 0, 0]}.$$

Because the denominator is source independent on the admissible sector, the reference theory acts only as a fixed normalizer and does not interfere with reflection positivity.

D.2 Bosonic and fermionic positivity on P_{adm}

The commutation relations

$$[\Theta, P_\Theta] = 0, \qquad \Theta P_{\mathrm{adm}} = P_{\mathrm{adm}}\Theta$$

ensure that reflection preserves the admissible subspace. The bosonic part then follows the standard Markov / reflection-stable Osterwalder–Schrader argument on admissible observables. For fermions, the admissible CAR kernel may be written as a reflected quadratic form; equivalently one may express the same condition as positivity of the associated Pfaffian structure on the admissible sector. Hence for every admissible observable O supported in $\tau > 0$,

$$\langle \Theta O, O\rangle_{\mathrm{rel}} = \frac{1}{Z_{\mathrm{ref}}[0,0,0]}\langle \Theta O, O\rangle_{\mathrm{adm}} \geq 0.$$

This appendix section is the mechanism behind Lemma 8.6 and theorems 8.15 and 8.16; it is recorded here so that the main-text reconstruction theorem is not supported only by a one-line axiom call.

E Yukawa kernels and positivity lemmas

This appendix is the proof motor for the transport positivity, strong-CP protection, and bounded-residual statements used in the main text. The transport program is based on the regularized kernel

$$\mathcal{Y}_f^{(\varepsilon)} = (D_f^\dagger D_f + \varepsilon^2)^{-1}.$$

Three remarks are important.

1. The regulator ε is not optional: without it the kernel becomes ill-defined exactly where the admissibility question is most delicate.
2. Positivity of the kernel belongs to the admissibility-closure layer rather than to the hard carrier kernel. The main text closes it through the finite hexagon gap and the lift to the full admissibility-projected operator; this appendix records the transport-side mechanism behind that closure.
3. Residual prefactors $\Lambda_{f,j}$ are bounded holonomy data inside the transport theorem. They must remain visible and must not be rhetorically hidden as if the mass map were diagonal without residual transport structure.

Remark (Positive-kernel determinant route to strong CP). If the physical mass map is written in the polar form

$$M_f = U_{f,L} H_f U_{f,R}^\dagger, \qquad H_f > 0, \qquad U_{f,L}, U_{f,R} \in SU(3)_F,$$

with H_f induced by the positive transport kernel $\mathcal{Y}_f^{(\varepsilon)}$, then the induced determinant phase vanishes:

$$\arg\det M_f = 0.$$

At the same time weak CP may survive through

$$V_{\mathrm{CKM}} = U_{u,L}^\dagger U_{d,L}.$$

Combined with the sheet-CP symmetry protection of θ_{sheet} in the relative action, this is the algebraic core of the strong-CP theorem stated in the main text. The role of the appendix remark is to display the polar-decomposition mechanics behind that theorem, not to reopen its status.

E.1 Optional strong-coupling sharpening

As an appendix-level sharpening of the closed confinement theorem, one may further require the admissible center-neutral Wilson functional to obey strong-coupling dominance. On that optional route the relative Wilson loop satisfies

$$\langle W(C)\rangle_{\mathrm{rel}} \leq \exp[-\sigma A_{\min}(C)], \qquad \sigma = c_3^2 \chi_{\mathrm{QCD}}^2,$$

and the physical Hilbert space contains no color-nonsinglet asymptotic states. This statement is kept in the appendix so that the main theorem surface does not depend on an extra strong-coupling sharpening beyond the closed admissible hadronic sector.

F Information-theoretic and cut-and-project readings

This appendix collects interpretive notes that help organize the algebraic structure of the carrier-first compiler. When an exact algebraic statement appears below, it refers back to a main-text proposition; the appendix itself does not upgrade any status label beyond the declared hard / closure / benchmark / conditional discipline.

F.1 Integrity-index overdetermination of the canonical branch

The canonical 41-chain admits a compact integrity-table reading:

Layer	Value	Role
Electroweak trace block $48\, I_1^{\mathrm{EW}}$	41	retained electroweak trace packet
Block exponent $32\, k_{\mathrm{EW}}$	41	E_8-side compression datum
UV hypercharge coefficient $10\, b_1$	41	one-loop electroweak closure datum
Occupancy-weighted carrier norm $\Omega_{\mathrm{adm}}\gamma + 1$	41	carrier / family compression identity

Table 18: Integrity-index overdetermination of the canonical branch. The common integer is read as an algebraic integrity marker, not as a cryptographic statement.

The value of this table is structural rather than rhetorical: four independently defined layers collapse onto one integer on the retained branch. This is exactly the sense in which the branch carries the integrity index $\mathcal{I}_{41}$.

F.2 Seed variance and surprisal readings

The bridge identities also admit a disciplined probabilistic reading. Since

$$\lambda_C^2 = \varphi_0^{\mathrm{ret}}(1 - \varphi_0^{\mathrm{ret}}),$$

the Cabibbo scale is exactly the standard-deviation scale of a Bernoulli variable with success probability $p = \varphi_0^{\mathrm{ret}}$:

$$\lambda_C = \sqrt{p(1-p)}.$$

Equivalently, the Fisher information of the Bernoulli family is

$$I(p) = \frac{1}{p(1-p)} = \frac{1}{\lambda_C^2},$$

so the Cabibbo bridge measures the inverse square-root Fisher scale of the retained seed. This is the precise information-theoretic statement. The present paper does *not* identify λ_C itself with Fisher information or with a Bures distance.

Likewise, whenever the electromagnetic closure is written with

$$\ln\big((\varphi_0^{\mathrm{ret}})^{-1}\big),$$

the exact probabilistic reading is the surprisal of the seed event. That term should not be called the Shannon entropy of the full Bernoulli distribution, which would instead be

$$H(\varphi_0^{\mathrm{ret}}) = -\varphi_0^{\mathrm{ret}} \ln \varphi_0^{\mathrm{ret}} - (1-\varphi_0^{\mathrm{ret}}) \ln(1-\varphi_0^{\mathrm{ret}}).$$

Finally, the neutrino bridge relation

$$\sin^2\theta_{13} = \varphi_0^{\mathrm{ret}} e^{-\gamma}$$

is naturally described as Boltzmann-like weighting by the carrier norm γ, but the present draft does not define a full thermodynamic partition function at that point. These readings are therefore organizational interpretations of already established formulas, not new closure theorems.

F.3 Exact parity-code and information-compression reading

The strongest information-theoretic statement is no longer a loose analogy. By Proposition 3.14, the one-family packet is exactly the even-parity subspace on five occupation bits,

$$S^+ \cong \mathrm{span}\Big\{|x\rangle : \sum_{i=1}^{5} x_i \equiv 0 \pmod 2\Big\},$$

so the carrier packet is kinematically the classical even-parity $[5,4,2]$ code. The split formula

$$S^+ = (\Lambda^{\mathrm{even}} E_3 \otimes \Lambda^{\mathrm{even}} E_2) \oplus (\Lambda^{\mathrm{odd}} E_3 \otimes \Lambda^{\mathrm{odd}} E_2)$$

shows that the coarse parity bits of the 3-sector and 2-sector are exactly locked. For the maximally mixed state on the 16 basis words, the two parity observables are perfectly correlated and therefore carry exactly one bit of mutual information.

The split weight enumerator in Proposition 3.15 is the corresponding code enumerator with hypercharge weights. It packages the six admissible occupancy sectors into the single polynomial

$$W(u,v) = 1 + 3u^2 + 6uv + v^2 + 2u^3v + 3u^2v^2.$$

The hypercharge-compression result of Proposition 3.16 sharpens the same picture further. On the uniform distribution over the 16 carrier basis states, the six Y-sectors have probabilities

$$\frac{1}{16}(1,3,6,1,2,3),$$

so the hypercharge entropy is

$$H(Y) = -\sum_i p_i \log_2 p_i = \frac{7}{2} - \frac{3}{4}\log_2 3 \approx 2.311 \text{ bits}.$$

Since the total carrier word takes 4 bits to specify, the unresolved nonabelian degeneracy is

$$4 - H(Y) \approx 1.689 \text{ bits}.$$

Thus hypercharge is an efficient family-sector compressor, but not a lossless label for the individual basis word.

Remark (Two code layers: kinematic and admissible). The code reading should be split into two layers. First, S^+ is exactly the classical even-parity $[5, 4, 2]$ code on five occupation bits. Second, on the commuting admissible branch the protected physical sector is

$$\mathcal{H}_{\mathrm{adm}} := \mathrm{Ran}(P_{\mathrm{adm}}) = \ker K_{\mathrm{adm}} = \ker \Delta_{\mathrm{adm}},$$

where the last equality uses the Hodge description of Lemma 8.2. In that second layer the penalties $K_\Sigma, K_c, K_\Theta, K_g$ act as stabilizer-like sector constraints, and P_{adm} is the Hodge projector onto the admissible physical subspace. This is the precise bridge to quantum-information language used in the manuscript. It is not yet a theorem of a self-correcting quantum memory, because no noise model, logical algebra, or lifetime analysis is derived here.

F.4 Conditional cut-and-project reading

The split

$$E = E_3 \oplus E_2$$

also admits a possible cut-and-project interpretation, with a three-dimensional visible sector and a two-dimensional internal sector. Such a reading would become mathematically sharp only after an explicit rank-5 lattice or module, a projection rule, and a seam/window selection theorem are written down. Until then, the quasicrystal language is only a conditional geometric reading of the carrier split rather than part of the hard core.

G Appendix elaboration on record algebra, horizons, and prediction semantics

This appendix elaborates the main-text record and prediction constructions while keeping the more speculative horizon extrapolations outside the core theorem stack.

G.1 Minimal admissible vacuum

At finite volume or with a declared cutoff, an admissible vacuum candidate is written as

$$\rho_B := \arg\min_{\rho \in \mathcal{V}_{\mathrm{adm}}} \mathrm{Tr}(\rho \mathcal{H}_{\mathrm{rel}}),$$

with lower boundedness and coercivity understood exactly as in the main-text closure axioms. The stronger generative theorem path described in the conclusion would refine this object to the global admissible vacuum functional $\mathfrak{V}[g, \chi_{\mathrm{geo}}, \Phi, \nabla_F, \delta, \{L_f\}]$ whose variations also select the transport branch. In this appendix, however, the same vacuum object is used only as the background state for the record-algebra elaborations; no stronger cosmological claim is added here beyond the closed existence statement already declared in the main theorem stack.

G.2 Record growth principle

One useful coarse-grained record algebra is

$$\mathcal{A}_{\mathrm{rec}}(t) := \bigoplus_{\Delta_i > H_{\mathrm{hub}}(t)} \mathcal{A}_i, \qquad S_{\mathrm{rec}}(t) = \log \dim \mathcal{A}_{\mathrm{rec}}(t).$$

As $H_{\mathrm{hub}}(t)$ decreases, more stable channels open. The intended arrow-of-time claim is therefore

$$\frac{dS_{\mathrm{rec}}}{dt} > 0,$$

but only as a record-growth principle, not as a replacement for detailed nonequilibrium dynamics.

G.3 Observer algebra and Born-type evaluation

An observer is modeled as a stable pointer subalgebra

$$\mathcal{A}_{\text{obs}} \subset \mathcal{A}_{\text{rec}}.$$

With orthogonal projectors $P_i \in \mathcal{A}_{\text{obs}}$ one writes the measurement channel

$$\mathcal{M}_O(\rho) = \sum_i (P_i \rho P_i) \otimes |i\rangle\langle i|.$$

In finite-dimensional sectors, positivity, additivity on orthogonal decompositions, and phase invariance lead to the standard Born-type assignment

$$p_i = \text{Tr}(\rho P_i).$$

The claim is therefore only a constrained evaluation principle on admissible sectors, not a new theorem about measurement in all settings.

G.4 Horizons as conjectural inner seams

If a horizon is modeled as an inner seam, the manuscript uses only an approximate factorization language,

$$\mathcal{H}_{\text{out}} \otimes \mathcal{H}_H \otimes \mathcal{H}_{\text{in}},$$

or, better, an algebraic split on an appropriate code subspace. This appendix intentionally avoids selling the factorization as exact fundamental structure.

G.5 Prediction semantics

The main-text prediction map can be decorated by explicit experimental metadata as

$$Q = (\rho_0, \mathcal{O}_Q, \Delta_Q, t, \mathcal{C}_Q, \mathcal{R}_Q), \qquad \text{Pred}(Q) = \mathcal{R}_Q(\text{Tr}(\rho_t E_Q(\Delta)), \mathcal{C}_Q)\,,$$

with

$$\rho_t = U(t)\rho_0 U^\dagger(t).$$

This is kept only as an optional formal extension.

H Extended benchmarks

This appendix turns the broader prediction surface into explicit experimental bookkeeping. The point is not to promote every row to main-text benchmark status, but to keep next test, kill criterion, and dependency structure visible wherever the theory already commits to a concrete target.

H.1 Operational prediction matrix

The appendix matrix keeps the same truth-mode discipline as the main text, but its final column adds a short row-level qualifier whenever an appendix-level auxiliary continuation needs to be made explicit.

Observable	TFPT target	Experimental reading next test	Kill criterion	Dependency class	Status note
Cosmic birefringence β	0.2424°	calibrated achromatic CMB rotation fit without EB self-nulling	calibrated $\beta = 0$ within $\pm 0.05^\circ$	seed	Intrinsic observation
δ_{CKM}	1.198 rad	global flavor fit	stable exclusion at $\geq 3\sigma$	flavor observation	Intrinsic observation exact phase
Rare kaons $K \to \pi\nu\bar{\nu}$	$\mathrm{BR}(K^+) = 8.22 \times 10^{-11}$ $\mathrm{BR}(K_L) = 2.65 \times 10^{-11}$	NA62 / KOTO Grossman–Nir test	K^+ outside $[6, 10] \times 10^{-11}$ or K_L mismatch	flavor observation	Intrinsic observation
PMNS phase / octant	$\delta_{\mathrm{CP}} = 240^\circ$ $\sin^2\theta_{23} = 0.4557$	global oscillation fit	exclude 240° or the lower octant at $\geq 3\sigma$	neutrino observation	Intrinsic observation
Σm_ν	5.8764×10^{-2} eV	cosmological mass-sum analyses	robust upper bound below the branch value	neutrino observation	Intrinsic observation
$m_{\beta\beta}$	1.516×10^{-3} eV	light-Majorana $0\nu\beta\beta$ searches	detection implying $m_{\beta\beta} \gtrsim 1 \times 10^{-2}$ eV	neutrino observation	Intrinsic observation
nEDM / strong CP	$\bar{\theta} = 0$	next-generation EDM searches	stable nonzero hadronic EDM signal	closure	Closure
Axion haloscope window	$m_a \approx 65.19$ μeV $\nu_a \approx 15.764$ GHz	haloscope scan at 15.764 GHz in the quoted window	exclusion in 15.764 GHz ± 50 MHz	cosmology observation	Intrinsic observation
m_{π^0}	134.979 MeV	hadronic theorem row	robust mismatch outside the stated uncertainty budget	closure	Derived hadronic row
η_B	5.97×10^{-10}	flavored leptogenesis consistency test	robust exclusion of the quoted branch value	cosmology observation	Intrinsic observation

Table 19: Operational prediction matrix extending the main-text prediction surface. Each row carries its own status note, experimental reading, kill criterion, and dependency class.

H.2 Target and kill-test rows

Test row	TFPT target or prohibition	Status reading	Failure mode
Second light Higgs doublet	no additional seam-even light doublet	kill test	robust discovery would falsify the bosonic-index route
Calibrated birefringence null	$\beta = 0.2424^\circ$ on the minimal branch	kill test	externally calibrated null consistent with zero at the stated tolerance would remove the minimal birefringence sector
Rare-kaon corridor	$\mathrm{BR}(K^+) = 8.22 \times 10^{-11}$ $\mathrm{BR}(K_L) = 2.65 \times 10^{-11}$	kill test	a stable kaon result outside the quoted corridor or incompatible GN-plane correlation would break the present flavor bridge
Strong-CP / EDM signal	$\bar{\theta} = 0$ on the admissible branch	kill test	stable nonzero EDM signal would break the present closure route
Cosmological neutrino sum	$\Sigma m_\nu = 5.8764 \times 10^{-2}$ eV from intrinsic neutrino closure	pressure test	a robust upper bound below the stated branch value would force a neutrino-sector revision
Haloscope window	$\nu_a \approx 15.76$ GHz from cosmology closure	target row	exclusion in the relevant coupled window would pressure the intrinsic axion row

Table 20: Target and kill-test rows. These are not current confirmations; they describe the most direct falsification or pressure points of the present branch.

H.3 Axion interface

The axion rows are read as intrinsic observation outputs of the cosmology block rather than as an independent hard theorem. On the closed cosmology branch,

$$f_a \approx 8.86 \times 10^{10}\,\mathrm{GeV}, \qquad m_a \approx 65.19\,\mu\mathrm{eV}, \qquad \nu_a \approx 15.764\,\mathrm{GHz}.$$

In standard haloscope units this gives

$$\left|g_{a\gamma\gamma}^{(\mathrm{phys})}\right| = \frac{|g_{a\gamma\gamma}|}{f_a} \approx 1.80 \times 10^{-12}\,\mathrm{GeV}^{-1},$$

so the practical scan prescription is a conservative window

$$\nu_a = 15.764\,\mathrm{GHz} \pm 50\,\mathrm{MHz}$$

at the quoted coupled sensitivity. This interface depends on the downstream PQ / scale-grammar branch and should therefore not be read with the same independence status as the hard carrier kernel or the closure theorems.

H.4 Supplementary mass-source rows

The compact main-text mass ledgers intentionally suppress explicit formula detail. The source rows collected here retain that information without forcing the main text to read like a mixed formula notebook.

Particle	TFPT formula or source row	TFPT value	Status note
e	$\frac{v_{\mathrm{geo}}}{\sqrt{2}}\,\frac{16\pi}{7}(\varphi_0^{\mathrm{ret}})^5$	0.52197 MeV	derived charged-lepton row uses v_{geo}, not $v_{\mathrm{EW}}^{\mathrm{ref}}$
μ	$\frac{v_{\mathrm{geo}}}{\sqrt{2}}\,\frac{4\pi}{3}(\varphi_0^{\mathrm{ret}})^3$	107.70 MeV	derived charged-lepton row uses v_{geo}, not $v_{\mathrm{EW}}^{\mathrm{ref}}$
τ	$\frac{v_{\mathrm{geo}}}{\sqrt{2}}\,\frac{7\pi}{6}(\varphi_0^{\mathrm{ret}})^2$	1.7723 GeV	derived charged-lepton row uses v_{geo}, not $v_{\mathrm{EW}}^{\mathrm{ref}}$
Intrinsic neutrino closure:			
ν_1	intrinsic transport closure	0	intrinsic observation
ν_2	intrinsic transport closure	8.614×10^{-3} eV	intrinsic observation
ν_3	intrinsic transport closure	5.015×10^{-2} eV	intrinsic observation
Σm_ν	intrinsic transport closure sum	5.8764×10^{-2} eV	intrinsic observation
u	ChPT ratio inversion anchored to m_s	2.195 MeV	derived; $\overline{\mathrm{MS}}$ at 2 GeV
d	$m_s/(m_s/m_d)$	4.67 MeV	derived; $\overline{\mathrm{MS}}$ at 2 GeV
s	strange-quark observation row in the light-quark solver	93.4 MeV	intrinsic observation; $\overline{\mathrm{MS}}$ at 2 GeV
c	up-type ratio ladder with $m_c/m_u = 486.92$	1.27 GeV	intrinsic observation after threshold matching
b	down-type ratio ladder with $m_b/m_s = 43.79$	4.18 GeV	intrinsic observation after threshold matching
t	up-type ratio ladder with $m_t/m_c = 133.01$	172.57 GeV	intrinsic observation; pole / $\overline{\mathrm{MS}}$ distinction remains external
π^0	GMOR relation from $(m_u + m_d)$ and $\langle\bar{q}q\rangle$	134.979 MeV	hadronic out-of-sample check
p	hadron mass ladder	935.03 MeV	derived hadronic scale check

Table 21: Supplementary mass-source rows referenced by the compact main-text mass surface.

For the main-text baryon-fraction comparison we use the Planck 2018 base-ΛCDM conversion

$$\Omega_b = \frac{\Omega_b h^2}{h^2}, \qquad h = \frac{H_0}{100},$$

so the quoted Ω_b row is not a free-floating cosmology number but an explicit reconstruction from the Planck parameter pair $(\Omega_b h^2, H_0)$ adopted in the suite ledger.

H.5 Extended compiler ledger

Row	TFPT source or target	TFPT value	Status note
m_γ	exact unbroken $U(1)_{\mathrm{em}}$ direction	0	exact structural row
m_g	exact color gauge direction before confinement	0	exact structural row
$W^\pm$	intrinsic threshold output at M_Z	80.369 GeV	intrinsic observation row
Z	intrinsic threshold output at M_Z	91.188 GeV	intrinsic observation row
H	intrinsic threshold output	125.25 GeV	intrinsic observation row
$\hat{s}_Z^2$ (M_Z)	local gauge closure plus seam-even matching	0.23116113	intrinsic observation row
α_s (M_Z)	local gauge closure plus singlet Wilson matching	0.11796213	intrinsic observation row
Σm_ν	intrinsic neutrino closure sum	5.8764×10^{-2} eV	intrinsic observation row
M_R	appendix-level seesaw scale note / stage-atlas node	1.31×10^{15} GeV	appendix-level auxiliary continuation
$\Sigma m_\nu^{\mathrm{ss}}$	appendix-level seesaw comparison from v_{geo} and M_R	5.764×10^{-2} eV	appendix-level auxiliary continuation

Table 22: Supplementary gauge, Higgs, and neutrino rows for the extended compiler ledger. Exact zero rows, intrinsic observation rows, and appendix-level auxiliary continuations are separated explicitly.

Field	Word-length data $L_{f,j}$	Derived transport matrix element	Status note
e	8	0.466	Derived
μ	5	1.085	Derived
τ	3	0.899	Derived
u	7	0.440	Derived
d	7	0.936	Derived
s	5	0.943	Derived
c	3	0.646	Derived
b	2	0.480	Derived
t	0	0.986	Derived

Table 23: Transport closure data as theorem outputs.

H.6 Novelty boundary and literature positioning

The paper uses standard tools from relative spectral geometry, APS index theory, transport/resolvent language, and conventional RG comparison. The claim of novelty is not that these background ingredients are new in isolation, but that the minimal $3+2$ carrier packet, the compression identities, and the carrier-first compiler architecture are organized into one tightly overdetermined program. Conversely, the manuscript does not claim novelty for standard benchmark conventions, nor does it claim that appendix-level information-theoretic or quasicrystal readings are part of the hard core.

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Symbol	First-duty meaning in the paper	Status class
$\mathfrak{T}_{\mathrm{prim}}$	primitive relative spectral datum $(X, S, \mathcal{H}_{\mathrm{int}}, D_{\mathrm{rel}}, \iota, f, \chi_{\mathrm{seed}})$ with $X = \widetilde{M} \setminus \Sigma$; appendix rewrite $\mathfrak{T}_{\mathrm{core}} = (\widetilde{M} \to M, \Sigma, S, \mathcal{H}_{\mathrm{int}}, \iota, D_{\mathrm{rel}}, f, \chi_{\mathrm{seed}})$	Derived
χ_{seed}	primitive seam-even scalar response reconstructed on the primitive layer before geometric reconstruction	Derived
$\sigma,\ \chi_{\mathrm{geo}}$	local Weyl field and derived local spectral scale $\chi_{\mathrm{geo}} = \Lambda e^{\sigma}$ on the geometric branch	Derived
$\mathrm{APS}_{\mathrm{dom}},\ \mathrm{APS}_{\mathrm{fam}}$	elliptic APS domain for primitive well-posedness and the canonically induced family-pairing data on the closed branch	Derived
$\mathfrak{T}_{\mathrm{min}}$	reduced theorem-path datum $(\mathcal{A}, \mathcal{H}, D, J, \Gamma, \iota, \nu)$ with positive spectral measure ν	Reduced datum
$E_3 \oplus E_2$	minimal carrier split into color and weak sectors	Derived
$b_1(N_{\mathrm{fam}}, N_{\Phi})$	carrier-side abelian coefficient function before canonical branch evaluation	Derived
$\mathsf{Adm}, K_{\mathrm{adm}}, P_{\mathrm{adm}}$	admissibility predicate, positive constraint operator, and operative selector on the admissible branch	Derived
F, Ω_{adm}	APS family local system and occupancy count after family closure; Ω_{adm} later feeds the observation reading of δ_{top}	Derived
$\mathcal{I}_{41}$	weighted integrity index $10\, b_1 = \Omega_{\mathrm{occ}}\gamma + N_{\Phi}$ on the canonical branch	Derived
δ_{ph}	exact transport stationary point on the C_6 hexagon	Derived
$\mathcal{R}_{\mathrm{SM}}$	practical finite-threshold numerical preconditioner for the scheme-specified observation map	Intrinsic observation support
$\mathcal{S}_{\mathrm{int}}, \mathcal{S}_{\mathrm{obs}}$	intrinsic closure surface and observation surface $\mathcal{R}_{\mathrm{TOE}}(\mathcal{S}_{\mathrm{int}})$ for the compiler interface	Derived / intrinsic observation
$\hat{m}_f,\ m_f^{\mathrm{obs}}$	internal compiler source masses and their observation-surface images after matching	Derived / intrinsic observation
$M_{\Sigma},\ g_{\Sigma}$	seam matching scale read from the first positive seam boundary mode and the gauge coupling evaluated there	Derived / intrinsic observation
Q_{Θ}	determinant-sector involution entering admissibility and the projector P_{Θ}	Closure
$\mathcal{C}_{\Sigma}$	sheet-CP symmetry operator $CP \circ \iota$ on the admissible branch	Closure
T	transport generator on the family / holonomy side	Closure
$\mathfrak{C}_{\Sigma}$	seam cycle data used by the matching and holonomy packages	Closure / benchmark support
ϖ_{spin}	spin holonomy weight entering the geometric package	Closure
$\mathcal{A}_{\mathrm{rec}}, \mathcal{A}_{\mathrm{obs}}, \mathrm{Pred}$	stable record algebra, observer algebra, and prediction map on the admissible sector	Closure

Table 15: Symbol guide with first-duty meaning and status class.

Symbol	Definition	Compiler role	First origin	Reappears in	Physical meaning
θ	seam-loop angle	core datum	seam data	a_θ	geometric rotation variable
a_θ	$\theta/(4\pi)$	hard normalization	spin lift	c_3, phase grammar	normalized cover rotation
c_3	$1/(8\pi)$	hard kernel constant	seam normalization	δ_{top}, β, σ	seam normalization constant
φ_{base}	$1/(6\pi)$	hard kernel constant	carrier geometry	φ_0^{ret}	base opening
δ_{top}	$\frac{3}{256\pi^4}$; observation reading $\Omega_{\mathrm{adm}} c_3^4$	hard kernel value with occupancy reinterpretation	retained numerical kernel	φ_0^{ret}, δ_2	topological surcharge
φ_0^{ret}	$\frac{1}{6\pi} + \frac{3}{256\pi^4}$	retained canonical seed	retained numerical kernel	λ_Y, β, flavor	minimal nontrivial opening
γ	$\mathrm{Tr}_E\, Y^2 = 5/6$	hard kernel invariant	carrier packet	κ_{E8}, δ_2	carrier norm
B	$\mathrm{rank}\, E_3/\mathrm{rank}\, E_2 = 3/2$	hard kernel invariant	carrier packet	δ_2	carrier rank ratio
Ω_{adm}	$3 \times 16 = 48$	closure quantity with bridge reinterpretation	APS family closure	δ_{top}	admissible chiral-state count on the family-closed branch
b_1	$\frac{4}{3} N_{\mathrm{fam}} + \frac{1}{10} N_\Phi$	hard kernel output	abelian index	benchmark map	hypercharge trace coefficient
κ_{E8}	$\gamma/\ln(248/60)$	conditional grammar	Appendix B	scale ladder	optional scale spacing

Table 16: Constants atlas for the present manuscript.